

Potter Engineering Center Renovation Study

Presented by American Structurepoint
in association with Applied Engineering and Blundall Associates

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PROJECT SUMMARY

The A.A. Potter Center is a classic example of a 1970's institutional building that could easily be dismissed.

From the exterior, large expanses of unbroken brick façade, the unyielding repetition of precast panels create a bland, dated, uninspiring image. On the interior, the triple combination of closed offices lining the perimeter, small deeply shaded windows, and square building footprint combine to create a severe lack of interior daylight. In its current state, it is not a structure fit to inspire students or staff.

These flaws, however, are not impossible to overcome. Moreover, the Potter Center has more qualities in its favor than it has negatives. Though unattractive, the exterior skin is in good condition and structurally sound. The structure itself is a concrete waffle slab system, which is not only in good condition but could be quite attractive if exposed. Floor-to-floor heights are generous. Its greatest asset though, is its location. The Potter Center sits in at the heart of campus, immediately adjacent to the Wilmeth Active Learning Center and Centennial Mall and a stone's throw from the Purdue Mall.

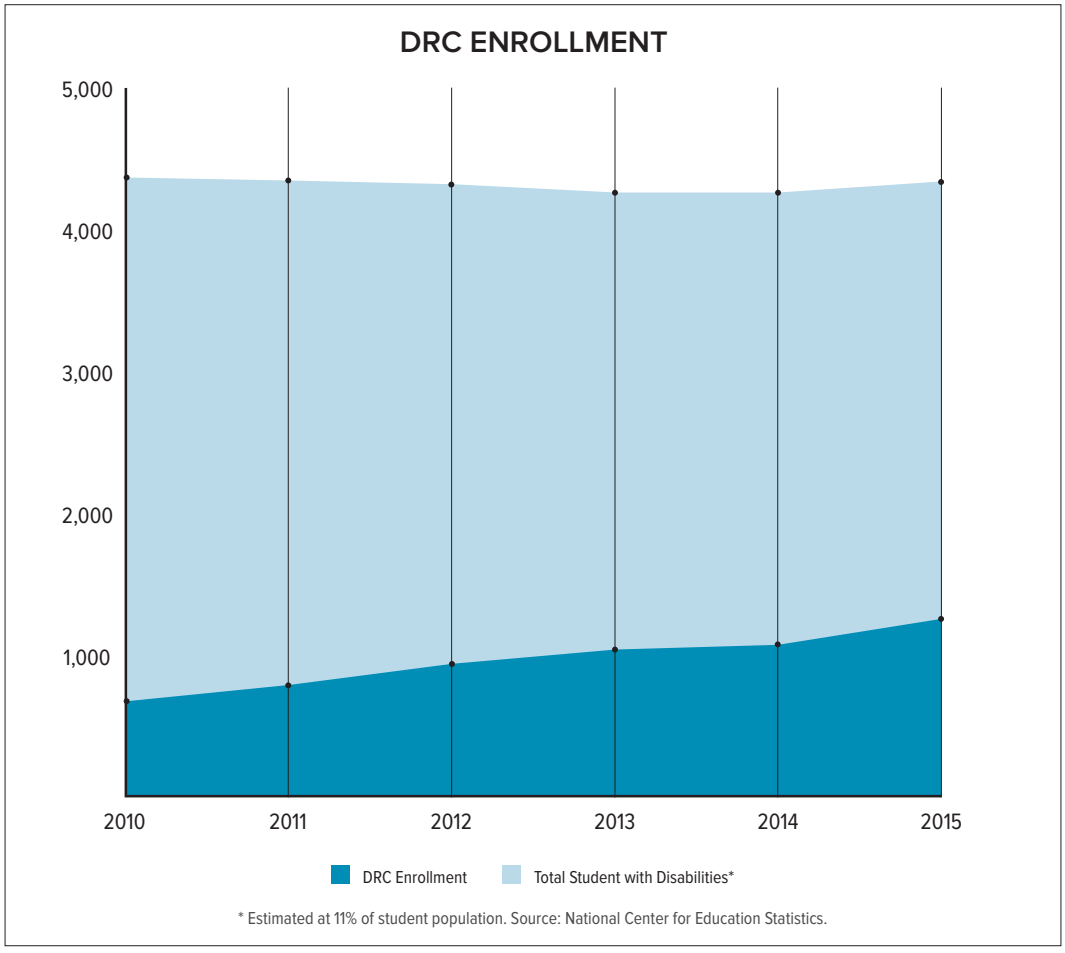
The Potter Center is the definitive example of a building in need of a renovation. Done thoughtfully, a renovation could overcome its current bad qualities and expand on its assets, easily breathing new life into an ideally located building.

PROGRAM

The programs/departments included in this study can generally be combined into three synergistic groups.

1. DISABILITY RESOURCE CENTER, ASSISTIVE TECHNOLOGY CENTER, INSTRUCTIONAL DATA PROCESSING, ONLINE COMPLIANCE

The Disability Resource Center is currently located on the eighth floor of Young Hall. As a department whose sole purpose is serving students with disabilities, this is not an ideal location. This study proposes to move the DRC into a more accessible, open, and welcoming first floor location. Additionally, this study proposes to expand the existing DRC program, which has expanded 85.5% over the last five years. While growing, these numbers appear to actually under-represent the number of students with disabilities. This number is expected to continue to increase as society continues to improve its understanding of disability as another form of diversity, civil rights, and social justice, allowing students to overcome the societal stigma of disability.



In addition to the DRC itself, the study proposes to co-locate several departments alongside the DRC to maximize efficiency, synergies, and productivity among those departments. The Assistive Technology Center provides students with disabilities the accommodations to do schoolwork, much in the same way the DRC currently provides students with disabilities the accommodations to take tests. This department has a very strong relationship and similar mission as the Alternate Format Production unit of the DRC.

Instructional Data Processing provides test scoring and evaluation services. IDP manages course evaluations, test proctoring services, Scantron grading, specialized gradebook work for certain departments, and Scantron scoring in support of research. The DRC's accommodated testing function primarily supports paper and pencil testing and is thus a major client of the IDP.

Online Compliance is a new unit, housed under Digital Education and led by an Associate Director of Digital Education. This department will ensure accommodation of digital information in various digital assets. Much of the work of this unit will support efforts of the DRC, although Digital Education will be another major user. The Associate Director position is funded and was recently filled. The remaining positions are yet to be funded. Therefore, these staff are new and not connected with the existing units.

2. CENTRAL TESTING CENTER (COMBINED DIGITAL EDUCATION TESTING CENTER, DRC ACCOMMODATED TESTING, FOREIGN LANGUAGE TESTING, PURDUE TESTING CENTER)

Rationale for Creation of a Centralized Testing Center

The Central Testing Center does not currently exist on campus. The University currently has several testing units that simultaneously provide similar testing services. These are the Disability Resource Center, Accommodated Testing, Purdue Testing Center, Digital Education Testing Center, and Foreign Language Testing. Each of these units has a small staff and small facility.

Because of the dispersion of testing programs, instructors must work with multiple offices for testing. A single exam might require working with the DRC for accommodations and perhaps an alternative format Instructional Data Processing (IDP) for proctoring and/or Scantrons, as well as Digital Education for online exams. There is currently no coordination among these different units.

Similarly, students work with multiple testing offices scattered across the campus. Students taking online courses may take their exam with Digital Education's testing unit. Students with disabilities may have to take some or all of their course-based exams through the services of the DRC's accommodated testing unit. Still other students may seek out the services of the Purdue Testing Center to take a professional certification, equivalency, or graduate entrance exam (among others). These are all separate units with differing reporting structures and very small staffs.

The existing facilities are not purpose-built. Therefore, each unit in isolation is limited in its ability to serve students. For instance, a two-person unit that is short one person cannot deliver a test (in some cases) due to the demand for two proctors. By bringing these units together in one unit and one space, the depth in staff and the purpose-built facility will be able to expand the ability of the comprehensive unit to serve more students in more ways.

In addition to the benefits to ease of use, there's an overall benefit to campus as well. By combining several small testing centers into one large robust center, an efficiency of scale can be realized, and more students can be serviced with less area. The net increase in space allocated for student testing is 4,145 square feet. However, the proposed testing center can service 247 percent more students in 44 percent less area per student than the existing, individual testing centers.

Overview of Combined Units

The primary accommodation offered by the Disability Resource Center is accommodated testing. Because the DRC will be a major contributor of students to the central testing center, there is a strong relationship between the DRC and the proposed CTC. During mid-term and final exam periods, the demand for accommodated testing exceeds the capacity of the DRC's testing space by a factor

greater than 10. The DRC is then forced to use libraries and other spaces to house accommodated exams. Additionally, the DRC staff must be augmented by volunteer and paid proctors from its parent unit (Student Success) and other units at Purdue. This Central Testing Center will create the ability for accommodated testing to occur in a much more efficient and controlled way and with a deeper overall testing staff (to oversee the inevitable volunteer proctors).

Foreign Language Testing is currently conducted in a general use computer laboratory. During summer orientation (STAR) this space becomes overcrowded, loud, and inefficient. This department would benefit greatly from a purpose built, high capacity testing center.

The Digital Education Testing Center is an on-campus proctored testing site that administers a variety of paper-based and computerized exams. Tests administered in this center are for Purdue-based distance learning courses, as well as CASA (Core Academic Skills Assessment) and Pearson Vue exams.

The Purdue Testing Center serves campus and patrons outside of campus by providing a variety of testing services. They are the only center on campus fully certified by national testing organizations and offer national registration and licensing exams. The tests administered by this department in addition to some tests given by the Digital Education Testing Center, represent the core of tests that will be administered in the high stakes testing area of the Central Testing Center.

Existing Testing Centers	Seats	Area	Tests/Day*	SF/Seat
Disability Resource Center, Accommodated Testing	24	1,515	15	63
Purdue Testing Center	16	952	30	60
Digital Education	25	2,790	20	112
Foreign Language Placement*	36	1,744	30	48
Total	**101**	**7,001**	**95**	**283**
Average	**25**	**1,750**	**24**	**71**

* Foreign Language Placement testing does not occur year round. Area shown does not include administration space.

Proposed Central Testing Center	Seats	Area	Tests/Day*	SF/Seat
Central Testing Center - Option B	250	11,871	TBD	47
Central Testing Center - Option C	237	11,192	TBD	47
Central Testing Center - Option D	250	11,416	TBD	46
Average	**246**	**11,493**		**47**

3. COLLEGE OF ENGINEERING

In 2014, the College of Engineering completed a campuswide master plan and identified the following goals.

- Provide appropriate office and office support space for the anticipated number of personnel
- Create a state-of-the-art teaching and learning environment
- Create a state-of-the-art research environment
- Incorporate state-of-the-art technology
- Create collaboration and interaction environments
- Enhance Purdue's close relationship with state and industry partners
- Create a community with a sense of place that proclaims the stature of the College of Engineering
- Align with the Campus Master Plan

This plan was driven by the need to provide more and greater quality space to meets the College's strategic growth initiative, including increased student enrollment and 30 percent personnel growth,

as well as the need to efficiently and strategically accommodate that growth. Within that plan, the A.A. Potter Center was identified as the ideal home to co-locate the COE's programs functions along with First-Year Engineering, since both of these programs focus on outreach toward, and engagement of, potential engineers. The Division of Environmental and Ecological Engineering was also identified as ideal for location in Potter due to its need to have a discrete home separate from Civil Engineering.

College of Engineering Programs

The Programs of the College of Engineering consist of Women in Engineering Program, Minority Engineering Program, the Office of Professional Practice, the Global Engineering Program, Honors, and the Office of Future Engineers. These programs share commonalities in the populations they serve and, in some cases, the types of facilities that are needed. Locating these programs together will develop synergies and generate efficiencies of space utilization that would not be possible if they were dispersed across the campus.

Engineering Education

Engineering Education has a critical need for additional space immediately. ENE is expected to grow over 200 percent in the future and will ultimately require over 70,000 nsf. ENE has an educational mission of serving all first-year engineering students and requires large and flexible instructional spaces. ENE's focus on first-year engineering students creates a strong synergy with the COE's Programs Functions.

The Division of Environmental and Ecological Engineering

The Division of Environmental and Ecological Engineering is expected to grow and become an independent School of the College of Engineering. The projected growth in space required for EEE is over 600 percent. Identifying a permanent home that can satisfy EEE needs for space and contribute to its independent identity was a major focus of the COE's master planning effort. Potter was identified as an ideal location due to the synergy between EEE and the School of Industrial Engineering in Grissom Hall.

RENOVATION OPTIONS

This study examines five scenarios for the renovation of the Potter Center. Each subsequent scenario represents an increase in scope and budget, but each scenario grows increasingly closer to realizing the goals of the departments proposed for relocation into Potter.

It is important to note that the proposed options are not phased. Thus, Scenario A could not expand into Option B, or B into C, without demolishing work done in a previous option. The exception to this rule is Option B2, which could represent a phase one of Option B. Note that "Scenario A" is not labeled as "Option A" because it was ruled non-viable by all those having input.

RENOVATION OPTIONS VERSUS OVERALL PROJECT GOALS

OPTIONS					INCLUDED DEPARTMENTS/RENOVATIONS
A	B2	B	C	D	
	○	●	●	●	Create a more accessible and user-friendly Disability Resource Center and increase efficiency and coordination between the DRC and related departments through co-location of those departments.
		●	●	●	Increase efficiency, cost control, academic integrity, testing flexibility and ease of use for students and staff by combining all the campus testing functions into one location to serve all of campus.
			○	●	Realize the expansion and growth goals of the College of Engineering by strategically re-locating departments identified by the College's master plan in accordance with their Strategic Initiatives Plan.

DEPARTMENTS AND RENOVATIONS INCLUDED PER OPTION

OPTIONS					
A	B2	B	C	D	INCLUDED DEPARTMENTS/RENOVATIONS
●	●	●	●	●	Disability Resource Center
●	●	●	●	●	Accessibility Upgrades
	●	●	●	●	Assistive Technology Center
	●	●	●	●	Instructional Data Processing
	●	●	●	●	Online Compliance Department
		●	●	●	Centralized Testing Center (Disability Resource Center Accommodated Testing, Purdue Testing Center, Digital Education, Foreign Language Placement)
			●	●	Business Center
			●	●	Division of Environmental and Ecological Engineering
			●	●	Women in Engineering
			●	●	Office of Future Engineers
			●	●	Office of Professional Practice
			●	●	Engineering Honors
			●	●	Global Engineering
			●	●	Minority Engineering Program
				●	First Year Engineering
				●	Engineering Education
				●	Reskinning of the South Façade
				●	Revision to the Centennial Mall Plan

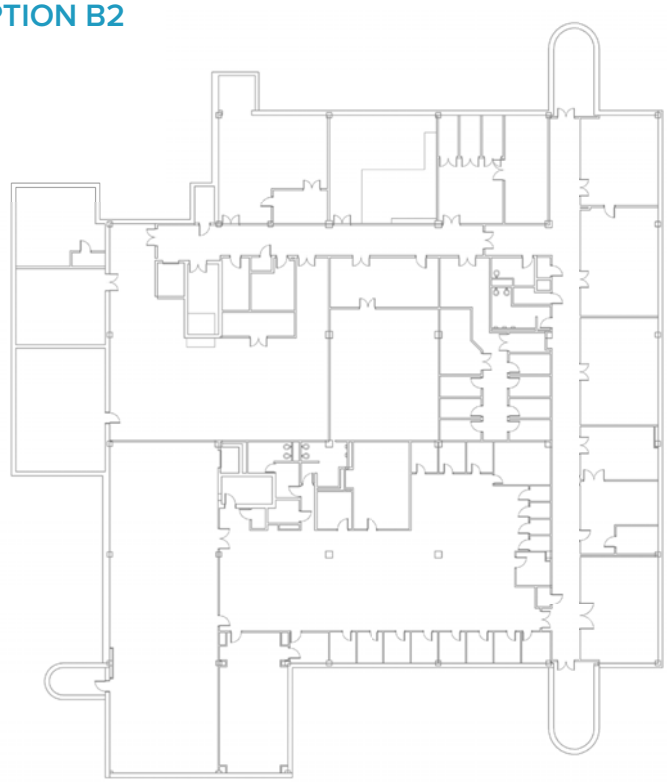
SCENARIO A



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

SCENARIO A - FIRST FLOOR PLAN

OPTION B2



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
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- WOMEN IN ENGINEERING

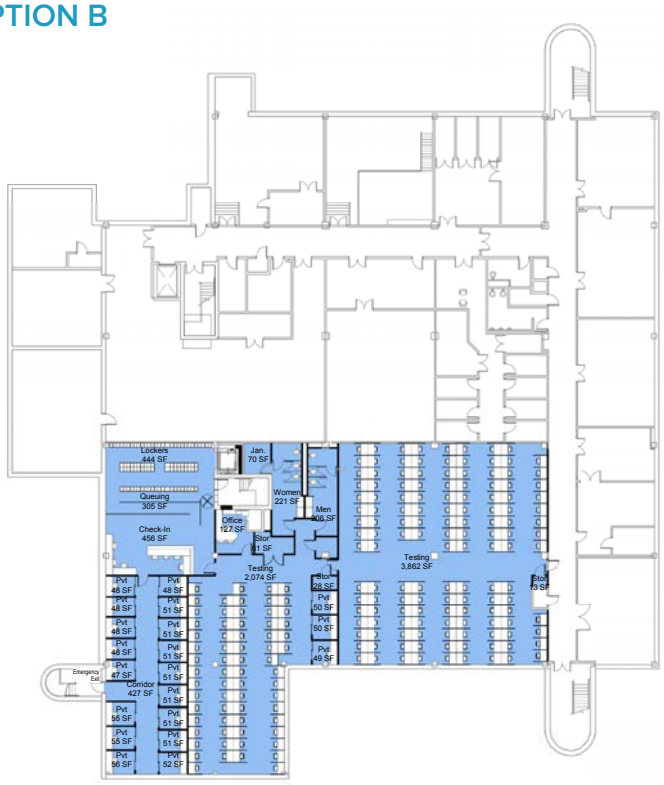
OPTION B2 - LOWER LEVEL



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
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- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B2 - FIRST FLOOR

OPTION B



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
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- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B - LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
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- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B - FIRST FLOOR PLAN

OPTION C



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
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- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION C - LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
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OPTION C - FIRST FLOOR PLAN

OPTION C



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
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- RESTROOM
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OPTION C - SECOND FLOOR PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
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- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION C - THIRD FLOOR PLAN

OPTION D



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
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OPTION D - LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
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- WOMEN IN ENGINEERING

OPTION D - FIRST FLOOR PLAN

OPTION D


i2i/Honors Teaching Lab
3,200 SF
Storage
466 SF
Grad/Thesis
1,651 SF
Grad/Thesis
2,051 SF
Corridor
4,558 SF
Conference
759 SF
Conference
307 SF
Conference
323 SF
Conference
319 SF
Conference
366 SF
Conference
230 SF
Conference
193 SF
Work/Cafe
587 SF
Teaching Asst
1,123 SF
Vis Schol
308 SF
Grad/Thesis
365 SF
PD Asst
737 SF
Director
191 SF
Mgmt Dir
120 SF
Dir Dev
117 SF
Dir PA
120 SF
Comm Dir
120 SF
Women
190 SF
Men
194 SF
Stor
47 SF
Stor
37 SF
Stor
25 SF
Stor Adv
173 SF
Files/Stor
85 SF
IDF
151 SF
Mail
Admin/Mail
213 SF
Inst/Lec
119 SF
Adj/Jnt Adj
119 SF
Inst/Lec
120 SF
Adj Ist Adj
120 SF
Emert
144 SF
Emert
120 SF
Focus
85 SF
Focus
44 SF
Adj Ist
123 SF
Advisor
120 SF
Advisor
124 SF
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123 SF
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120 SF
Advisor
120 SF
Advisor
120

PRO/CON ANALYSIS OF RENOVATION OPTIONS

Scenario A	**PROS** » The DRC is located in an accessible first-floor location. » Space is available in spring 2016. » This is the lowest-cost option.
	CONS » This option does not realize any of the goals of the College of Engineering. » There is insufficient area to fit the proposed DRC departmental program and accommodated testing function. » This plan does not allow for future DRC growth. » There is a lack of interior daylight. » The most direct wheelchair access path to the DRC is via a loading dock ramp. This presents a demeaning, rather than gracious and welcoming, sense of entry. » There is insufficient space to co-locate the ATC with the DRC. » This option does not realize any of the goals for the College of Engineering. » Future renovation of the remainder of Potter would be extremely detrimental to the DRC, very likely requiring it to temporarily relocate. The DRC has very specific spatial requirements and could be costly and difficult to temporarily move. » The only upgrade for the proposed space in comparison to the existing DRC space is its first-floor location; there is very little to be gained in comparison to the money spent.
Option B2	**PROS** » The DRC is located in an accessible first-floor location. » Space is available in summer 2017. » This is the lowest-cost option to effectively accommodate the DRC and most related functions. » This option realizes some spatial and synergistic goals resulting from co-location and adjacencies of the DRC and related departments.
	CONS » This option does not realize any of the goals of the College of Engineering. » Future renovation of the remainder of Potter would be extremely detrimental to the DRC, very likely requiring it to temporarily relocate. The DRC has very specific spatial requirements and could be costly and difficult to temporarily move. » This option does not create the CTC. Removing the potential for efficiency, cost control, academic integrity, testing flexibility, and ease of use for students and staff.
Option B	**PROS** » The DRC is located in an accessible first-floor location. » Space is available in summer 2017. » The majority of spatial and synergistic goals resulting from co-location and adjacencies of departments have been realized. » Nearly all proposed departments included in this option have been successfully accommodated in an effective and efficient arrangement. The Central Testing Center would likely be more efficient if not split between floors. » The majority of spatial and synergistic goals resulting from co-location and adjacencies of departments have been realized.
	CONS » This option does not realize any of the goals of the College of Engineering. » Future renovation of the remainder of Potter would be extremely detrimental to the DRC, very likely requiring it to temporarily relocate. The DRC has very specific spatial requirements and could be costly and difficult to temporarily move. » High security and general testing is split between two floors, possibly reducing the efficiency of the testing center.

Option C	**PROS** » A large majority of costly research lab space and lab support space is preserved. » The COE programs can be co-located in one floor, creating easy access to students. » The DRC is located in an accessible first floor location.
	CONS » Area likely needs to be created elsewhere on campus for displaced instructional and research lab space. » Does not fully realize the goals of the College of Engineering by excluding the Engineering Education and First-Year Engineering departments. » Insufficient queuing and storage area for high-stakes testing. » Insufficient area to properly distribute shared/community space throughout the building, resulting in the majority of community spaces being on the second floor. » Most departments cannot be ideally laid out due to constraints imposed by existing lab spaces. Many office spaces are without windows. » While lab space is preserved, it is inferior in function and safety to new research lab space (see Lab Assessment for more detail). » A future renovation of the remainder of Potter would be extremely detrimental to the DRC and CTC, likely requiring these departments to move elsewhere. Both departments have very specific spatial requirements and could be costly and difficult to temporarily relocate. » The current condition of the existing lab spaces preserved in this study are likely counter-productive to the mission of the COE programs/departments included in the renovation, as these programs are focused on recruiting and retaining COE students. » New mechanical systems will be required to serve the existing laboratories and it is recommended the existing laboratories be converted to 100% exhaust. This imposes a higher first cost on the project since the air handler will be sized to handle large volumes of outside air that will not be required if and when the laboratories are moved. » This option requires renovation of the building to occur around existing lab spaces. At a minimum, this will be very disruptive to the labs and will likely require some, if not all, labs to relocate or shut down during some or all of the renovation work. » Due to space constraints, restrooms are not added or renovated on the second or third floor. » A new elevator is not added in this option, and the existing elevator must remain a freight elevator to serve remaining labs.
Option D	**PROS** » All goals of all departments are met. » The renovation will produce expansion/flex space, allowing departments within Potter Center to grow in place. » This is the most effective and efficient of the proposed options measured by space utilization. Departments are ideally located in the building and among each other to effectively and efficiently operate. » Reskinning of the façade permits a far greater amount of daylight into the building, creating a more desirable working environment and exterior appearance fitting for recruitment and retaining students and staff. This parallels the overall goal of many of the departments proposed for inclusion in Potter. » Revision of the south plaza integrates the Potter Center into Centennial Mall, creating a welcoming and fully accessible building, as well as functional outdoor space. » Moves the College of Engineering closer to realizing their 2014 Master Plan. » Creates 5,200 sft of flex space on the second and third floors.
	CONS » 28,267 sft of research lab space is displaced from Potter and may require relocation. » Research lab space is displaced for less costly office and teaching lab space. » This is the most expensive of the proposed options.

CONSTRUCTION COST SUMMARY

	GSF	COST PER SFT	TOTAL CONSTRUCTION COST
SCENARIO A	5,109	$221.01	$1,129,125
OPTION B2	16,779	$199.87	$3,353,619
OPTION B	25,333	$161.80	$4,098,818
OPTION C	61,212	$178.34	$10,916,799
OPTION D	104,943	$197.80	$21,003,788
ALTERNATES			TBD

CAMPUS SPACE USE IMPACT ANALYSIS

The chart on the following page assesses the impact of each proposed option on space utilization within the A.A. Potter Center and Purdue campus as a whole.

All areas affected are categorized using the FCIM Classification system and are accounted for in this chart. It should be noted that the "310 – Office" column shows space allocated to that specific category for informational purposes. The area shown in this column is already contained in the "300 – Offices" category; therefore, this column is not included in calculations.

SCENARIO A

Examining spatial efficiency only, this scenario is quite efficient as it squeezes the DRC into an area smaller than their current home. However, it accomplishes none of goals of this department and could be detrimental to the functioning of the DRC.

OPTION B2

A loss of assignable area appears in this option as well as in Option B. This can be attributed to the increase in general building area. This increase comes from the increased size and quantity of restrooms (expanded for accessibility and use) and an increase in circulation area required to navigate around the new departmental suites. The large quantity of space released can be attributed to the libraries moving to the Active Learning Center, but no programmed area replaces the basement library spaces in this option.

OPTION B

By comparing Option B to Option B2, the spatial efficiency of the testing center can be examined. Option B captures the release of space by the existing testing centers moving to the Central Testing Center in Potter. In Option B2, these centers stay in their current homes and the lower level of Potter is not renovated. Thus the area of renovated space in Option B2 is reduced by 9,322 sft. However, reducing the renovated area only results in an increase in open area on campus of 4,015 square feet. By "spending" this 4,015 sft, the campus can service 247 percent more students in 44 percent less area than would be possible with the existing testing centers.

OPTION C

This is the only option showing a loss (though small) of net square footage in the Potter Center. This can generally be attributed to small leftover rooms and storage areas that are not lab spaces or lab support but were not demolished as part of the renovation because the proposed program did not have functions that could effectively utilize these spaces. A greater decrease in assignable area over Option D is shown because large hallways could not be narrowed or repurposed as in Option D. Though the goal of this option was to preserve research lab space, 5,295 sft of lab space was lost in order to accommodate the spaces requested.

OPTION D

This option removes all existing research lab space, essentially replacing it with instructional lab and office space. Thus, 28,267 sft of research lab space would need to be accommodated elsewhere on campus. According to the College of Engineering's master plan, this space should be shifted to the South Campus.

The large loss of General Building space is an effect of the current departments leaving their suites, but not "leaving" the hallways that served those suites, thus no general building space was released

elsewhere on campus. A loss of assignable square footage is due to the classification of circulation space relative to open office area. Because the College of Engineering has moved to an open office concept, away from departmental suites, the circulation serving the open office area is classified as general building circulation rather than departmental service, as it normally would have been. This pushes a large amount of area from assignable to net area.

CAMPUS SPACE USE IMPACT ANALYSIS

Scenario A

	100 - Classroom	200 - Instructional Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassified	000 - General Building	TOTAL NSF	TOTAL ASF
Space Renovated/Created in Potter	0	0	0	3,096	1,509	1,278	0	0	0	0	0	0	342	4,716	4,374
Space Displaced from Potter	3,115	0	0	0	0	1,049	0	0	0	0	0	0	0	4,164	4,164
Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**0**	**0**	**3,096**	**1,509**	**229**	**0**	**0**	**0**	**0**	**0**	**0**	**342**	**552**	**210**
Space Released on Campus	0	0	0	2,349	1,626	1,078	0	0	0	0	0	0	0	3,427	3,427
Space Displaced from Potter	-3,115	0	0	0	0	-1,049	0	0	0	0	0	0	0	-4,164	-4,164
Space Accommodated Elsewhere	3,115	0	0	0	0	1,049	0	0	0	0	0	0	0	4,164	4,164
Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**0**	**0**	**2,349**	**1,626**	**1,078**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**3,427**	**3,427**

Option B2

	100 - Classroom	200 - Instructional Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassified	000 - General Building	TOTAL NSF	TOTAL ASF
Space Renovated/Created in Potter	0	0	0	6,698	3,291	3,098	0	3,616	178	0	0	0	1,920	15,510	13,590
Space Displaced from Potter	3,115	0	0	1,204	1,204	10,329	0	0	0	0	0	0	403	15,051	14,648
Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**0**	**0**	**5,494**	**2,087**	**-7,231**	**0**	**3,616**	**178**	**0**	**0**	**0**	**1,517**	**459**	**-1,058**
Space Released on Campus	0	0	0	4,473	2,520	9,226	202	0	1,624	0	0	0	372	15,897	15,525
Space Displaced from Potter	-3,115	0	0	-1,204	-1,204	-10,329	0	0	0	0	0	0	-403	-15,051	-14,648
Space Accommodated Elsewhere	3,115	0	0	1,204	1,204	10,329	0	0	0	0	0	0	403	15,051	14,648
Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**0**	**0**	**4,473**	**2,520**	**9,226**	**202**	**0**	**1,624**	**0**	**0**	**0**	**372**	**15,897**	**15,525**

Option B

	100 - Classroom	200 - Instruction-al Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassi-fied	000 - General Building	TOTAL NSF	TOTAL ASF
Space Renovated/Created in Potter	0	0	0	8,080	4,802	10,630	0	3,529	178	0	0	0	2,427	24,844	22,417
Space Displaced from Potter	3,115	0	0	2,621	1,414	17,629	202	0	0	0	0	0	775	24,342	23,567
Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**0**	**0**	**5,459**	**3,388**	**-6,999**	**-202**	**3,529**	**178**	**0**	**0**	**0**	**1,652**	**502**	**-1,150**
Space Released on Campus	0	1,744	0	5,792	5,161	2,722	0	0	1,624	0	0	0	0	11,882	11,882
Space Displaced from Potter	-3,115	0	0	-2,621	-1,414	-17,629	-202	0	0	0	0	0	-775	-24,342	-23,567
Space Accommodated Elsewhere	3,115	0	0	2,621	1,414	17,629	202	0	0	0	0	0	775	24,342	23,567
Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**1,744**	**0**	**5,792**	**5,161**	**2,722**	**0**	**0**	**1,624**	**0**	**0**	**0**	**0**	**11,882**	**11,882**

* This value is also contained in "300 - Offices"

CAMPUS SPACE USE IMPACT ANALYSIS (CONT.)

		100 - Classroom	200 - Instructional Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassified	000 - General Building	TOTAL NSF	TOTAL ASF
Option C	Space Renovated/Created in Potter	0	2,963	0	22,548	18,611	11,051	0	6,313	0	0	0	0	18,521	61,396	42,875
	Space Displaced from Potter	3,115	4,690	5,764	11,841	7,221	18,182	202	732	1,947	0	0	0	15,709	62,182	46,473
	Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**-1,727**	**-5,764**	**10,707**	11,390	**-7,131**	**-202**	**5,581**	**-1,947**	**0**	**0**	**0**	**2,812**	**-786**	**-3,598**
	Space Released on Campus	0	1,744	469	14,869	11,869	3,792	0	0	1,624	0	0	0	0	22,498	22,498
	Space Displaced from Potter	-3,115	-4,690	-5,764	-11,841	-7,221	-18,182	-202	-732	-1,947	0	0	-15,709	-15,709	-62,182	-46,473
	Space Accommodated Elsewhere	3,115	0	0	3,597	2,390	17,629	202	0	0	0	0	775	775	25,318	24,543
	Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**-2,946**	**-5,295**	**6,625**	7,038	**3,239**	**0**	**-732**	**-323**	**0**	**0**	**-14,934**	**-14,934**	**-14,366**	**568**

		100 - Classroom	200 - Instructional Labs	250 - Research Labs	300 - Offices	* 310- Offices	400 - Study	500 - Special Use	600 - General Use	700 - Support	800 - Health Care	900 - Residential	050 - Unclassified	000 - General Building	TOTAL NSF	TOTAL ASF
Option D	Space Renovated/Created in Potter	0	12,540	0	39,796	34,116	11,911	0	10,502	0	0	0	0	20,329	95,078	74,749
	Space Displaced from Potter	3,115	4,690	28,736	19,748	13,641	18,182	202	732	1,947	0	0	0	16,514	93,866	77,352
	Net Change in Potter Total = (Renovated) - (Displaced)	**-3,115**	**7,850**	**-28,736**	**20,048**	20,475	**-6,271**	**-202**	**9,770**	**-1,947**	**0**	**0**	**0**	**3,815**	**1,212**	**-2,603**
	Space Released on Campus	0	7,587	469	25,376	19,891	3,792	0	21	1,624	0	0	0	0	38,869	38,869
	Space Displaced from Potter	-3,115	-4,690	-28,736	-19,748	-13,641	-18,182	-202	-732	-1,947	0	0	0	-16,514	-93,866	-77,352
	Space Accommodated Elsewhere	3,115	0	0	2,621	1,414	17,629	202	0	0	0	0	0	775	24,342	23,567
	Net Change on Campus Total = (Released) + (Displaced) + (Accommodated)	**0**	**2,897**	**-28,267**	**8,249**	7,664	**3,239**	**0**	**-711**	**-323**	**0**	**0**	**0**	**-15,739**	**-30,655**	**-14,916**

* This value is also contained in "300 - Offices"

EFFICIENCY ANALYSIS PER DEPARTMENT

	Option	% Project	ASF	Faculty/Staff	Grad	Office ASF	Total Headcount	FTE	Office ASF/ Headcount	Office ASF/FTE
Disability Resource Center	Existing		3,427	14		1,626	14	14.0	116	116
	A	100.00%	4,374	15	2	1,509	17	15.9	89	95
	B2	67.25%	6,424	15	2	1,785	17	15.9	105	112
	B	20.59%	3,927	15	2	1,867	17	15.9	110	118
	C	11.51%	3,761	15	2	1,839	17	15.9	108	116
	D	7.08%	3,761	15	2	1,839	17	15.9	108	116
Assistive Technology Center	Existing		998	1		150	1	1.0	150	150
	B2	8.96%	856	1		120	1	1	120	120
	B	4.49%	856	1		120	1	1.0	120	120
	C	2.47%	808	1		137	1	1.0	137	137
	D	1.52%	808	1		137	1	1.0	137	137
Central Testing Center [1] (all departments)	Existing		6,564	13		2,851	13	13.0	219	219
	B	62.24%	11,871	18		1,357	18	18.0	75	75
	C	34.26%	11,192	18		1,344	18	18.0	75	75
	D	21.50%	11,416	18		1,058	18	18.0	59	59
Instructional Data Processing	Existing		2,230	5		534	5	5.0	107	107
	B2	14.29%	1,365	6		666	5	6.0	133	111
	B	7.57%	1,444	6		656	6	6.0	109	109
	C	3.49%	1,141	6		742	6	6.0	124	124
	D	2.09%	1,108	6		760	6	6.0	127	127
Online Compliance [2]	Existing									
	B2	9.50%	908	8		720	8	8.0	90	90
	B	5.11%	974	8		786	8	8.0	98	98
	C	1.64%	537	8		537	8	8.0	67	67
	D	0.95%	502	8		502	8	8.0	63	63

EFFICIENCY ANALYSIS PER DEPARTMENT (CONT.)

	Option	% Project	ASF	Faculty/Staff	Grad	Office ASF	Total Headcount	FTE	Office ASF/ Headcount	Office ASF/FTE
Business Office [2]	Existing									
	C	1.93%	631	7		631	7	7.0	90	90
	D	1.19%	630	7		630	7	7.0	90	90
Engineering Education	Existing		16,845	77		8,022	77	77.0	104	104
	D	26.48%	14,060	46	146	11,207	192	109.9	58	102
Minority in Engineering	Existing		1,525	7		680	7	7.0	97	97
	C	4.84%	1,582	6		560	6	6.0	93	93
	D	3.26%	1,731	6		560	6	6.0	93	93
Office of Future Engineers	Existing		694	9		694	9	9.0	77	77
	C	2.18%	712	4	3	600	7	5.3	86	113
	D	1.24%	656	4	3	600	7	5.3	86	113
Women in Engineering	Existing		1,072	13		849	13	13.0	65	65
	C	5.20%	1,699	10	11	1,580	21	14.8	75	107
	D	3.42%	1,818	10	11	1,673	21	14.8	80	113
Engineering Honors	Existing		829	4		474	4	4.0	119	119
	C	2.68%	875	10		875	10	10.0	88	88
	D	13.94%	7,400	10		891	10	10.0	89	89
Environmental & Ecological Engineering	Existing		3,738	25		2,412	25	25.0	96	96
	C	27.36%	8,937	30	54	6,042	84	53.6	72	113
	D	15.74%	8,357	34	52	4,991	86	56.8	58	88
Office of Professional Practice	Existing		1,592	8		1,016	8	8.0	127	127
	C	2.42%	790	9		678	9	9.0	75	75
	D	1.59%	844	9		677	9	9.0	75	75

[1] This department is the combination of several existing departments and is offering an expanded program in the proposed design.
[2] New Department

AREA ANALYSIS PER DEPARTMENT

	Option	Class-rooms	Inst. Lab	Res. Lab	Offices	Study	Special Use	General Use	Support	Health Care	Residential	Total ASF	ASF % Change
Disability Resource Center	Existing	0	0	0	2,349	1,078	0	0	0	0	0	**3,427**	
	A	0	0	0	3,096	1,278	0	0	0	0	0	**4,374**	28%
	B2	0	0	0	3,936	2,362	0	0	0	0	0	**6,298**	84%
	B	0	0	0	3,927	0	0	0	0	0	0	**3,927**	15%
	C	0	0	0	3,761	0	0	0	0	0	0	**3,761**	10%
	D	0	0	0	3,761	0	0	0	0	0	0	**3,761**	10%
Assistive Technology Center	Existing	0	0	0	150	848	0	0	0	0	0	**998**	
	B2	0	0	0	120	736	0	0	0	0	0	**856**	-14%
	B	0	0	0	120	736	0	0	0	0	0	**856**	-14%
	C	0	0	0	137	671	0	0	0	0	0	**808**	-19%
	D	0	0	0	137	671	0	0	0	0	0	**808**	-19%
Central Testing Center [1] (all departments)	Existing	0	1,744	0	2,946	1,874	0	0	0	0	0	**6,564**	
	B	0	0	0	1,357	9,894	0	0	0	0	0	**11,251**	71%
	C	0	0	0	1,510	9,064	0	0	0	0	0	**10,574**	61%
	D	0	0	0	1,461	9,229	0	0	0	0	0	**10,690**	63%
Instructional Data Processing	Existing	0	0	0	606	0	0	0	1,624	0	0	**2,230**	
	B2	0	0	0	1,365	0	0	0	0	0	0	**1,365**	-39%
	B	0	0	0	1,445	0	0	0	0	0	0	**1,445**	-35%
	C	0	0	0	1,141	0	0	0	0	0	0	**1,141**	-49%
	D	0	0	0	1,108	0	0	0	0	0	0	**1,108**	-50%
Online Compliance [2]	Existing												**N/A**
	B2	0	0	0	733	0	0	0	0	0	0	**733**	
	B	0	0	0	733	0	0	0	0	0	0	**733**	
	C	0	0	0	537	0	0	0	0	0	0	**537**	
	D	0	0	0	502	0	0	0	0	0	0	**502**	

ARE ANALYSIS PER DEPARTMENT (CONT.)

	Option	Class-rooms	Inst. Lab	Res. Lab	Offices	Study	Special Use	General Use	Support	Health Care	Residential	Total ASF	ASF % Change
Business Office [2]	Existing												N/A
	C	0	0	0	631	0	0	0	0	0	0	631	
	D	0	0	0	630	0	0	0	0	0	0	630	
Engineering Education	Existing	0	5,843	0	10,507	0	0	21	0	0	0	16,371	
	D	0	2,648	0	11,412	0	0	0	0	0	0	14,060	-14%
Minority in Engineering	Existing	0	0	0	1,008	517	0	0	0	0	0	1,525	
	C	0	0	0	680	902	0	0	0	0	0	1,582	4%
	D	0	0	0	722	1,009	0	0	0	0	0	1,731	14%
Office of Future Engineers	Existing	0	0	0	694	0	0	0	0	0	0	694	
	C	0	0	0	712	0	0	0	0	0	0	712	3%
	D	0	0	0	656	0	0	0	0	0	0	656	-5%
Women in Engineering	Existing	0	0	0	1,072	0	0	0	0	0	0	1,072	
	C	0	0	0	1,699	0	0	0	0	0	0	1,699	58%
	D	0	0	0	1,818	0	0	0	0	0	0	1,818	70%
Engineering Honors	Existing	0	0	0	829	0	0	0	0	0	0	829	
	C	0	0	0	875	0	0	0	0	0	0	875	6%
	D	0	5,936	0	1,464	0	0	0	0	0	0	7,400	793%
Environmental & Ecological Engineering	Existing	0	0	469	2,716	553	0	0	0	0	0	3,738	
	C	0	2,963	0	6,219	0	0	0	0	0	0	9,182	146%
	D	0	2,911	0	5,413	0	0	0	0	0	0	8,324	123%
Office of Professional Practice	Existing	0	0	0	1,592	0	0	0	0	0	0	1,592	
	C	0	0	0	790	0	0	0	0	0	0	790	-50%
	D	0	0	0	844	0	0	0	0	0	0	844	-47%

[1] This department is the combination of several existing departments and is offering an expanded program in the proposed design.
[2] New Department

POTTER CENTER RENOVATION IN RELATION TO THE "PURDUE MOVES" CAMPUS INITIATIVE

> This is no ordinary time for higher education, and we are determined to seize the opportunity to increase our investment in areas that have the greatest potential for impact. Our students will grow, not just academically, but in ways that truly prepare them to effectively work and live in today's world. Our research will deliver still greater benefits to our state and to all corners of the globe. And we will achieve this in a way that is fiscally responsible.
>
> – Mitch Daniels

Any major project undertaken on any campus must be examined in light of the overall goals and direction for that campus. Here we will examine the A.A. Potter Center Renovation through the lens of the “Purdue Moves” campus initiative.

Initiatives	Strongly Support	Support	No Impact
Affordability and Accessibility			
Increasing Value	●		
Expanding Opportunity		●	
STEM Leadership			
Expanding Engineering	●		
Transforming Technology			●
Strengthening Computer Science			●
World-Changing Research			
Drug Discovery			●
Plant Sciences Research			●
Research Communication			●
Transformative Education			
Transforming Teaching and Learning	●		
International Experiences	●		
Living and Learning		●	
Year-Round University		●	

AFFORDABILITY AND ACCESSIBILITY

Increasing Value

By renovating an existing building rather than constructing a new building, Purdue will substantially reduce construction costs. Additionally, more efficient space utilization in the proposed plans reduces the demand for additional area on campus. Renovations that increase spatial efficiency also support a key principal of the campus' master plan—the promotion of compact growth.

By creating a larger, combined central testing center and capitalizing on the economy of scale, the University will realize large efficiencies of space and staff within these departments. This will release space on campus for other uses and decrease the need for the University to construct new space. This effect will be multiplied during final exam periods by greatly reducing or eliminating the need to create improvised testing centers across campus. These impromptu centers result in the inefficient use of a large quantity of space and resources.

Expanding Opportunity

The expansion and relocation of the Disability Resource Center will create a greater ease of use and access to students with disabilities, allowing more students to successfully attend and navigate the university and focus on success.

STEM LEADERSHIP

Expanding Engineering

The College of Engineering's programs/departments focus on the recruitment and retention of students. Co-location and expansion of these programs will result in greater efficiency and effectiveness, which will directly correlate with Purdue's goals of increasing engineering graduate student enrollment and ultimately increase the number of graduating engineers.

Construction of a 21st century learning environment will attract, recruit, and retain students and faculty. This will directly contribute to the success and growth of the College of Engineering.

WORLD-CHANGING RESEARCH

Drug Discovery, Plant Sciences Research, and Research Commercialization

Because the full renovation option proposes to displace research lab space for classroom and office space, the Potter renovation may be at odds with this goal. In this instance, the goal of expanding engineering directly affects the goal of investing in research. It should be noted that it is important to consider the age and condition of the research lab space in Potter when assessing its true value to campus. A full assessment of the existing space is attached in the appendix to this study.

TRANSFORMATIVE EDUCATION

Transforming Teaching and Learning

The College of Engineering strongly believes in the i2i or Ideas in Innovation style educational model. Additional i2i spaces will be created in the renovated Potter Center. Additionally, the Potter Center will be home to a multitude of flexible non-traditional study spaces—study lounges, meeting rooms, conference rooms, work cafés, and tutorial centers—supporting a 21st century educational model.

International Experiences

A critical driver to improving the number of students studying aboard is increasing student awareness of these opportunities. This is the primary focus of the College of Engineering's Global Engineering Program. This program is one of many of the COE's programs proposed for location on the ground floor of Potter. This location will create a synergy between the GEP and all the COE's programs focused around the common goal of student success and advising. This synergy is not currently realized.

Living and Learning

The proposed Potter Center renovation does not contain a student housing component, so it cannot be a direct contributor to the integration of learning space into residential space. However, the center is located in a key location along the student success corridor. Therefore, its utilization could play a major role in the success of the corridor, which is a major contributor to the Living and Learning initiative. The uses proposed for the building appear to be ideally suited to integrate with this corridor as they focus on the retention and attraction of students, access to testing, ability for disabled students to interface with the University, and general study and educational spaces. Additionally the proposed redesign of the Potter south terrace creates an outdoor classroom and gathering space, a function not currently found within the corridor.

Year-Round University

By concentrating the College of Engineering's Programs in one location, students will have enhanced access to advisors, who can help guide students toward summer programs, shortening student's time to graduation.

Summer online courses have been identified as a focus area for the expansion of summer education. The creation of the central testing center will greatly support online courses by creating an efficient, effective, and academically rigorous testing environment. It will also expand students options for testing by creating a greater ease of testing access.

CONCEPTUAL PLAZA REDESIGN - OPTION D



CONCEPTUAL AERIAL OF SOUTH FACADE AND PLAZA - OPTION D



CONCEPTUAL SOUTH FACADE AND PLAZA - OPTION D



INTRODUCTION

INTRODUCTION

PURPOSE AND OBJECTIVES

American Structurepoint, in conjunction with Applied Engineering and Blundall Associates, has been invited to conduct the following study regarding the future use of the A.A. Potter Engineering Center.

This study will holistically examine the building's potential to accommodate six different functions:

- The Disability Resource Center (DRC)
- The Assistive Technology Center (ATC)
- The New Online Compliance Department
- A new campus Centralized Testing Center (CTC)
- The Instructional Data Processing (IDP) Department
- Relocation of nine College of Engineering (COE) departments in accordance with their overall master plan

GOALS

The intent of this study is to propose a renovation that accomplishes three distinct goals. Note that not all of these goals can be accomplished with most options presented here. Only Option D fully meets these aims:

- Create a more accessible and user-friendly DRC and increase efficiency and coordination between the DRC and related departments through co-location of those departments.
- Increase efficiency, cost control, academic integrity, testing flexibility, and ease of use for students and staff by combining all the campus testing functions into one location to serve all of campus.
- Realize the expansion and growth goals of the College of Engineering by strategically relocating departments identified by the College's master plan in accordance with their Strategic Initiatives Plan.

It is important to note that the building is currently occupied by several departments of the College of Engineering (COE). The COE recently completed a holistic master plan to study the current and future spatial needs of the college in order to accomplish the goals of the COE's Strategic Plan. Much of the space within Potter Center was allocated for COE use as part of that master plan. The new departments being proposed for inclusion in Potter as part of this study were not accounted for in the COE master plan. It is the overarching goal of this study to propose a design solution that simultaneously respects the needs of and supports the future goals for the College of Engineering while co-locating the proposed functions within the building and capitalizing on inter-departmental synchronicities where possible.

This study will examine the work associated with the proposed renovations/relocations in five separate renovation scenarios outlined below, each with sequentially increasing scope and budget.

Scenario A

- Relocation of the Disability Resource Center and Assistive Technology Center
- Building accessibility upgrades to accommodate disabled students

Option B2

- Relocation of the Disability Resource Center and Assistive Technology Center
- Building accessibility upgrades to accommodate disable students
- Relocation of the Instructional Data Processing Department

Option B2

- Relocation of the Disability Resource Center and Assistive Technology Center
- Various building access upgrades for disabled students
- Relocation of the Instructional Data Processing Department
- Creation of a Centralized Testing Center

Option C

- Relocation of the Disability Resource Center and Assistive Technology Center
- Various building access upgrades for disabled students
- Creation of a Centralized Testing Center
- Relocation of the Instructional Data Processing Department
- Selective renovation to accommodate portions of the College of Engineering's Master Plan while simultaneously preserving the greatest amount of research lab space possible

Option D

- Relocation of the DRC and Assistive Technology Center
- Creation of a Centralized Testing Center
- Relocation of the Instructional Data Processing Department
- Renovation of the entirety of the remaining building to accommodate the College of Engineering's Master Plan
- Creation of a new building entrance and selective "reskinning" of the façade.
- Selective revision of the currently proposed plan for Centennial Mall to holistically integrate the Potter Center into campus

PROJECT PARTICIPANTS

PURDUE UNIVERSITY CONTRIBUTORS

The design team would like to thank the following Purdue University Faculty and Staff for their time and contribution to this study.

Donna Ahlen – Manager, Facilities Planning and Construction
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Jason Baumer – Continuing Lecturer in German; Course Coordinator
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Brad Bowen – University Engineer and Senior Director of Asset Management
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Frank Dooley – Vice Provost for Teaching and Learning
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Cindy Fields – Purdue Testing Center Manager
Kevin Flanagan – Digital Education
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Caren Hanley – Disability Resource Center
Madeleine Henry – Head, School of Languages and Cultures
Aaron Hunley – Capitol Project Manager
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Laura Miller – Instructional Data Processing
David Nelson – Associate Director of the Center for Instructional Excellence
Kim Rechkemmer – Assistant Director for Space Management
Byron Reed – ITaP Director of Learning Spaces
David Schwarte – Assistive Technology Specialist Consulting and Training
Randall Ward – Disability Resource Center Director

DESIGN TEAM

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Rajesh Patel
Ralph Power
Rachel Romines

Blundall Associates | COST ESTIMATING
Martyn Blundall
Phillip Salisbury
Bryan Smedberg

BUILDING HISTORY AND CURRENT USE

The A.A. Potter Engineering Center (POTR) is a multi-purpose teaching and research facility administered by the Institute for Interdisciplinary Engineering Studies. The Center's mission is to develop and promote innovative technologies in response to society's needs. Its goal is to enhance quality of life.

The Center is named for Andrey Abraham Potter, the dean of engineering at Purdue from 1920 to 1953. Dean Potter built Purdue into the largest and one of the most respected engineering colleges in the country. From 1945 to 1946, he served as acting president of Purdue; he was offered the presidency but declined.

The Potter building is centrally located on the north campus, immediately to the east of the Wilmeth Active Learning Center (currently under construction). The building currently houses the Siegesmund Engineering Library, the Office of Professional Practice, Visual Analytics for Command, Control, and Interoperability Environments (VACCINE), Laboratory of Renewable Resources Engineering (LORRE), VLSI Design and Test Laboratory, the Center for Materials Under eXtreme Environment (CMUXE), the offices of the Division of Environmental and Ecological Engineering, and the CSOPS Research Entity for School Research.

The building houses two named spaces. It is critical the following spaces be accommodated for and named in the newly renovated space.

K.S. FU ROOM FOR CROSS-DISCIPLINARY RESEARCH

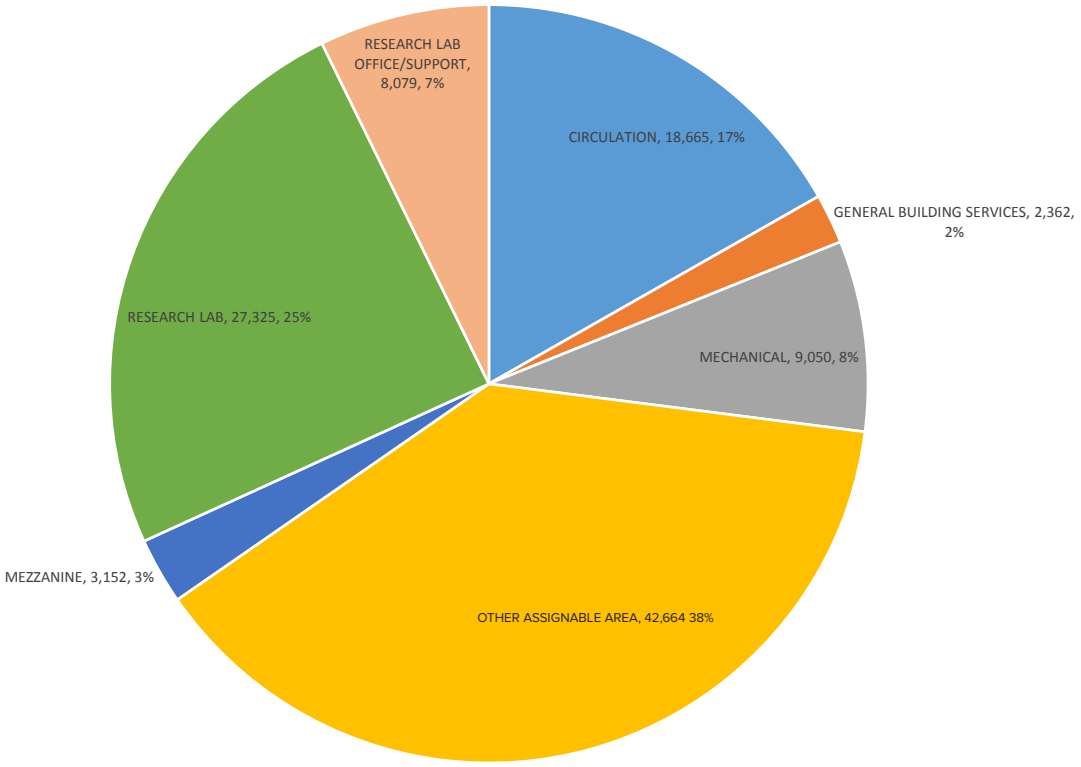
The K.S. Fu Room for Cross-Disciplinary Research (Room 234) is a meeting room that comfortably seats up to 50 people and has been equipped for cross-disciplinary work, special research forums, site visits, cross-disciplinary seminars, presentations, workshops, roundtables, and the display of publications, posters, and audio/visual materials.

This conference room was named in recognition of the late Professor King-Sun Fu because of his many accomplishments as Goss Distinguished Professor of Engineering. Professor Fu was a member of the faculty of the Purdue School of Electrical Engineering from 1960 to 1985.

THE MALOTT COMMONS

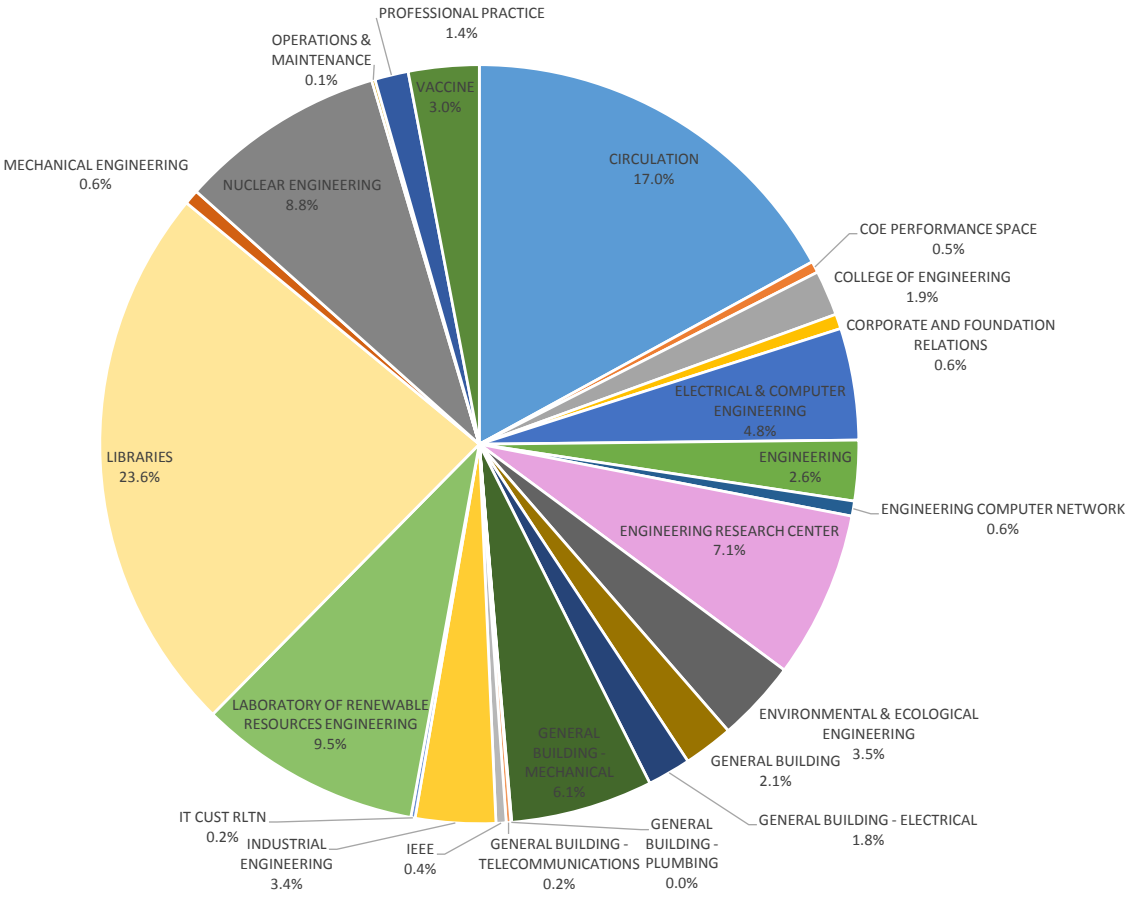
This common area honors Thomas J. Malott (BSME 1962, HDR 2002) and Sandra H. Malott, who donated in support of the Office of Professional Practice. The commons is an entity shared by the entire Potter Center.

POTTER CENTER EXISTING USE



Mezzanine – Area excluded from all options
Research Labs – Area excluded from options A, B, and C
Research Lab Office/Support – Area excluded from options A, B, and C
Mechanical – Area excluded from all options
Other Assignable Areas – Varying amount included in all options
Circulation – Varying amount included in all options

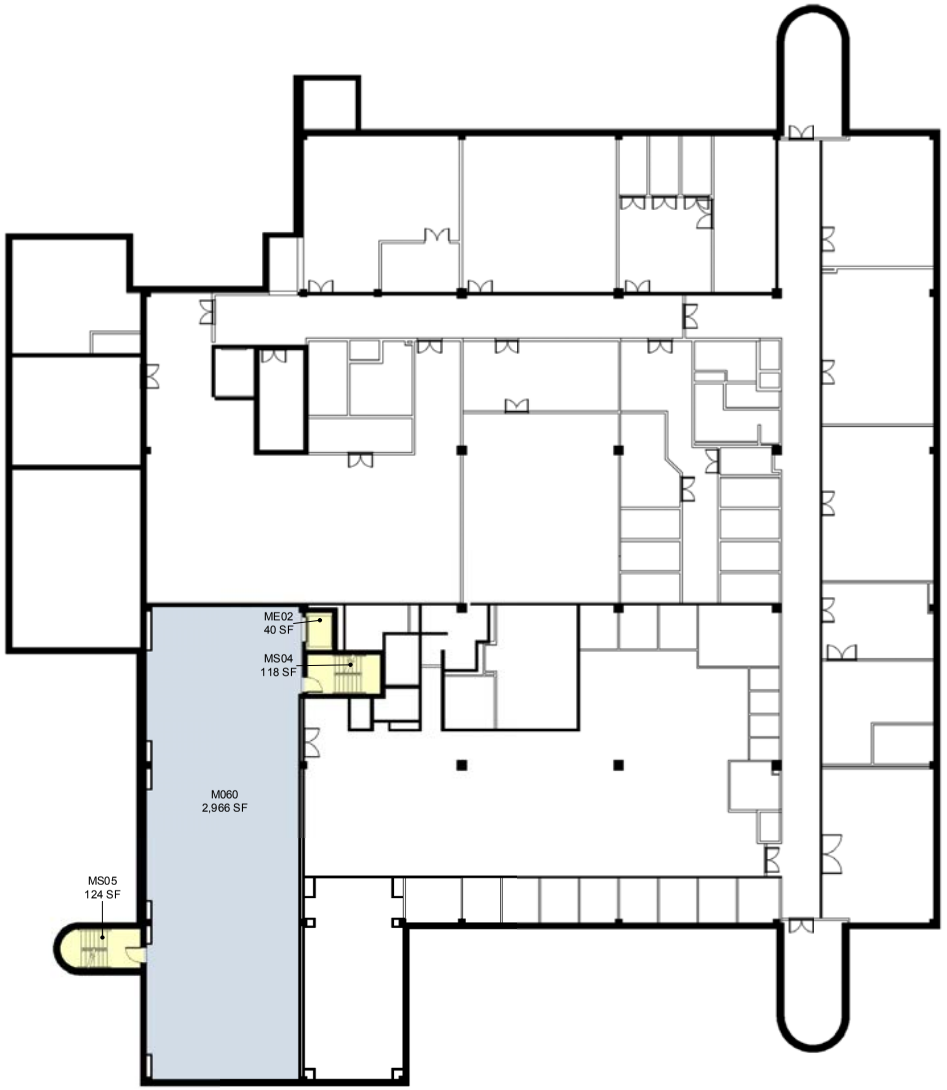
POTTER CENTER EXISTING DEPARTMENT USE





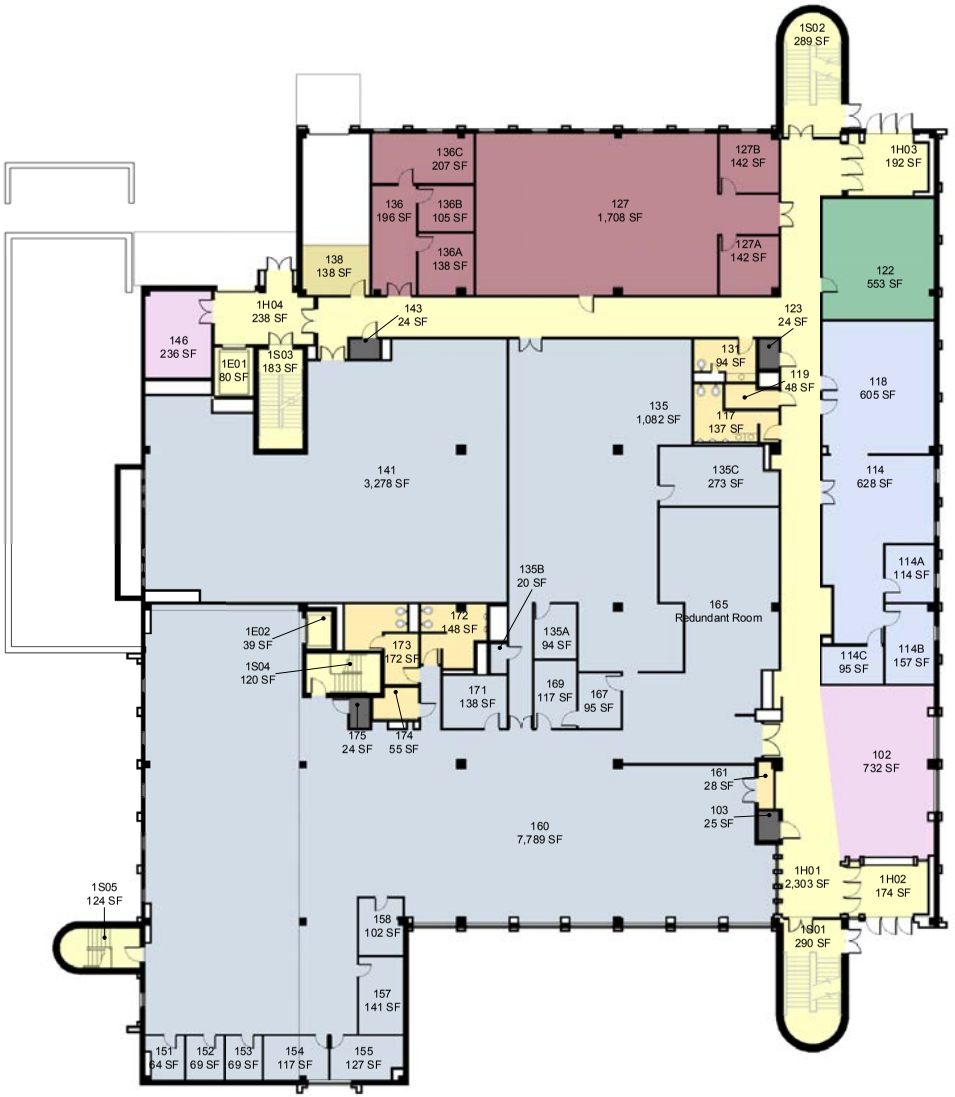
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- COE FLEX SPACE
- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENGINEERING RESEARCH CENTER
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- IEEE
- INDUSTRIAL ENGINEERING
- IT CUST RLTN
- LABORATORY OF RENEWABLE RESOURCES ENGINEERING
- LIBRARIES
- MECHANICAL ENGINEERING
- NUCLEAR ENGINEERING
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- VACCINE

EXISTING LOWER LEVEL



- CIRCULATION
- COE FLEX SPACE
- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
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- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- VACCINE

EXISTING MEZZANINE



- CIRCULATION
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- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
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- NUCLEAR ENGINEERING
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- VACCINE

EXISTING FIRST FLOOR



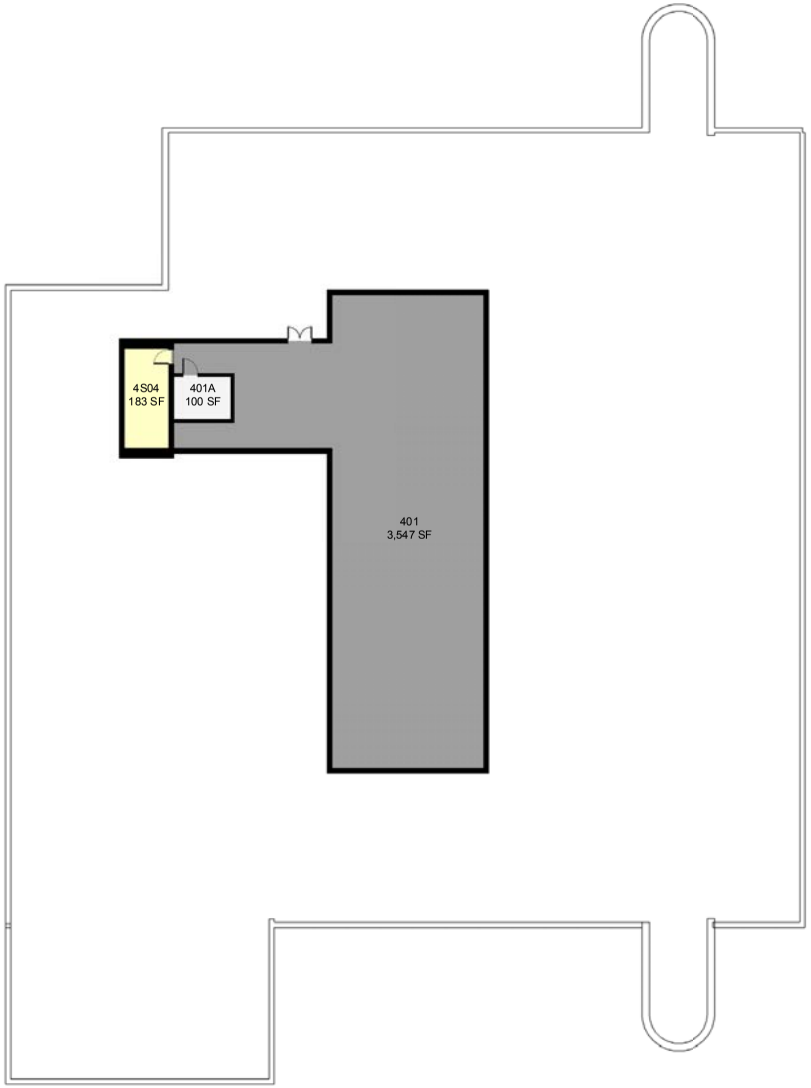
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EXISTING SECOND FLOOR



- CIRCULATION
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- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
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- VACCINE

EXISTING THIRD FLOOR



- CIRCULATION
- COE FLEX SPACE
- COLLEGE OF ENGINEERING
- CORPORATE AND FOUNDATION RELATIONS
- ELECTRICAL & COMPUTER ENGINEERING
- ENGINEERING COMPUTER NETWORK
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EXISTING PENTHOUSE

PHYSICAL ANALYSIS

Date of Construction	1974
Number of Stories	3 above grade 1 below grade 1 partial mezzanine 1 penthouse
Area	Basement: 30,629 sft Mezzanine: 3,712 sft First Floor: 28,003 sft Second Floor: 27,952 sft Third Floor: 27,776 sft Penthouse: 4,236 sft TOTAL AREA: 122,308 sft TOTAL OCCUPIABLE AREA: 114,360 sft
Sprinkler System	Fully sprinklered
Floor-to-Floor Heights	Basement to First: 15’-4” Basement to Mezzanine: 7’-8” Mezzanine to First: 7’-8” First to Second: 14’-0” Second to Third: 14’-0” Third to Roof: 14’-0”
Cladding	Exterior
Typical Brick Cavity Wall Veneer Assembly	3 5/8” brick 1 7/8” airspace 1/2” impregnated sheathing board 6” 20-gauge metal stud at 16” on center with 6” batt insulation Vapor barrier 1/2” gypsum board
Typical Precast Concrete Panel Exterior Wall Assembly	6” to 8” precast concrete panel (thickness varies) 1/2” Airspace 1/2” impregnated sheathing board 6” 20-gauge metal stud at 16” on center with 6” batt insulation Vapor barrier 1/2” gypsum board
R Value:	11.14 (wall construction does not meet current code requirements for insulation)
Current Replacement Value	$54,756,601*
Deferred Maintenance Value	$7,930,500*

*as reported according to the 2014 study produced by The SLAM Collaborative

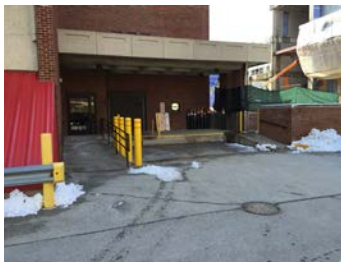
ACCESSIBILITY

Though examination of accessibility to disabled persons is always important, it is especially important to this study because of the proposed relocation of the Disability Resource Center to the Potter Center.

Because the DRC serves Purdue's disabled student population, this building will be among the most heavily trafficked buildings on campus by students with special needs. While the entire building should be remodeled to be ADA compliant, areas specifically serving the DRC should be designed above and beyond code requirements to accommodate these students to the greatest extent possible.

As outlined in the following analysis, in its current condition, the Potter Center is not an accessible building. Therefore, it is the opinion of this study that the Center would not be fit for use by the DRC unless the DRC's relocation occurred concurrently with accessibility upgrades.









PARKING

The Potter Center is not served by a designated parking lot. However, there are three parallel parking spaces along the east side of the alley immediately east of Potter. These spaces are not in an ideal location close to an existing ramp to access the building; restriping and labeling these as ADA spaces appears to be the best available solution. Because the building is bound to the west and south by pedestrian-only roads, there is limited opportunity to create alternate parking solutions. A sidewalk should be constructed connecting these spaces to the sidewalks and entrance at the southeast entrance. There is currently no sidewalk along this side of the building.

BUILDING ENTRANCES AND ENTRANCE ACCESS

The finished floor level of the Potter Center is approximately 2' above grade level; therefore, the only accessible routes to the building are by ramp. Three ramps are present—one each at the northwest, northeast, and southeast corners. The ramp at the northwest corner is adjacent to the building's loading dock and is primarily intended to function as a service ramp. The door at the end of this ramp leads toward the freight elevator, which is currently the building's only elevator.

The ramp near the southeast corner entrance is 106' from the southeast door. It is somewhat hidden behind a wall along the far west edge of the exterior terrace. A student in a wheelchair wishing to use the southwest entrance (the building's main entrance) would need to travel between 138 and 274' farther than an able-bodied peer. Because the south side of the building is along a pedestrian only path, the ramp itself is easily accessible.

ELEVATORS

The building currently contains a single freight elevator at the northwest corner of the building. Aesthetically, the freight elevator is not welcoming and is not a pleasure to use. On an emotional level, relegating wheelchair-bound students to ride with the building's cargo is not hospitable.

To examine a worst-case scenario, starting at a point at the southeast corner outside the building, an able-bodied student wishing to use the building's main entrance and travel to room 205 at the southeast corner of the second floor would travel 186'. A wheelchair-bound peer would travel 727'. That is 541' (a tenth of a mile) farther to reach the same location. A new elevator, close to the building's main entrance, would help eliminate this discrepancy.

RESTROOMS

No ADA-compliant restrooms exist in the Potter Center. Additionally, doors leading into restrooms do not have the necessary adjacent area to allow a wheelchair-bound student to operate the door. Restrooms on the second and third floor contain a stall with the width to accommodate ambulatory accessible use (38" wide), but the 4'-6" depth of these stalls is not compliant with the required 5' minimum.

Without significant restroom renovations, disabled students will likely need to leave the building to use the restroom.

Recommended Accessibility Upgrades

In order for the Potter Center to house the Disability Resource Center, we strongly recommend accessibility upgrades be performed at a minimum on and around areas that would serve the DRC, but ideally throughout the building. The following upgrades should occur where applicable:

- ADA-compliant door handles
- ADA-compliant signage
- ADA fire system notification
- Proper spacing around doors
- Proper clearance requirements in bathrooms
- Proper door closure forces
- Proper counter heights
- Proper outlet and switch heights

ARCHITECTURAL ASSESSMENT

Though in respectable condition for an over 40-year-old structure, the Potter Center has had very few improvements or cosmetic upgrades over its lifespan and aesthetically appears quite dated.

The building is generally designed in the “brutalist” or “brick brutalist” style popular for university, government, and high-rise housing projects from the 1950s to the mid-1970s. The façade is a combination of large areas of unadorned, blank, brick and unbroken, repetitive facades of narrow 5'-4" precast concrete panels. The panels themselves feature very deep protrusions and offer quite a bit of depth to the façade. However, the continuous repetition of the panels tends to temper the visual interest created by their depth and instead lends the precast areas a sense of monotony. The building’s most visually intriguing exterior feature are its protruding brick stair towers with curved outer walls.

On the interior, the building’s structural frame has the potential to be a highlighted feature. Its two-way concrete beam system (essentially a large waffle slab) is rather attractive. The deep beams running in perpendicular directions give the ceiling a highly textured aesthetic. Unfortunately, the slabs are currently hidden behind dropped acoustic ceilings in most locations. The floor system, in addition to the generous floor-to-floor heights, creates a space with the potential to become an attractive interior. The terrazzo flooring along corridors on the first floor is also an asset. Terrazzo is a very tough and durable flooring material that can be refinished and restored to nearly as-new condition.

With the exception of a row of seven large windows along the first story of the south face, window openings generally occur within the precast panels and are thus quite small at 5.4 sft each. The small window size, in conjunction with a 1'-4" precast concrete overhang directly above each window, allows only a limited amount of daylight into the building’s interior. This, in combination with very limited corridor glazing, results in an interior that lacks natural light. Replacing areas of precast panels with glazing and opening up corridor walls would certainly allow more daylight into areas of the building; however, achieving a desirable amount of daylight across the entire floorplate will be complicated by the building’s geometry. An “ideal” footprint for a well day-lit building is approximately 60' deep in the north-south direction and longer across the east-west direction. The Potter Center, however, is approximately a 160' square. Without cutting an internal atrium through the core of the building, it would be unreasonable to expect high levels of daylight to be present near the core, even after a substantial renovation.

The dated interior and exterior appearance of the building is not likely to affect the DRC or Central Testing Center; however, one could argue it may be a hindrance to the objectives of the College of Engineering. The COE departments slated to move into this building are student-centric departments focused on recruiting students to the college and encouraging the next generation of engineers. Engineering is a field of study focusing on innovation, research, and technology. This message may be diluted when delivered to potential COE students in a facility that projects an image of being outdated.

STRUCTURAL ASSESSMENT

Based upon our review of existing drawings, we were able to ascertain information defining the structural systems for the building.

During our walkthrough of the building, we visually verified to the greatest extent possible the existing structural systems. In general, the building structure appeared to be in good condition and generally as depicted in the original construction documents. In general, the structural system (both gravity resisting and lateral resisting) is intended to remain unchanged throughout the building as a result of the renovation of the facility.

FOUNDATION SYSTEM

The building has a full basement below its ground floor footprint. The basement is bounded by reinforced concrete walls supported on continuous reinforced concrete wall footings. The interior concrete columns are supported on a system of shallow reinforced concrete spread footings. The basement has a concrete slab on grade for its entire footprint.

FLOOR AND ROOF FRAMING SYSTEM

The floor and roof system is framed with a reinforced concrete waffle slab. The waffle slab is supported at regular intervals by reinforced concrete columns that continue down to the basement foundation level. At column locations, the concrete waffle slab construction transitions to solid slab.

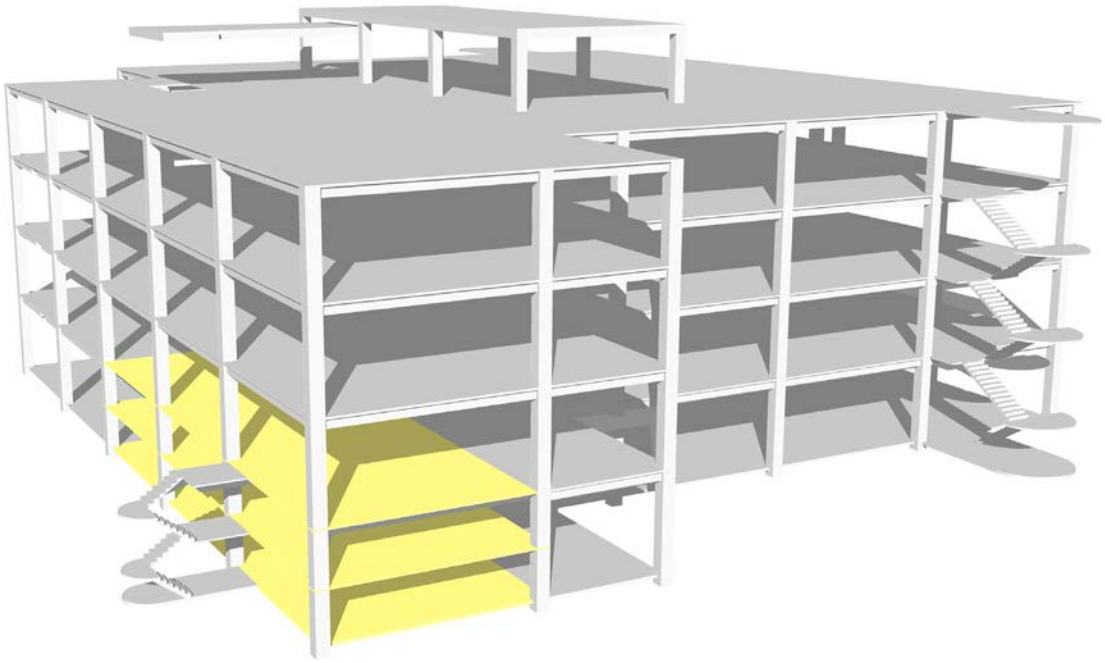
Special framing exists in the southwest corner of the building from the foundation level up to the second-floor level. In this area, steel framing exists that serves as library stacks. The steel structure frames the first-floor level as well as an intermediate level that occurs between the basement level and the first-floor level. The concrete waffle slab structure that occurs typically throughout the building does not occur at the first-floor level in this location.

LATERAL SYSTEM

Although it could not specifically be identified on the existing plans nor definitely confirmed during the walkthrough, it can reasonably be assumed that the lateral force resisting system for the building is accomplished through reinforced concrete shear walls at each of the stair shafts in the building (four total).

EXISTING MEZZANINE LEVEL

There is currently a steel stack framing system in the southwest corner of the building that partially vertically splits the 15’-4”-high basement level into two shorter 7’-8” floors. The headroom in these floor levels is approximately 7’-5”, with sprinkler heads at 7’-0”. These dimensions are lower than the 7’-6” head height requirement permitted by code. Therefore, the mezzanine level and basement area under the mezzanine level cannot be utilized if left unchanged. Removing the mezzanine framing would allow the basement level underneath the mezzanine to be utilized; however, because the framing system also infills an opening on the first floor, the first-floor area would be lost and there would not be a net gain of area. By removing the mezzanine and infilling the 3,000-sft opening on the first floor, the project would gain the square footage on the basement level.



EXISTING MEZZANINE LEVEL

CIVIL ASSESSMENT

The building's main entrance is currently located in the southeast corner of the building. This entrance's elevation is substantially above the grade of the adjacent alley to the east as well as the current hardscape on the south side of the building.

Access is gained to the main entrance from the surrounding lower grade by a series of stairs at the southeast corner of the building as well a ramp system that is located west of the stairs. There is currently no designated ADA-accessible parking areas that serve the building.

To improve the access to the main entrance of the building, ADA-accessible parking should be considered near the main entrance of the building. This parking would preferably be located adjacent to the ramp system that leads to the main entrance or as parallel parking with a clearly defined accessible route to the ramp system. The feasibility of providing the ADA-accessible parking will need to be studied in further detail with the current improvements being constructed for the Centennial Mall Streetscape and Utility Tunnel Repair 2016 project. Analysis of the existing grades and available space along the existing alley to provide parallel parking should also be investigated to ensure the best accessible option is pursued.

CENTENNIAL MALL SITESCAPE AND UTILITY TUNNEL REPAIR

This site and landscaping project will occur immediately south of the Potter Center and around the Wilmeth Active Learning Center, immediately west of Potter. It will fully reimagine Centennial Mall.

As a whole, the proposed design treats the Potter Center as a background building and is designed solely in deference to the Active Learning Center. Though this project directly affects the Potter Center's "front yard," it generally ignores the Potter building. There is some attention paid to installing new landscaping at the southeast stair, which is currently the building's main entrance. This is especially concerning with regard to the Disability Resource Center, a department whose sole purpose is to accommodate students with special needs, since the main entry stair is inaccessible. Students who need to use a ramp to access Potter from the south must travel to the west, to a ramp hidden in a small break in the landscaping. Students who cannot use the stairs are forced to travel a minimum of 138' and a maximum of 274' farther than their peers.

It is the strong recommendation of this study that if the Potter Center is ultimately remodeled to create a new home for the Disability Resource Center, then accessibility to the building should be simultaneously addressed so all students can use a common entry path to the building. If the primary intent of the relocation of the DRC is to move it to a more accessible location, that goal will not be met without a concurrent exterior renovation.

The lack of integration of Potter's circulation flow into the Centennial Mall project should be concerning for the College of Engineering as well. The primary goal of the COE's reimaging of Potter is to house all of the student-centered engineering departments under one roof. Many of these departments are focused on recruiting and encouraging the next generation of engineers—functions that will actively benefit from student foot traffic and a high degree of visibility to campus. These goals are not supported by the current Centennial Mall design.

LEGEND
1 Centennial Mall
2 Seat Elements
3 Bicycle Path
4 Service Drive
5 Cafe Terrace
6 Potential Sculpture
7 Existing Trees
8 Proposed Trees
Mechanical Engineering
Potter Hall
Active Learning Center
Brown
Wetherill
Stanley Coulter Hall
GRANT STREET
CENTRAL DRIVE
Purdue University, West Lafayette Campus
CENTENNIAL MALL & POTTER QUAD
SCHEMATIC SITE PLAN | 03 March 2015

MATCH LINE - REFER TO SHEET L101
POTTER ENGINEERING CENTER
WILMETH ACTIVE LEARNING CENTER
MECHANICAL ENGINEERING ADDITION
BROWN LAB OF CHEMISTRY

MEP ASSESSMENT

MECHANICAL SYSTEMS, HVAC

- The building uses campus steam and chilled water for heating and cooling.
- Beginning in circa 2000, a heating water system was added to support the VAV terminal unit replacement projects.
- Between approximately 2000 and 2013, all of the original terminal units were replaced with VAV reheat terminal units.
- A list of obsolete VAV terminal unit controllers was obtained, and in general, all but the second-floor units are obsolete.
- The building is served by eight air handlers.
- In general, the casings of all of the air handlers observed were in reasonably good shape. No observations could be made on the condition of the fans and the coils.
- There are three main air handlers.
  - ACB-1 serves most of the basement and first floors.
  - ACP-7 serves the third floor.
  - ACP-8 serves the northern portions of the first, second, and third floors.
  - During a site visit, it was observed all three of these air handlers were running in the 45 to 55 Hz, meaning there is little additional capacity available. This was between 8 and 9 AM on a day when the outside air temperature was approximately 25° F and the loads on the systems should not be near peak conditions.
- There are four small air handler/fan coil units that serve small functional areas.
  - ACB-2 provides perimeter heating and cooling for the first, second and third floors.
  - ACB-3 serves primarily the northern portion of the basement.
  - ACB-21 is a 2 to 3-year-old unit that serves labs in the basement adjacent to the mechanical room.
  - AC3-17 serves a suite of small rooms.
- In 2013, a study of the controls systems was conducted, and most of the energy savings options recommended were accomplished leaving little room for additional system temperature control improvements.
- A spot check of supply duct air velocities and static pressure drop per 100' indicates the ductwork was most likely sized using the static regain method. This generally leads to smaller ducts with higher operating duct static pressures and higher horsepower requirements, which are not acceptable under the current energy code.
- The existing air handler systems are intertwined, making it difficult to repurpose areas of the building without affecting more than one system.

MECHANICAL SYSTEM, PLUMBING

- The sanitary waste system incorporates both a gravity drain main and a lift station.
- The building is served by a 3" domestic water line at the northwest corner of the building.
- The domestic hot water system is served by a water softener in the basement mechanical room and utilizes a vertical storage water tank with a steam-fired tube bundle. The system has a return water pump.

ELECTRICAL SYSTEMS

- The transformer vault is located in Room B049, with transformers L-1 and P-1 (primary 12.47 kV, secondary 208V) and P-2 (primary 12.47 kV, secondary 480V).
- The switchboards, ATS, and emergency panel are in Room B048. This consists of 208V distribution switchboards M-L-1 and M-P-1, the 480V distribution panel MSB-1, and panel EM-1.
- Fed from M-L-1 are four bus ducts, one each to the northwest, northeast, southwest, and southeast corners of the building. From the basement, these ducts run vertically and serve the

208/120V distribution and/or lighting and receptacle panels on each floor.

- Room B046 has a motor control center MCC-G-1 that serves mechanical loads and a 208/120V panel L-G-23 for serving the lighting, receptacle, and other smaller loads in this area.
- B003, B023, and B043 each have one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panel board. Basement Room B075 has one 208/120V lighting and receptacle fused-style panel board.
  - SPD-G-SE and L-G-SE in B003
  - SPD-G-NE and L-G-NE in B023
  - SPD-G-NW and L-G-NW in B043
  - L-G-SW in B075
- Room 143 has one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panelboard. Rooms 103, 123, and 175 each have one 208/120V lighting and receptacle fused-style panel board.
  - L-1-SE in 103
  - L-1-NE in 123
  - SPD-1-NW and L-1-NW in 143
  - L-1-SW in 175
- Rooms 203, 223, 243, and 275 each have one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panel board.
  - SPD-2-SE and L-2-SE in 203
  - SPD-2-NE and L-2-NE in 223
  - SPD-2-NW and L-2-NW in 243
  - SPD-2-SW and L-2-SW in 275
- Rooms 303, 323, 343, and 375 each have one 208/120V distribution panel with fused switches and one 208/120V lighting and receptacle fused-style panel board.
  - SPD-3-SE and L-3-SE in 303
  - SPD-3-NE and L-3-NE in 323
  - SPD-3-NW and L-3-NW in 343
  - SPD-3-SW and L-3-SW in 375
- Room 401 has a motor control center MCC-PH that serves mechanical loads and 208/120V panel L-PH-1 for serving the lighting, receptacle, and other smaller loads in this area.

TELECOM SYSTEM

- BDF is located in Room B045. This room was very warm when we did our site survey. We have since confirmed with ITaP that the BDF is cooled by a fan coil unit in need of maintenance.
- The building is currently fed with a 250-pair copper cable from the Tel building (CENT02, 1301-1550 with bad pairs on 1548, 1526, 1725, and 1726) as well as a 24 single-mode fiber fed from the Tel Building (CENT FO1204, 97-120)
- IDF is located in Room 401.
- There is a telecom switch with some wall-mounted patch panels in Room 175.
- All other basement, first, second, and third floor rooms in the northeast and southwest corners of the floors that are noted as IDF on the existing plans are just pass-through spaces with no telecom equipment. These spaces are actually electric closets.
- Existing cabling is CAT 5e.
- Potter has a 600pr copper cable and two 12-strand SM fiber-optic OSP cables serving the building.

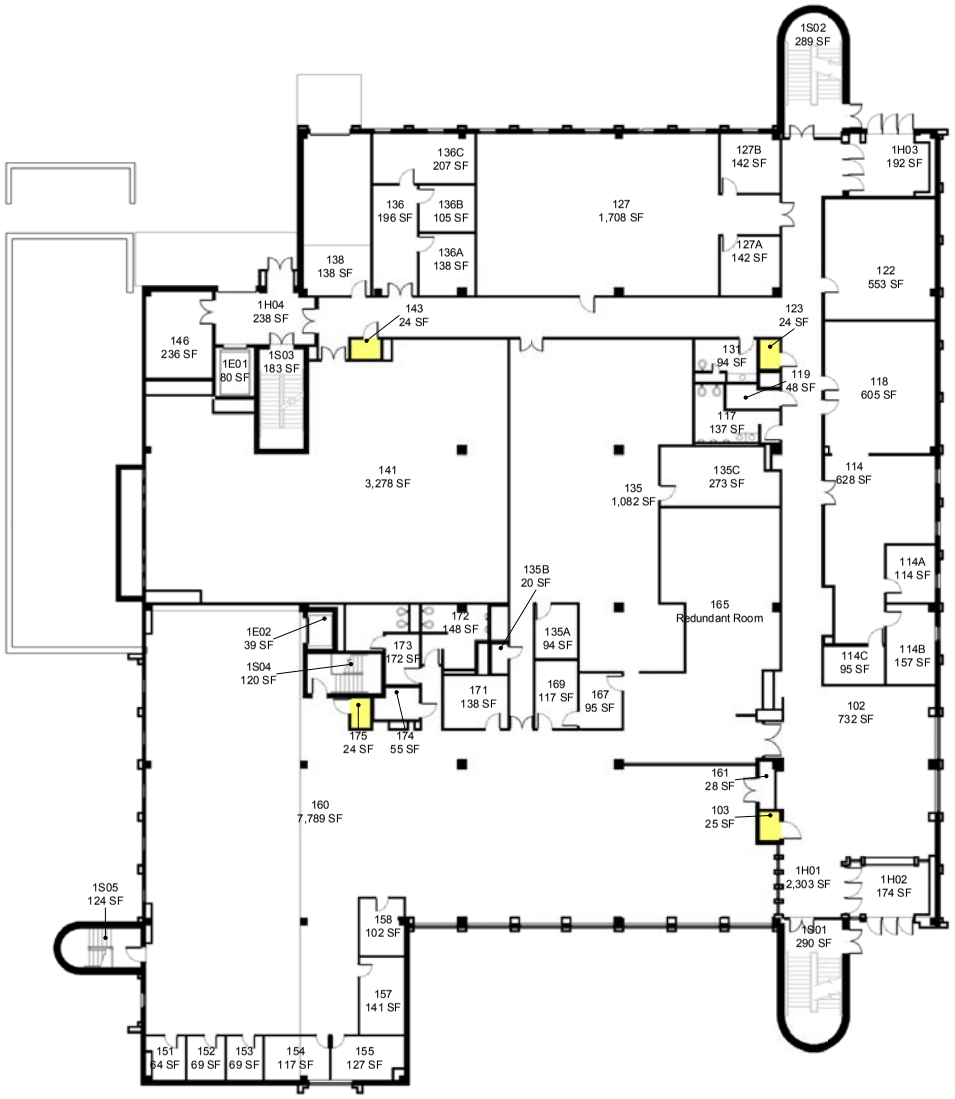
FIRE ALARM SYSTEM

- Appears to be a fairly new Simplex 4100U-ES system installed.



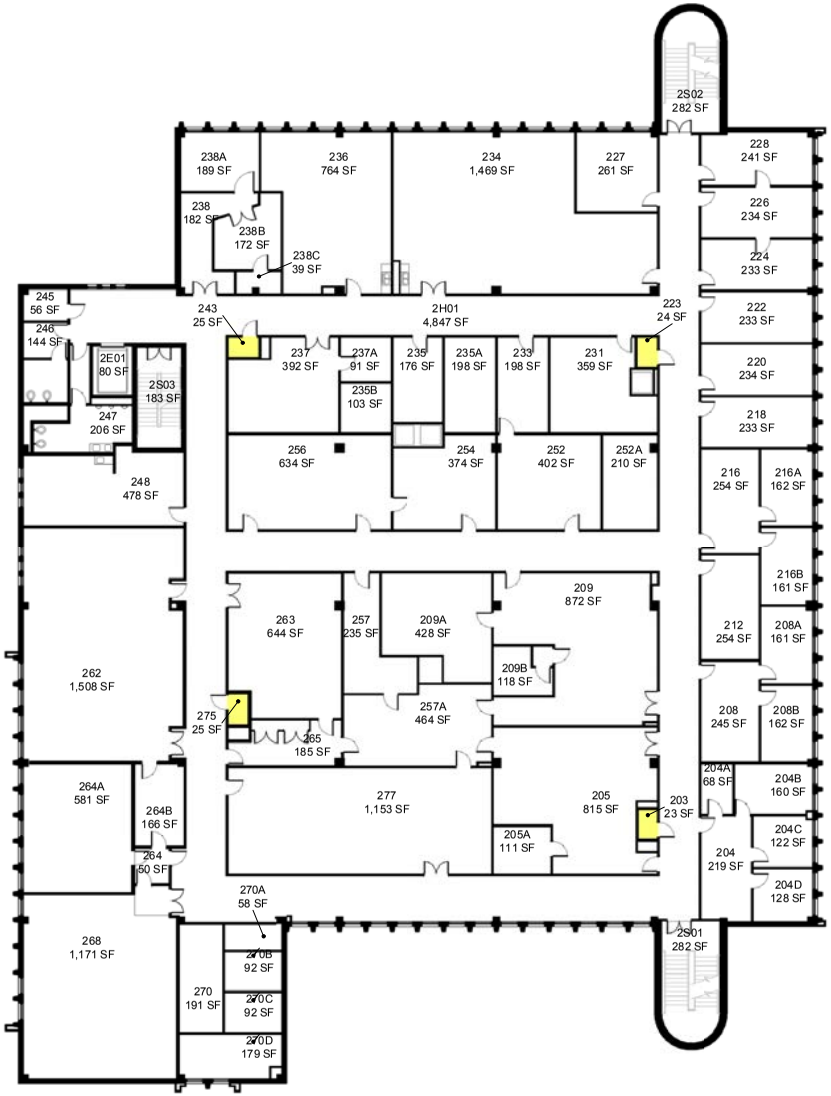
GENERAL BUILDING - MECHANICAL
GENERAL BUILDING - ELECTRICAL
GENERAL BUILDING - PLUMBING
GENERAL BUILDING - TELECOMMUNICATIONS

LOWER LEVEL MECHANICAL



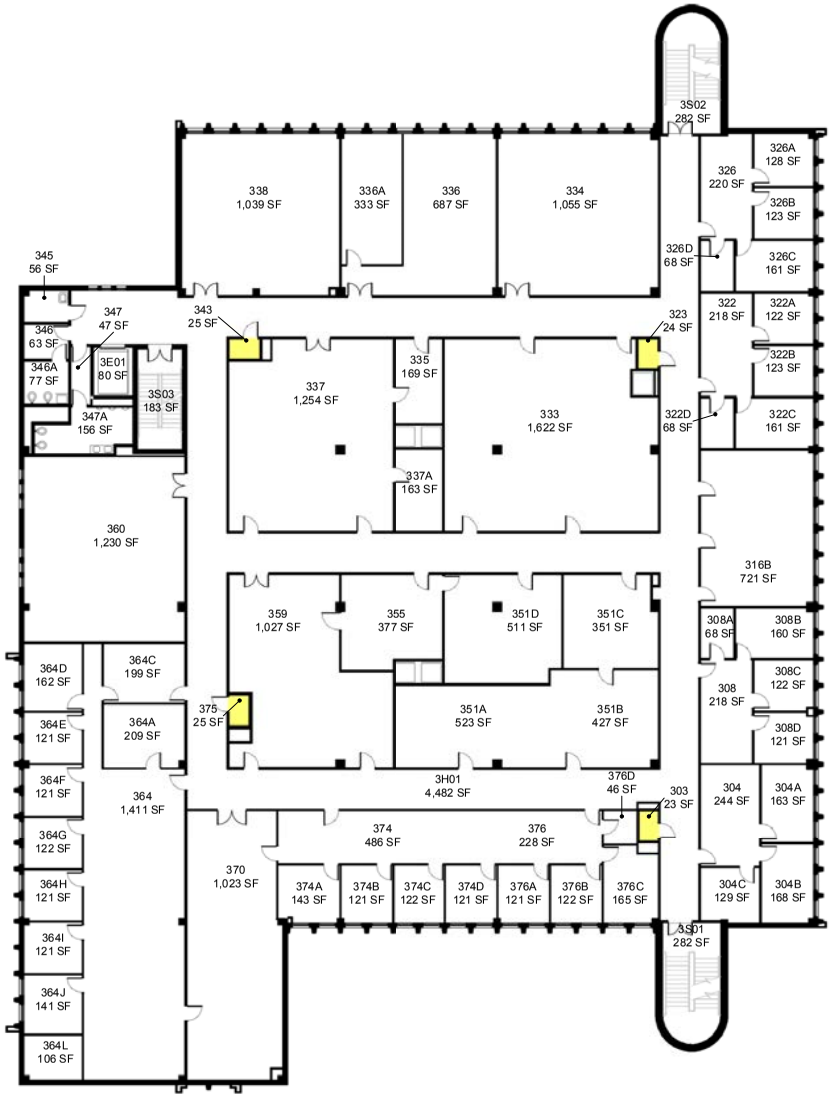
GENERAL BUILDING - MECHANICAL
GENERAL BUILDING - ELECTRICAL
GENERAL BUILDING - PLUMBING
GENERAL BUILDING - TELECOMMUNICATIONS

FIRST FLOOR MECHANICAL

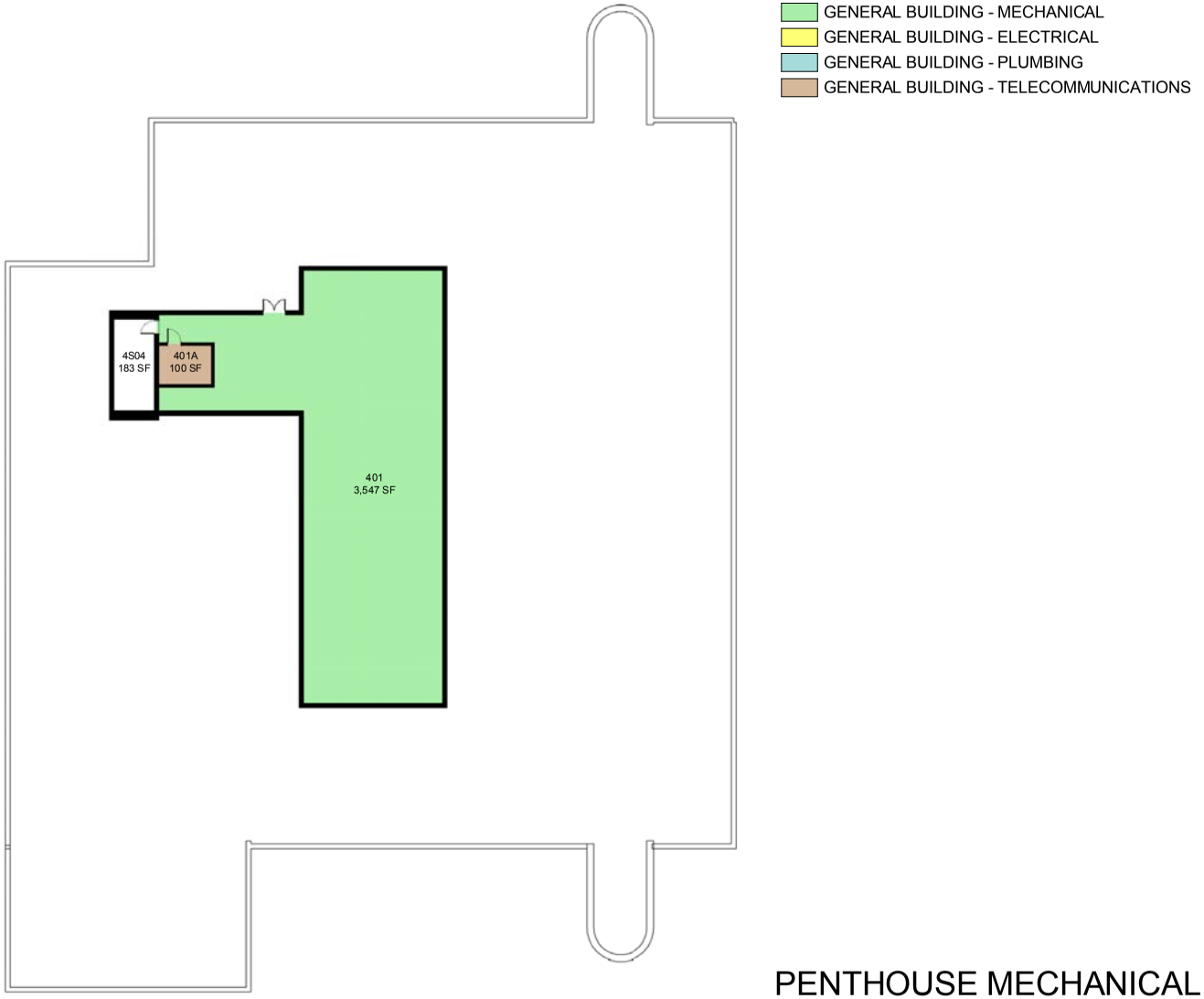


GENERAL BUILDING - MECHANICAL
GENERAL BUILDING - ELECTRICAL
GENERAL BUILDING - PLUMBING
GENERAL BUILDING - TELECOMMUNICATIONS

SECOND FLOOR MECHANICAL



THIRD FLOOR MECHANICAL

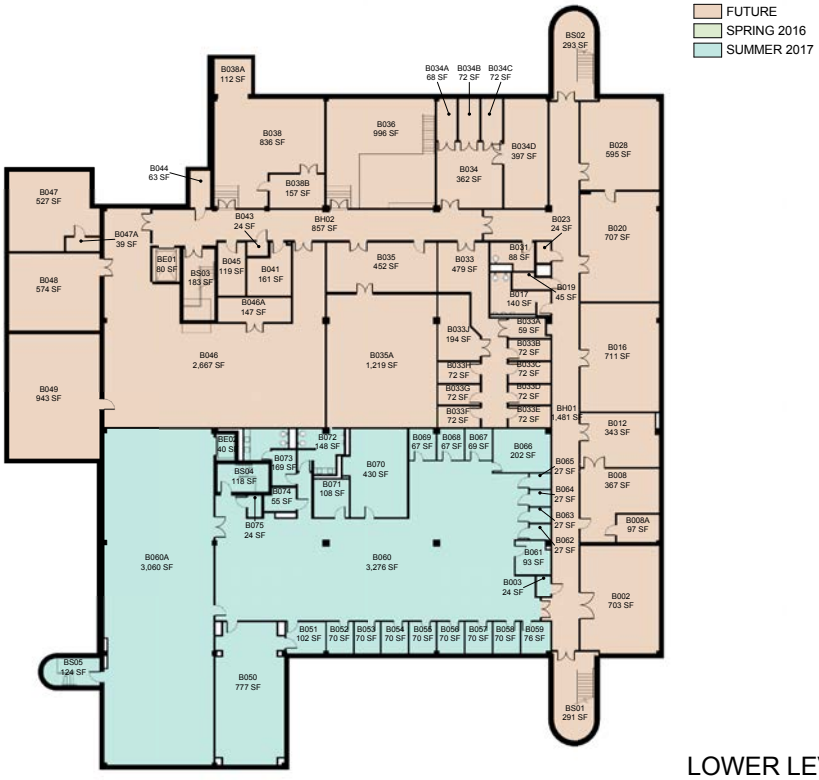


PENTHOUSE MECHANICAL

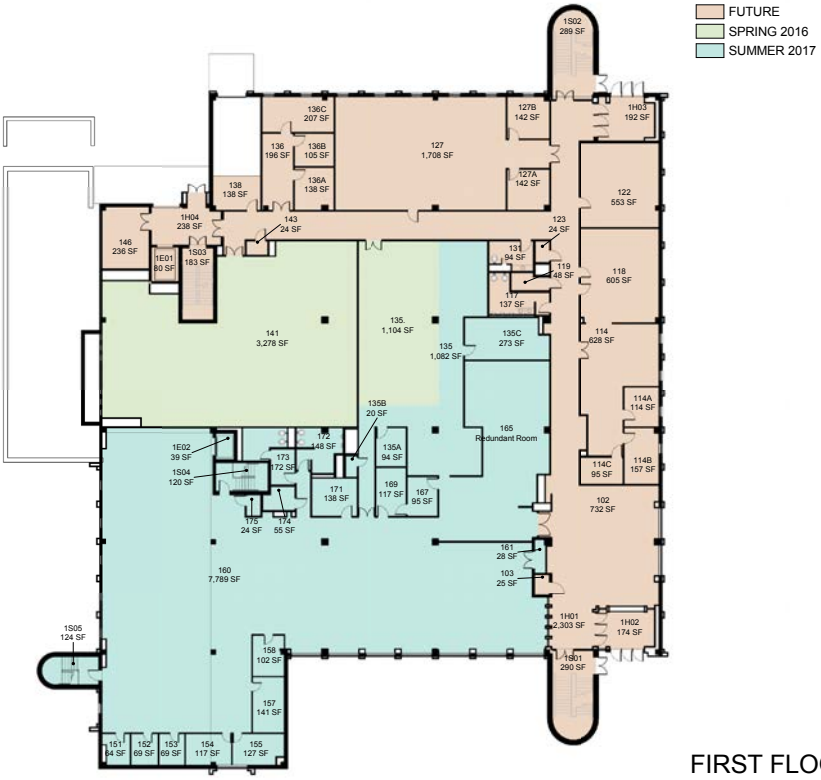
EXTENT OF PROPOSED RENOVATIONS

We have examined five options for renovation of the Potter Center.

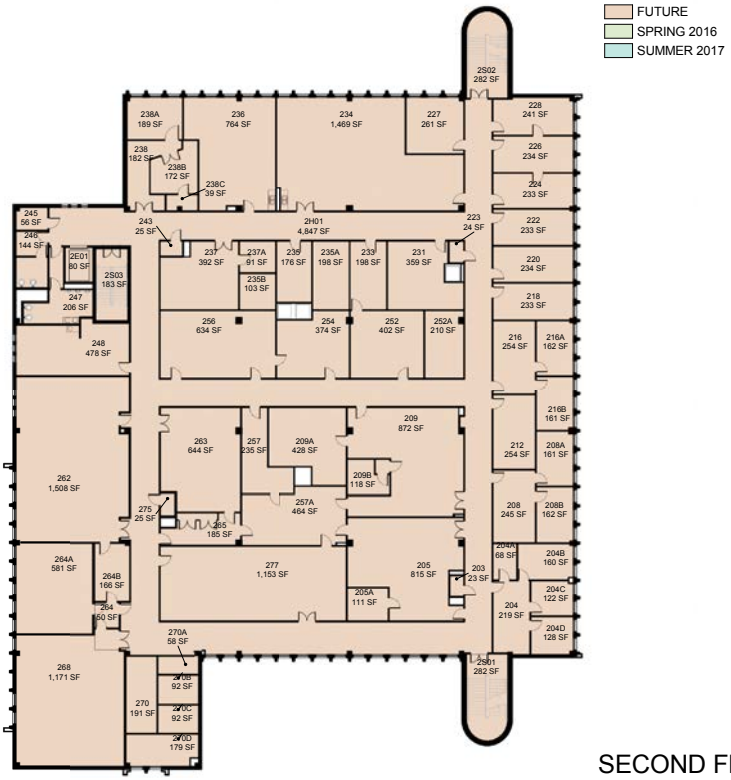
Option A corresponds to space available in the spring of 2016. Options B2 and B correspond to space available in summer of 2017. Option C utilizes the entire building, leaving research labs unchanged. Option D represents renovation to the entire Potter Center.



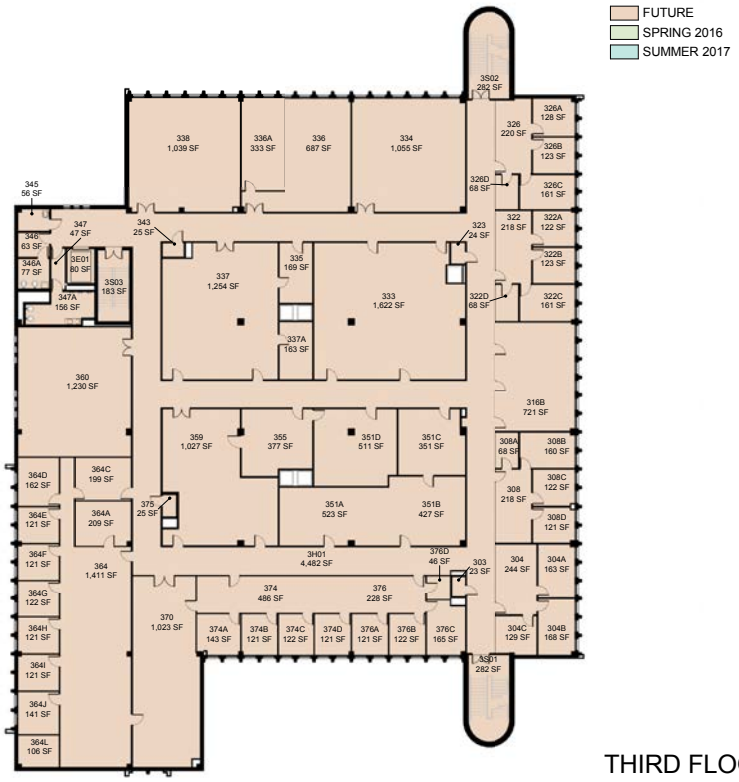
LOWER LEVEL AVAILABILITY



FIRST FLOOR AVAILABILITY



SECOND FLOOR AVAILABILITY



THIRD FLOOR AVAILABILITY

CENTRAL TESTING CENTER

The goals of the CTC are to create a new campus-wide resource that will present faculty with a new option for testing and increase testing flexibility for courses across campus; to increase cost control by efficiently combining testing services and departments; to increase efficiency of resources through greater utilization of staff, testing equipment, and testing space; to increase academic integrity by creating purpose-built space to effectively proctor exams administered by a robust and specifically trained staff; and to greatly increase ease of use both for students and faculty by creating a single destination for testing needs.

CURRENT TESTING CENTER USAGE

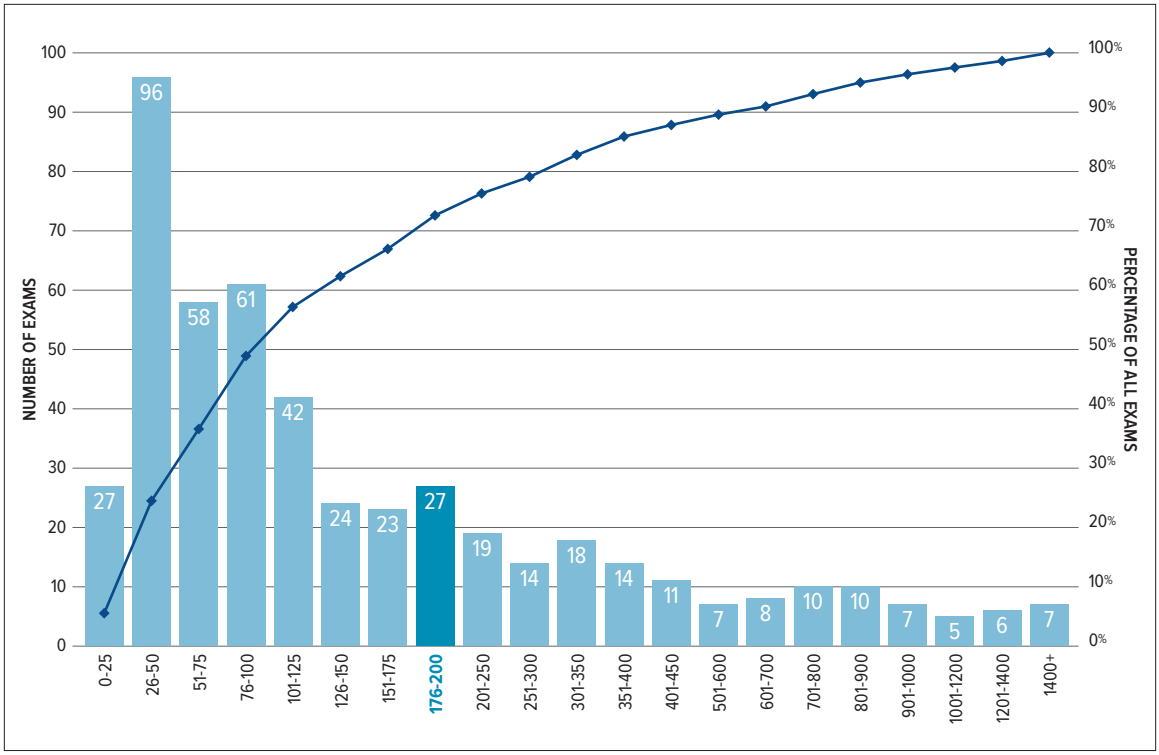
The four testing centers being consolidated into the Potter Center facility have a total of 101 testing seats. However, the ultimate goal of the testing center is not solely to create a larger and more efficient aggregated testing center, but to create a new resource for testing that can be used by instructors across campus. Therefore, the testing center has been sized to serve campus as a whole.

Testing Center	Seats	Tests/Day*
Disability Resource Center	24	15
Purdue Testing Center	16	30
Digital Education	25	20
Foreign Language Placement	36	30
SUM	**101**	**95**

* Foreign Language Placement testing does not occur year round.

Beyond the need to bring units together is the need, as shared by University leadership, to create better conditions for exam delivery at Purdue. A task force on this topic, under the leadership of the director of Digital Education, has been meeting for about a year. The result has been to encourage the bringing together of testing units and to find ways to address University concerns about test integrity and proctor services. One specific result is the conclusion that a contained, flexible, and carefully managed testing space that can accommodate up to 200 students at one time would be a definite asset to faculty and instructional staff. This new center will additionally support the expansion of proctored testing for Purdue's flipped/hybrid classes that are redeveloped as part of IMPACT and the growth of online courses likely to flow from the new strategic plan for Digital Education. The center will also reduce the demand for academic and non-academic spaces during periods of volume testing. Furthermore, faculty are presented with a facility where their exams could be delivered in multiple ways, using the testing center or their own personnel or both. The center will likewise serve as a crucial asset during the 25 days in the summer during STAR and BGR.

The following chart illustrates the number of evening exams given on campus in the fall of 2014. Evenings are the time of day in which most tests are administered. As evidenced by the chart, a testing center with between 176 and 200 seats would serve approximately 74 percent of campus.



CURRENT TESTING CENTER FACILITIES AND CONDITIONS

There are currently four small testing centers spread throughout campus—Foreign Language Placement, Digital Education, the Disability Resource Center Accommodated Testing Function, and the Purdue Testing Center. Each center focuses on a specific testing need and has a dedicated staff. Each also has a unique way of administering their specific test(s), as well as specific rules and regulations for the administration of that test. The intent of the Central Testing Center is to combine these functions into one purpose-built location. Analysis of the four testing centers follows here.

* Items marked with this symbol appear to be the most stringent of this line of requirements and will be labeled as a "controlling" requirement. The testing center, in part or in whole, will then be designed toward meeting this requirement.

** This area presents an interesting synergy with some of the functions housed within the College of Engineering that focuses on introducing students to the discipline of engineering and may benefit from the exposure to incoming freshman.

Foreign Language Placement Testing

The goal of the foreign language testing center is to effectively place students in foreign language courses appropriate for their current comprehension of that foreign language.

Physical Test Administration Requirements

- Windowless and distraction-free environment
- Computer at each testing station
- Large line for queueing students during peak demand that is acoustically separated from the testing center*
- Gathering area for students waiting for others to complete testing**
- Pre-screening area to conduct individual interviews with students prior to testing*
- Semi-private post-testing area where students are debriefed on their test results*
- Storage area for headphones

Testing Personnel

- Three TAs are used during the year
- Five TAs are used during STAR

» Some TAs must be specially qualified to effectively speak the language being tested in order to verify that students are sitting for the correct test*

Testing Load

» Number of Testing Seats: 36
» Total Number of Students Tested Per Regular Session: 90 to 100
» Total Number of Students Tested During STAR: 150 to 300 per day for 19 days
» Typical Duration: 15 to 60 minutes

Current Testing Hours

The testing lab is open during five sessions per year.

» The month of August
» Three days during November
» Two days during January
» Three days during April
» STAR - Last two weeks of June through first two weeks of July (19 days) from 10 AM to 3 PM

Current Conditions

The Foreign Language Testing Lab currently operates out of a standard computer lab in Stanley Coulter Hall, Room 231, which is repurposed for testing during five sessions per year. Computers are not purpose built for testing. TAs must log in and manually set up each station before testing begins. Tests are web based. There are no storage lockers for jackets and backpacks; therefore, students bring all their items with them to the testing center. The testing stations do not have partitions between students. Both the pre-test screening interview and post-test debriefing are conducted inside the testing room.

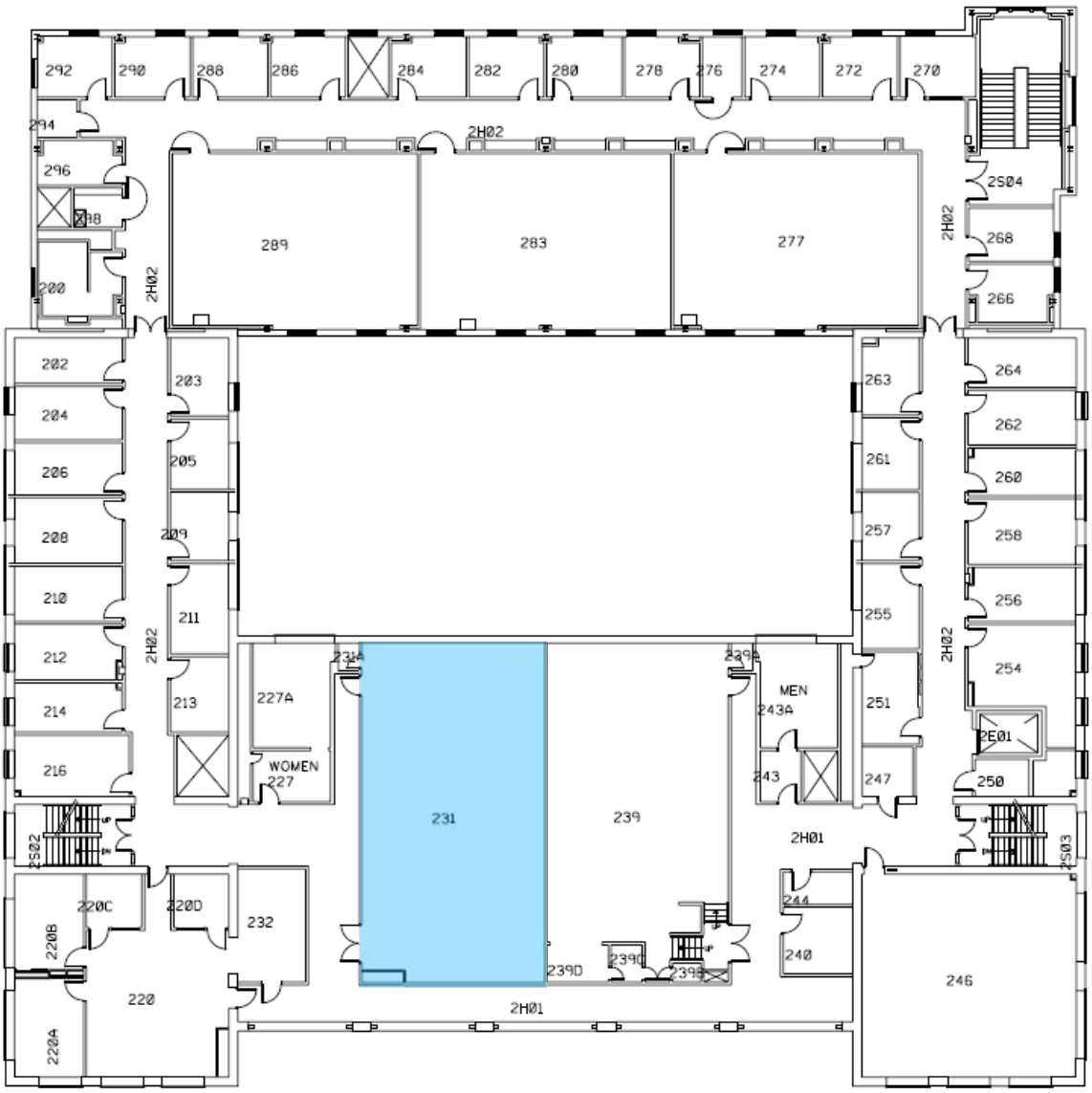
Current Sequence of Operations for Testing

» Students do not schedule exams in advance; they simply arrive at the testing site carrying a permission form (Form 231).
» Students line up in the hallway leading up to the testing center. The line can be very long, and the noise created by large numbers of students talking can be distracting for students taking the test, as well as for other occupants of the building.
» Students sit down at a table to go through the screening process. Note that the testing staff feels strongly that this interaction with students is critical to the efficiency and academic rigor of the testing.
» Each student hands TA Form 231 and their student ID for TA to verify their identity.
» The duration of the screening process is 7 to 10 minutes.
» Once the student has finished the screening process, they are directed to sit at an open computer to take their exam. It is up to the student to choose the correct exam. Students bring jackets, backpacks, and any other materials they have with them to the test station. TAs monitor students in person to discourage cheating.
» The duration of an exam is 15 to 60 minutes. Tests do not have scheduled breaks. Due to the short duration of the test, it is very rare for a student to ask to leave to use the restroom. Some tests require headphones for listening comprehension.
» Students receive results immediately following completion of the exam.
» Once a student completes the exam, the student raises a hand and a TA walks over to explain the results.
» Students exit the exam and often wait for others to complete their exam.

Potential Areas for Improvement

» Acoustically separated waiting area for students
» Mechanism to scan student IDs prior to testing
» Ability to scan Form 231 on site during the screening process
» Ability to test more frequently throughout the year

Current Floor Plan STANLEY COULTER HALL (SC) - SECOND FLOOR



Digital Education Testing Center

The Digital Education Testing Center is an on-campus proctored testing site that administers a variety of paper-based and computerized exams. All testing is scheduled ahead of time by individual students. Tests administered in this center are for Purdue-based distance learning courses, as well as CASA (Core Academic Skills Assessment) and Pearson Vue exams.

Physical Test Administration Requirements

- Some requirements vary by test. When proctoring tests for professors as make up exams, the requirements are dictated by that professor.
- Lockers for student item storage
- White noise system in testing area
- Business office in close proximity to handle cash or check payments by students for certain tests*

Testing Personnel

- One full-time staff

Testing Load

- Number of Testing Seats: 25
- Number of Students Typically Tested Simultaneously During the Week: 5-6
- Number of Students Typically Tested Simultaneously on Saturday: 20
- Typical Duration: 90 Minutes

Current Testing Hours

- Monday, Tuesday, Friday: 8:30 AM to noon and 1:30 to 4:30 PM
- Wednesday, Thursday: 10 AM to noon and 1:30 to 4:30 PM

Current Conditions

The Digital Education Testing Center is located in the lower level of the Stewart Center in Room G59. The test proctor sits outside the room and can view students through windows in the wall located approximately 4' above ground. The proctor can also monitor students through video cameras mounted in the ceiling. All materials students bring to the testing center are stored in lockers, unless the testing center has received specific instructions to allow students to bring items with them to the test. The testing room is currently not acoustically well isolated from the outside, and noise can cause an issue. A white noise system is installed in the ceiling to create background noise. All testing is taped. The video is stored for 30 days.

Current Sequence of Operations for Testing

- Student arrives within 15 minutes of their scheduled testing time.
- Student shows ID to the staff for identity verification.
- Student signs login form.
- Staff assigns student a locker to store any materials not permitted in the testing room. Student keeps the key during the exam.
- Students must pay a fee for certain tests. Most pay fees online. Students with cash or checks are sent to the business office and then return to the testing center with a receipt.
- For certain tests, students must have their picture and thumbprint taken.
- The student is directed toward a specific station to take their exam. Only certain computers can administer CASA tests, and separate computers can administer Pearson tests. The computer setup is dictated by the testing company. Paper exams account for 40% of tests; 60% are online.
- Exams do not have scheduled breaks. If students ask to leave to use the restroom, the clock continues running on their exam. It is very rare for students to ask to leave during an exam.
- Once the student completes an exam, he/she exits the room.
- Each student signs the logout form.
- Student retrieves materials from an assigned locker and leaves the testing center.
- Completed paper exams are then mailed, scanned, or picked up by the professor.

Potential Areas for Improvement

- Greater acoustic isolation for the testing room
- Better view into the testing room for the proctor

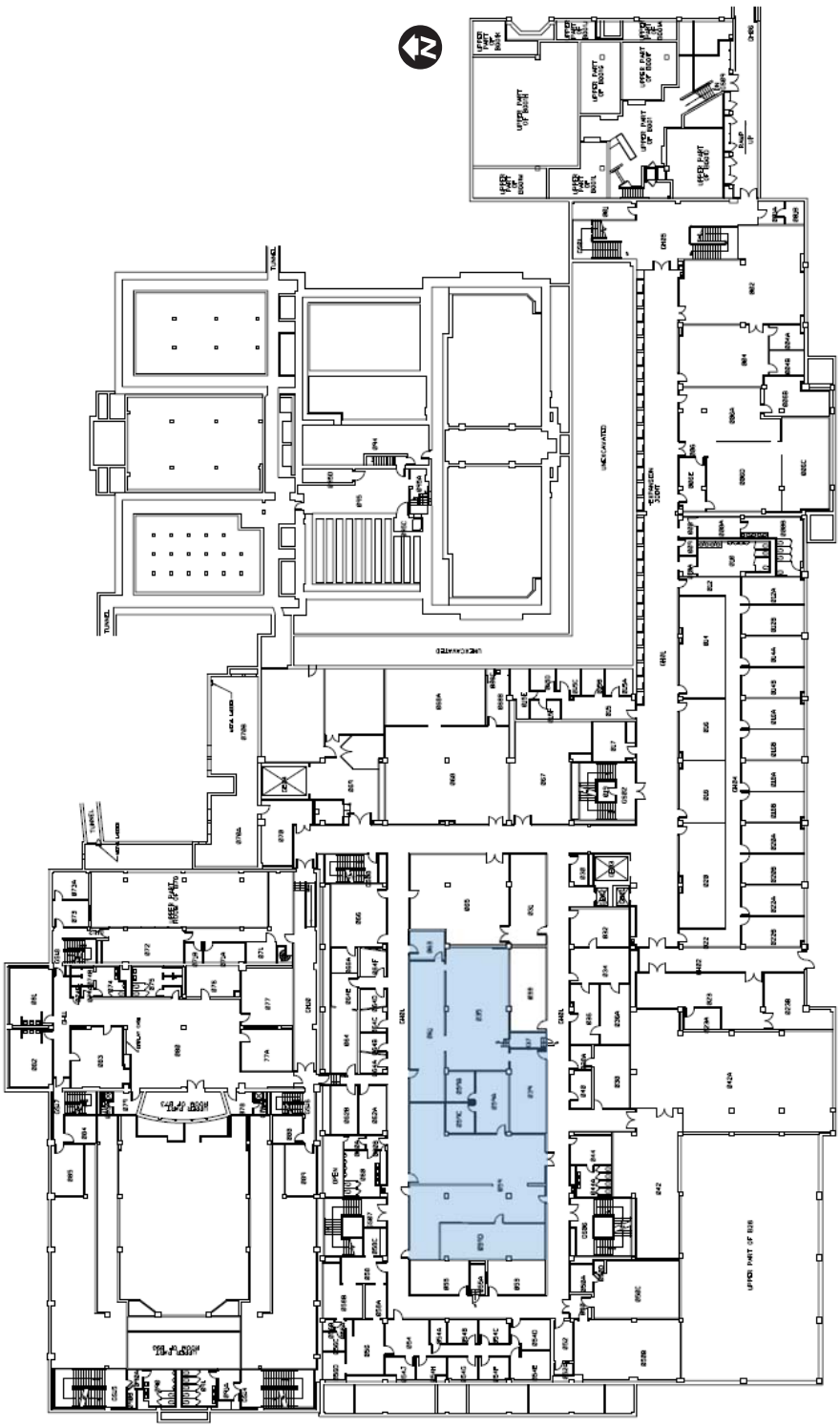
Online Compliance

This new unit is being created per a requirement set by Purdue's vice president for Ethics and Compliance, who will oversee compliance with digital materials. The proposal for the unit is currently under review by Purdue's in-house legal council. The unit will be housed within Digital Education and be led by an associate director of Digital Education. While the associate director position is funded and was recently filled, the remaining positions are yet to be funded. These staff (and the corresponding demand for space) are therefore new and not connected with the existing units. As part of this project, the staff for this unit will be accommodated. This unit will work in conjunction with the Disability Resource Center, and locating these offices in proximity will enhance collaboration and increase efficiency.

Current Photos



Current Floor Plan STEWART CENTER (STEW) - GROUND FLOOR



Disability Resource Center (Accommodated Testing Function)

The goal of the accommodated testing function of the Disability Resource Center is to create an environment that accommodates students with special needs when taking exams during typical coursework.

Physical Test Administration Requirements

- Windowless, very quiet, distraction-free environment
- Lockers for student item storage
- Testing stations must be clear of all items, including computer monitors*
- Private, isolated testing rooms (used for certain students who require extreme isolation from distraction and for students who must take a test with another person working as a scribe/reader)*

Testing Personnel

- Two full-time staff

Testing Load

- Current Number of Testing Seats: 24
- Number of Students Typically Tested During the Semester: 2,500
- Number of Students Typically Tested During Finals Week: 650

Current Testing Hours

- Weekdays from 8 AM to 8 PM

Current Conditions

The Disability Resource Center and its associated testing center is located in Young Hall, Room 830. The testing function of the DRC is among the critical functions provided by the center. One-third of all students registered with the DRC use the testing center. Unlike the majority of tests given at other test centers, testing at the DRC is nearly all paper-and-pencil testing. The primary accommodation provided by the DRC is to allow more time for test taking. Tests are scheduled by students. Students primarily take tests in a large room with testing stations divided by study carols, similar to the computer testing centers. A small number of students requiring additional isolation or assistance from a scribe or reader use private testing rooms. Up to 12 private testing rooms can be used at one time. The testing center's greatest demand occurs during finals week. During this time, the DRC sets up four testing locations across campus, and proctors administrate approximately 650 exams.

Unfortunately, during this period, the demand for accommodated testing far exceeds the capacity of the DRC's testing space. The DRC makes use of libraries and other spaces to house accommodated exams. More importantly, the DRC staff must be augmented by volunteers and paid proctors from its parent unit (Student Success) and other units at Purdue. The Central Testing Center will create the ability for accommodated testing to occur in a much more efficient and controlled way and with a deeper overall testing staff (to oversee the inevitable volunteer proctors).

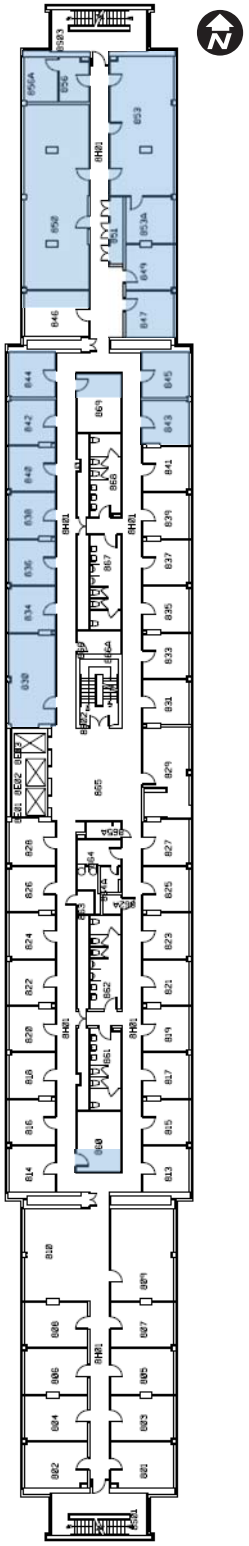
Current Sequence of Operations for Testing

- Student arrives at the DRC at the scheduled testing time.
- Student checks in with staff.
- Student shows ID to staff for identity verification.
- Staff assigns student a locker to store any materials not permitted in the testing room.
- Staff assigns the student a seat number where he/she goes to take their exam.
- Student leaves DRC.
- Completed tests are picked up from the DRC. A large amount of these tests then travel to IDP for grading.

Potential Areas for Improvement

- Need 10 private testing rooms
- Need 30 to 35 testing seats
- Video recording for private testing rooms

Current Floor Plan ERNEST C. YOUNG HALLL (YONG) - EIGHTH FLOOR



Purdue Testing Center

The goal of the Purdue Testing Center is to serve campus and patrons outside of campus by providing a variety of testing services. It is the only center on campus fully certified by national testing organizations and offer national registration and licensing exams.

Physical Test Administration Requirements

- Lockers for student item storage
- White noise system in testing area
- Secure server stored in a locked location to store testing video*
- Locked room for national test storage*
  - No windows permitted
  - Must have a hard lid ceiling
- Security Equipment*
  - Three screens for video monitoring
  - Three DVRs
  - Thumbprint scanner
  - Student card swipe
  - Area to take pictures
  - Signature pad
  - Palm vein scanner
  - Wand-type metal detector

Testing Personnel

- One full-time staff

Testing Load

- Highest Number of Students Per Day: 28 to 32
- Number of Testing Seats: 16

Current Testing Hours

- Monday, Tuesday, Friday: 8:00 AM – 5:00 PM
- Wednesday, Thursday: 8:00 AM – 9:00 PM

Current Conditions

The Purdue Testing Center is located in Schleman Hall, Room B42. Of all the testing centers on campus, the Purdue Testing Center has the most stringent testing requirements and accountability standards. This is driven by the standards required by the national testing companies. Videotaping of testing is critical, and footage is stored on a locked server. The tests given in this center can be quite long (some are six hours) and include mandated breaks. Due to the high number of students taking advanced credit exams, those are given outside the center, typically in Hampton hall, which seats 40 to 80 at a time.

Current Sequence of Operations for Testing

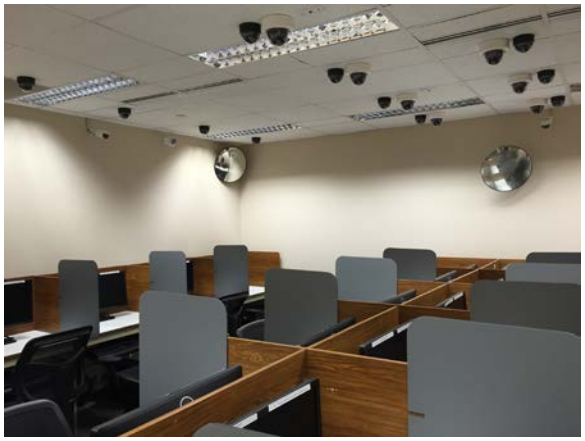
- Students line up to be checked in for testing.
- Three to four students at a time load their belongings into a locker.
- Some students are directed to read certain material before they are admitted.
- Students line up for admittance.
- Students go through security procedures required by that test (signature pad, palm vein scanning, thumbprint, picture taken).
- Proctor walks student to outside the testing room.
- Proctor pats the student down or wands the student.
- Student waits until a station is open and sits down.
- When student completes exam, he/she signs out and leaves the center.

Potential Areas for Improvement

- Break area with restrooms and vending machines inside the “security zone.”
- Secure restrooms should have multiple fixtures for quick use. Some students only receive an 8-minute break during their test.
- Need a larger testing center with 30 to 40 seats, possibly two 20-person rooms.

Current Photos

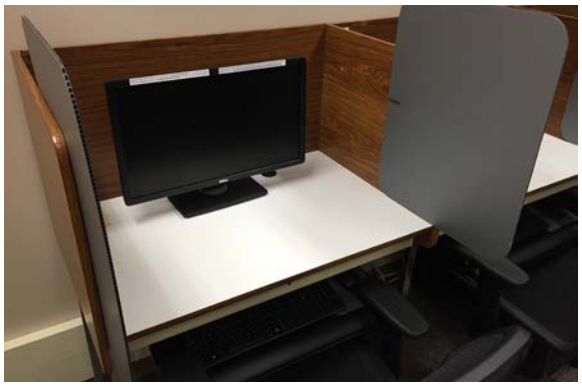








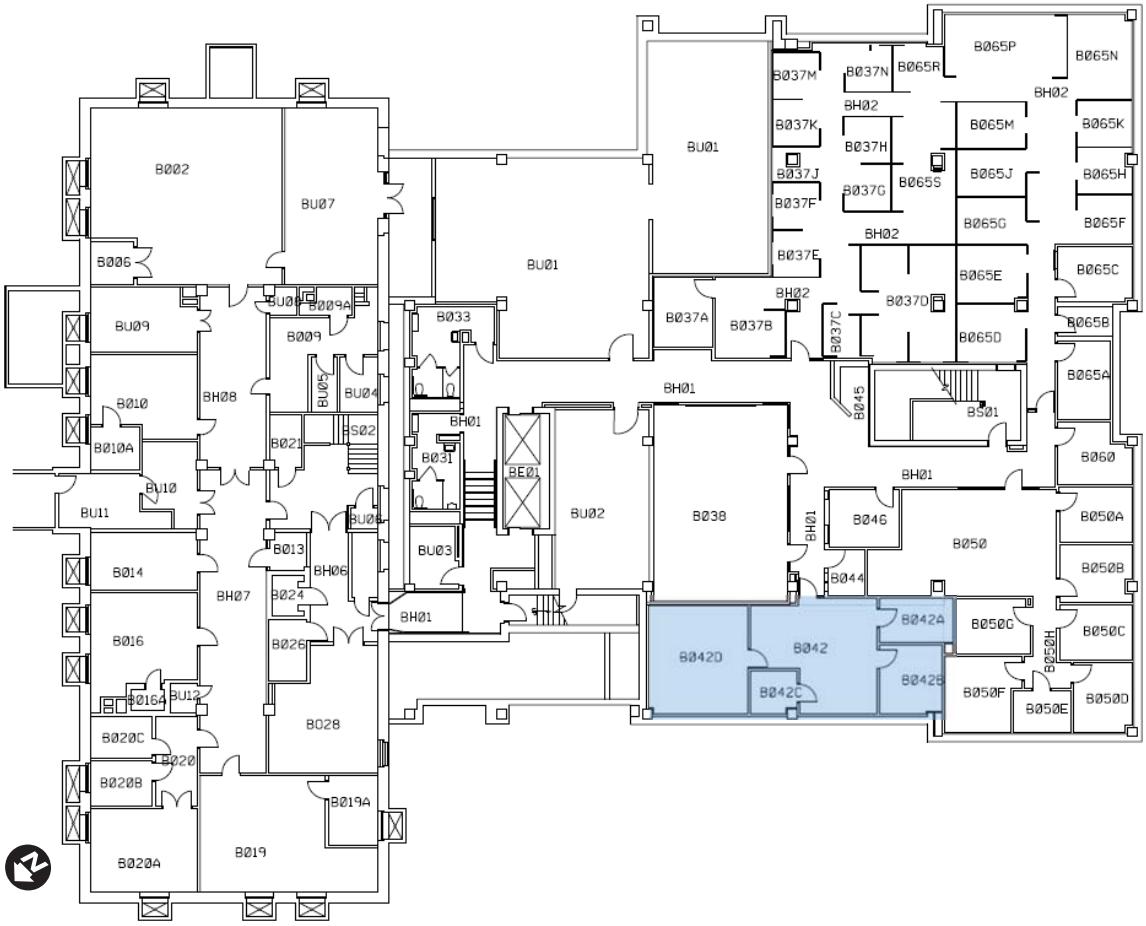








Current Floor Plan HELEN B. SCHLEMAN HALL OF STUDENT SERVICES (SCHL) - BASEMENT



SPACE PROGRAM

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

CENTRAL TESTING CENTER DESIGN

The overall goal of the Central Testing Center design is to combine these existing testing centers in one purpose-built location. The testing center will also accommodate general “course” tests. In this instance, professors will direct entire classes to the testing center for administration of tests as required by that course. In general, the testing center will focus solely on testing as administered by computer. Professors administering traditional pencil-and-paper tests will be encouraged to continue to administer those tests in their classroom spaces. The exceptions to this rule will be for students with special needs and for make-up testing. In these instances, the center will accommodate pencil-and-paper testing.

The majority of tests administered in the center will be in service of a Purdue University course. However, the center will also handle specialized national licensing and certification exams, such as those currently offered through the Purdue Testing Center and the Digital Education Testing Center. These national exams are offered through outside testing vendors that give franchise locations the

ability to administer these tests providing they follow strict administration guidelines. Due to the stringent nature of these requirements, creating a secondary testing space for the administration of these "high-security" tests appears to be the most prudent course of action. Not applying these rules to the entire testing center will allow the larger "general" testing center a higher degree of freedom and flexibility. Therefore, as evident in the design of the space, the centralized testing center will have two distinct spaces—one for general testing and another smaller space for high-security testing.

CENTRAL TESTING CENTER SEQUENCE OF OPERATIONS

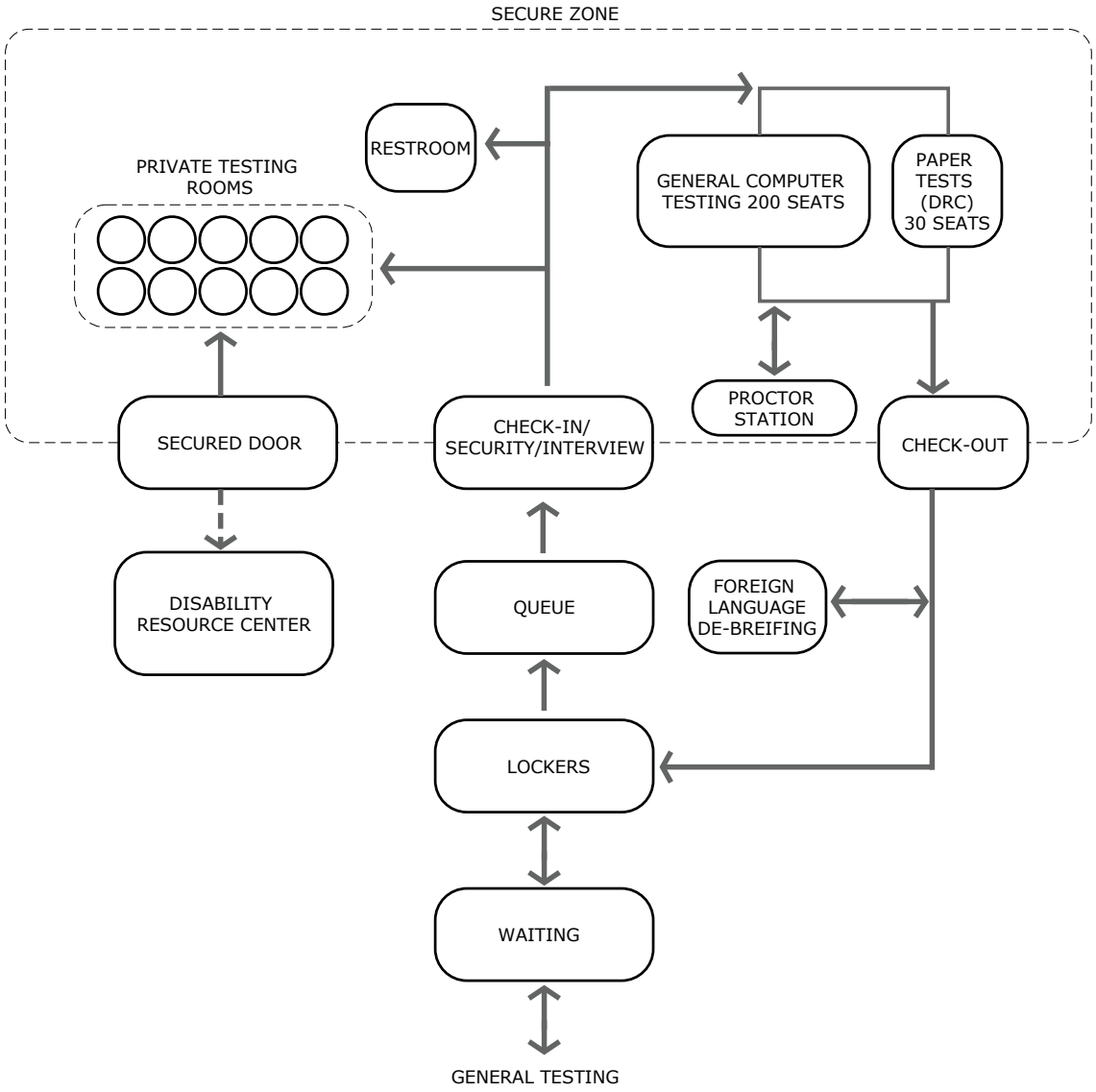
Central Testing Center - General Testing

1. All general testing students (including students taking accommodated testing through the DRC) enter the testing center through a waiting room/lounge space.
2. Students are instructed through signage to place all their items in lockers before checking in.
3. Students wait in line to check in.
4. Students check in and staff ensures compliance with the rules for their particular test (i.e. what items students can bring with them, or if they have to sit for an interview)
5. Students taking foreign language placement testing sit for the pre-test interview.
6. Students are assigned a testing station and directed toward the testing area appropriate for their test, or to a private testing room.
7. Proctors monitor tests live and via video monitor from the proctoring station.
8. Upon completion of the test, students check out.
9. Students taking foreign language testing go to the debriefing station to review their test results.
10. Students go to their lockers to retrieve their items.
11. Students exit through the waiting area.

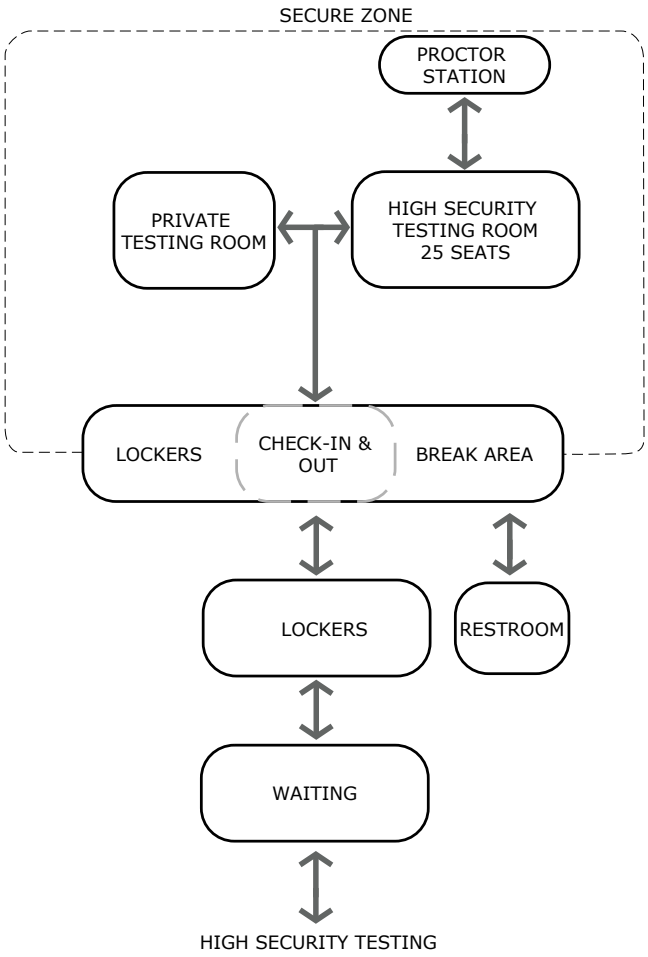
Central Testing Center – High-Security Testing

1. All high-security testing students (including students taking accommodated testing through the DRC) enter the testing center through a waiting area.
2. Students wait in line to check in and are assigned a locker.
3. If required by the test, students complete the pre-test requirements (i.e., reading a pamphlet)
4. Students place all their items in lockers.
5. Students go through the security procedures required for that test (signature pad, palm vein scanning, thumbprint, picture taken).
6. Students are individually walked to their testing station or private room.
7. During the course of the test, students may freely take a break or use the restroom in the secured break area. This area will be separated from the testing center via a one-way door. After taking a break, students will check back in and be walked to their testing station.
8. Proctors will monitor tests live and via video monitor from the proctoring station.
9. Upon completion of the test, students raise their hand and are escorted from their testing station.
10. Students will go to their lockers to retrieve their items.
11. Students will exit through the waiting area.

GENERAL TESTING CIRCULATION DIAGRAM



HIGH-SECURITY TESTING CIRCULATION DIAGRAM



HIGH-SECURITY TESTING CENTER REQUIREMENTS

A testing center must meet strict requirements to receive certification to administer a vendor's tests. The following requirements are a compilation of the requirements of testing vendors Pearson VUE, CLEP, and Prometric. In each case, the most stringent of each vendor's requirement has been compiled in an effort to meet the requirements of all vendors. It will be important for this information to be confirmed before proceeding with actual construction of the testing space. The individual requirements of each vendor can be found in the appendix of this document.

Staff

Based on 25 testing stations, a staff of two people is required to be physically on site to operate the center when in use. The total testing center staff must contain at least two certified TAs. One certified TA must be present in the testing center at all times.

Admissions Desk

This is the area where students first interact with a testing center representative. It also is where students check out during breaks and completion of the test.

Requirements

- Web camera
- Palm vein device
- Signature pad
- Hand-held metal detector
- Chair for candidate photographs, 2' from counter
- Area monitored by a camera with audio

Proctor/Surveillance Station

This workstation is where the test proctor will monitor testing in progress.

Requirements

- Viewing window into room
- Monitors to view video surveillance of testing in progress
- Telephone
- Area monitored by a camera with audio

Lockers

Students will store all their belongings in a locker during testing.

Requirements

- One locker per work station
- Must be outside the exam delivery room
- Key must be given to student

Testing Room

This is the physical space that houses the individual testing stations.

Requirements

- Individual test station screens positioned perpendicular to entrance so monitors cannot be seen from entrance
- Stations should not be at right angles to each other
- A Pearson VUE video camera should be centered above the partition between every other workstation
- A Prometric video camera should be centered above the partition between every other workstation
- No more than 25 test stations
- Must contain a wall clock, mounted to be visible to exam candidates
- Room must be entered directly from reception area with a lockable door
- Room must be lit using glare-free lighting
- A microphone must be centrally located to record room audio
- Palm scans are required at each entrance and exit of the room
- A white noise system should be designed into the space
- Surfaces should be finished in acoustically absorptive materials

Testing Stations

The station where test candidates will be seated to take tests.

Requirements

- Comfortable chair with back and height adjustment
- Test cannot be administered on a laptop
- Single 19-24" monitor, non-touchscreen, aspect ratio of 16:9 or 16:10. Note that Pearson VUE testing requires GPU scaling with this aspect ratio
- Four-foot-wide desk surface
- Partitions between stations, 5' high, projecting 4' from back of desk

Private Testing Room

Individual testing room where a single candidate can take a test in isolation, with space for an accommodation assistant when necessary.

Requirements

- Sound-isolated room
- Door with viewing window
- Two comfortable chairs with back and height adjustment
- Test cannot be administered on a laptop
- Single 19-24" monitor, non-touchscreen, aspect ratio of 16:9 or 16:10. Note that Pearson VUE testing requires GPU scaling with this aspect ratio
- Five-foot-wide desk surface

Server Room

The area to house the local server/cache proxy for storing test delivery information and test packages as well as temporary storage of candidate results.

Requirements

- Secure space
- Connected to all exam delivery workstations through a LAN
- Server should never be turned off
- Can be in a remote location
- Server cannot be a laptop

INSTRUCTIONAL DATA PROCESSING

The goal of Instructional Data Processing (IDP) is to support the educational process at Purdue University by providing fast, reliable, accurate test scoring and evaluation services while embracing the most current technologies. IDP manages course evaluations, test proctoring services, Scantron grading, specialized grade book work for certain departments, and Scantron scoring in support of research.

CURRENT FACILITIES AND CONDITIONS

Space Program

- Area for a high-capacity scanner (Scantron iNSIGHT 150)
  - Requires precise humidity control
  - Requires a dust-free environment
  - An acoustically isolated space is ideal as the scanner produces a high volume of noise
- Transaction counter for interacting with customers and taking Scantron sheets
- Storage area to keep one month's worth of Scantron sheets
- Secure office with a meeting space
- Desks for four staff members in an open area
- Incoming/outgoing Scantron test storage area

Space Requirements

- Independently secured department space
- The department should be in an easily accessible location for faculty but not in a high student traffic zone to lessen security concerns

Current Test Grading Load

- 400,000 to 500,000 tests graded per semester
- 75,000 to 85,000 tests graded during finals week

Current Personnel

- Four staff members and one director

Hours of Operation

- Standard Hours
  - Monday, Tuesday, Thursday, Friday: 8 AM to 5 PM
  - Wednesday: 8 AM to 1 PM and 2 PM to 5 PM
- Final Exam Week Hours
  - Monday through Thursday: 8 AM to 8 PM
  - Friday: 8 AM to 10 PM
  - Saturday: 10 AM to 6 PM
  - Sunday: Noon to 2 PM

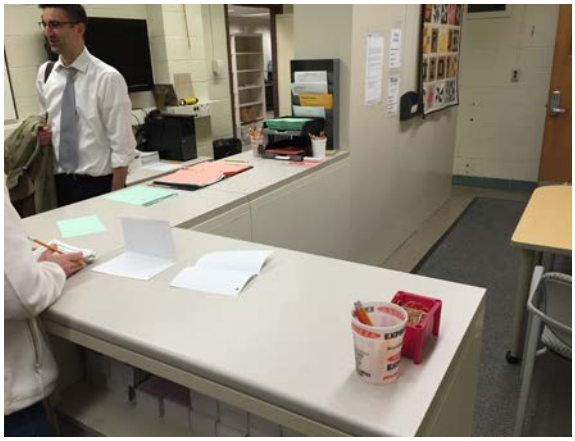
Current Conditions

Instructional Data Processing is located in the Steward Center in Room G-39. This location generally works well for the department as it is somewhat centrally located, yet not in an area of the building with a high amount of student traffic. Spatially, the department's area is somewhat oversized; however, the current space is not functioning well in several ways. Because the grading machine is not acoustically isolated, it makes adjacent spaces difficult to conduct meetings, and because the HVAC is not properly regulating humidity for the area, the grading machine has issues operating at maximum efficiency. Co-locating this department with the DRC would result in increased test-grading efficiency as the DRC is one of IDP's major clients. Professors with students utilizing the testing center in the DRC must currently travel to the DRC first to pick up completed tests to then combine with tests completed in their classroom before dropping them off at IDP to be graded. There may be another opportunity for efficient use of staff between the test proctoring services administered by IDP and the need for test proctors at the proposed centralized testing center.

Potential Areas for Improvement

- Need a six- to seven-person meeting space for proctor training
- Need a better conditioned and humidity controlled space to keep the grading machine running smoothly
- Need an acoustically isolated space to separate noise from the grading machine from the rest of the office
- Would like windows into their office
- A centralized Scantron pick-up/drop-off location near the front reception desk of the renovated Potter Hall may increase efficiency and ease of use.

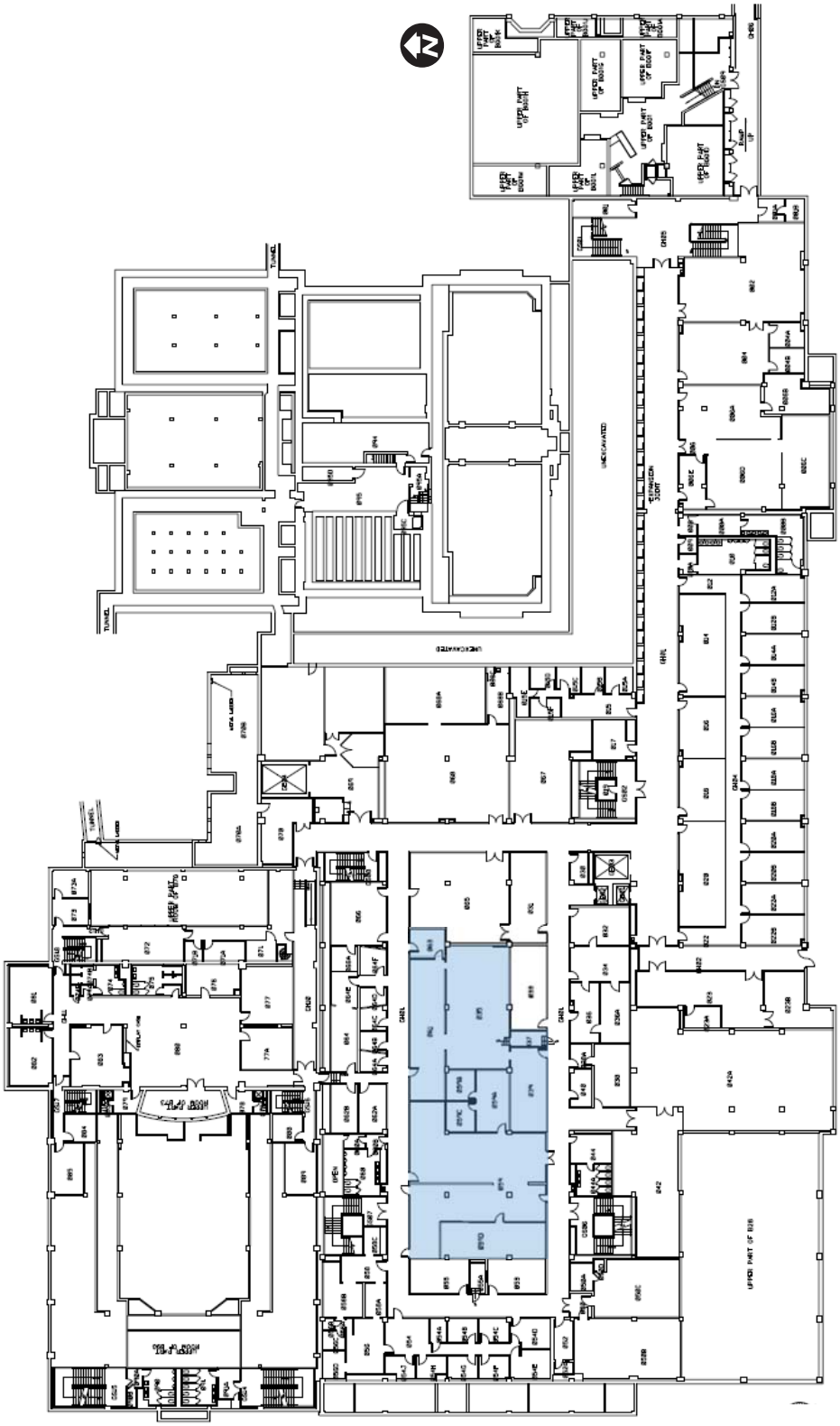
Current Photos







Current Floor Plan STEWART CENTER (STEW) - GROUND FLOOR



Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

DISABILITY RESOURCE CENTER

The Disability Resource Center (DRC) ensures qualified students with disabilities equal access to all University programs, services, and activities. DRC supports students with disabilities by coordinating reasonable and appropriate accommodations, educating the University community about disability issues and rights, and promoting an informed and inclusive campus environment that actively addresses individual and institutional barriers.

CURRENT FACILITIES AND CONDITIONS

Space Requirements

- Easily accessible first-floor location
- Panic button under all staff desks
- All desks, doorways, and furniture should be universally accessible, beyond ADA where applicable.
- Scooters/wheelchairs should be considered.
- Accommodation specialist offices should contain a separate meeting table for four and be acoustically isolated.
- Two to three ADA-accessible parking spaces should be adjacent to the building.

Current Conditions

The Disability Resource Center is currently located in Young Hall on the eighth floor in Room 830. The location on the eighth floor presents a challenge to the primary clients of this department, who are often students in wheelchairs/scooters. At present, Purdue is one of only two schools in the Big Ten that does not locate its DRC facility on the first floor. The DRC department is currently operating in limited space and could use more room. This will become an even larger issue as the department continues to grow over time.

Future Growth

The DRC at Purdue has seen a substantial growth in the number of students that self disclose and request reasonable accommodations. The most frequently utilized accommodation is accommodated testing. DRC data reflects 677 students were registered with the DRC for fall 2010. That number has grown to 1,256 students registered for services for fall 2015. This reflects an 85.5 percent growth over this time period.

A published article in the Journal of Vocational Rehabilitation reports that as a result of current legislation and labor market trends, the number of students with disabilities has risen to 17% students attending postsecondary education programs.[1] The results of the "American Community Survey" estimates the overall rate of disability in the US population in 2014 was 12.6%[2]. From the National Center for Educational Statistics Fast Facts sheet, data reflects for 2011-2012 the percentage distribution of disabled students is 11.1% of the student body.

If we consider Purdue's total undergraduate enrollment to be at 39,000 students, it would appear the DRC's presently reported numbers underrepresent disabled student numbers actually attending the university. As society continues to improve its understanding of disability as another form of diversity, civil rights, and social justice, we can expect students to overcome the societal stigma of disability. This will result in the continued increase of the number of students self-reporting.

Potential Areas for Improvement

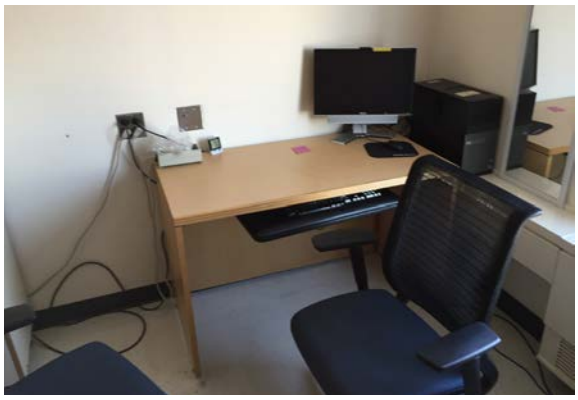
- Relocate department to a first-floor location
- Provide more room in the reception/waiting areas
- Greater access to daylight and a more welcoming environment
- Alternate Format Production area needs additional space

[1] Journal of Vocational Rehabilitation, vol. 16, no. 3,4, pp. 189-198, 200 by Stodden, Whelley, Chang, Harding reports, 42
[2] www.DisabilityCompendium.org

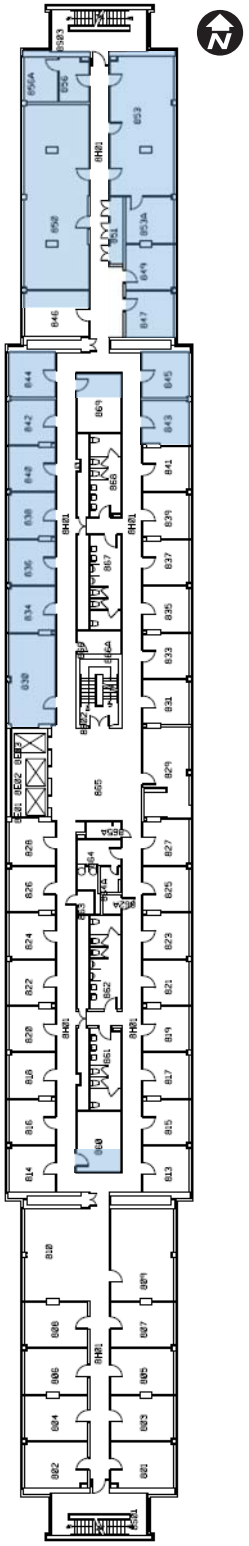
Current Photos








Current Floor Plan ERNEST C. YOUNG HALLL (YONG) - EIGHTH FLOOR



Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Grand Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Grand Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

ASSISTIVE TECHNOLOGY CENTER

The Assistive Technology Center (ATC) promotes an accessible information technology infrastructure at Purdue that is usable by all students, faculty, and staff regardless of disability status. The ATC provides students with disabilities the accommodations to do schoolwork, in much the same way the DRC currently provides students with disabilities the accommodations to take tests.

CURRENT FACILITIES AND CONDITIONS

The ATC is located in Stewart Center Room 111. The space has recently been outfitted with new furniture, and the current size is sufficient for its function. Much like the DRC, it is a space regularly accessed by students with disabilities, so the space should be designed beyond code minimums and instead be designed with universal accessibility in mind.

Hours of Operation

- Monday through Thursday: 7 AM to 11 PM
- Friday: 7 AM to 11 PM
- Saturday: 9 AM to 6 PM
- Sunday: Noon to 11 PM

Space Program

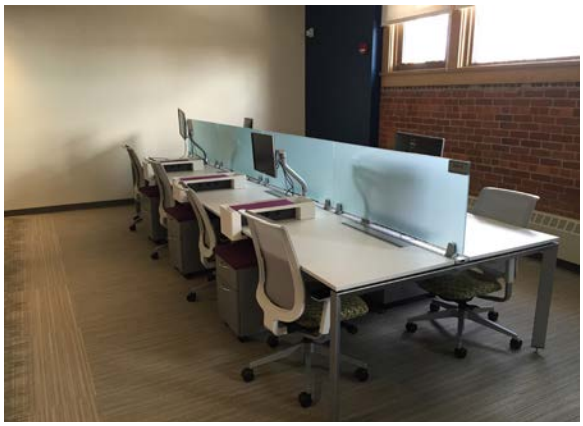
Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Grand Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Grand Total	**1**	**850**	**850**
		TOTAL	**970**

COLLEGE OF ENGINEERING

In 2014, the College of Engineering at Purdue conducted a campus-wide facilities master plan that outlined the future plans for growth of the College in accordance with the College's Strategic Growth Initiatives. The plan identified the Potter Center as a home for the College's Student Resource Programs. The plan did not, however, include the new CTC, DRC, ATC, or IDP. The goal of this study will be to find synergies and efficiencies among these programs in order to co-locate these departments within the Potter Center while simultaneously respecting the goals of the College of Engineering.

In addition to housing the new departments, it is the goal of the College to create an efficient space suitable for 21st century education. Using Grissom Hall as a benchmark, the College would like to re-image the building, bringing closed office areas in toward the center and creating open office areas around the perimeter. This strategy will allow the maximum amount of daylight penetration into the building, and efficiency strategies such as "hotdesking" will ensure maximum utilization of the space.





RELOCATED COLLEGE OF ENGINEERING DEPARTMENTS

Environmental and Ecological Engineering (EEE)

The EEE program pursues a new and unique approach to environmental engineering. In addition to considering conventional waste emission control and treatment, students learn to use systems engineering techniques to proactively avoid environmental problems. This approach adopts an integrated perspective to ensure engineering systems are created that can exist in harmony with the natural world and serve society.

EEE builds upon the strong community of faculty and students engaged in environmentally related research at Purdue. EEE is an interdisciplinary program that is structured to be inclusive of all efforts across the College related to environmental and ecological engineering.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Academic Dean / Department Head	1	120	120
Admin/Operations Staff, Workstation	2	41	82
Clerical-Workstation	1	40	40
Grad-Thesis Desks (64 Students)	45	26	1170
Hoteling Offices	4	120	480
Management	2	120	240
Professional Staff - Office	3	120	360
Research - Visiting Faculty	4	120	480
Tenured/TT Faculty	10	120	1200
Visiting Scholars	4	64	256
Grand Total	**76**	**891**	**4,428**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Open Lab - Project Engineering	2	300	600
Reception	1	180	180
Student Organizations	1	120	120
Teaching Lab - Wet	1	1200	1,200
Teaching Lab - Wet/Dry	1	1200	1,200
Teaching Support - 25%	1	300	300
Mediascape Collaboration Space	1	150	150
Grand Total	**9**	**3570**	**3,870**
		TOTAL	**8,298**

Engineering Education (ENE)

The School of Engineering Education envisions a more inclusive socially connected and scholarly engineering education. This implies radically rethinking the boundaries of engineering and the purpose of engineering education. Their mission is to transform engineering education based on scholarship and research rests on three pillars: Reimagining engineering and engineering education, creating field-shaping knowledge, and empowering agents of change.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Adjuncts/Joint Appointments	2	120	240
Communications Director	1	120	120
Director of Project Assessment	1	120	120
Director, Development	1	120	120
Emeritus	2	120	240
Grad-Thesis Desks (160 Students)	112	26	2,912
Instructor/Lecturer	2	120	240
Managing Director	1	120	120
Post Doctoral Assistants - Workstation	9	40	360
Teaching Assistant	20	40	800
Tenured/TT Faculty	28	120	3,360
Visiting Scholars	4	64	256
Grand Total	**183**	**1,130**	**8,888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	14	100	1,400
Student Organizations	1	120	120
Student Societies	3	120	360
MS Distance Learning Classroom (Opt.)	1	1,440	1,440
Workroom	1	120	120
Grand Total	**20**	**1,900**	**3,440**
		TOTAL	**12,328**

Engineering Advising (ENE Adv.)

Housed within the Engineering Education program, First-Year Engineering has a staff of professional advisors, available to help students in a variety of academic success, registration, personal, and professional issues. The advisors are here to be a student's definitive source of information and advice; students can consult with their advisors any time they have a question or problem, no matter how small.

Global Engineering Program (GEP)

Purdue's Office of Global Engineering Programs firmly believes that immersive, transformative international experiences such as studying, working, or researching abroad lead to more effective engineers and graduates who can work effectively across cultural, ethnic, and national boundaries and help businesses succeed in an increasingly interconnected world.

The GEP program can play a key role in translating innovations to scalable solutions for eradicating extreme poverty, hunger, improving global health, building resilient societies, and empowering vulnerable populations. The program focuses on educating globally competent engineers and creating global impact.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Assistant Director	1	120	120
Associate Director	2	120	240
Director	1	120	120
Grad-Thesis Desks (160 Students)	1	26	26
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Program Manager	1	120	120
Undergrad Assistant	2	40	80
Grad Student Admin. Workstation (Hotdesk)	3	64	192
Grand Total	**14**	**891**	**1,179**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Storage	1	120	120
Student Ambassador Workroom	1	360	360
Workroom	1	120	120
Grand Total	**4**	**720**	**720**
		TOTAL	**1,899**

Office of Professional Practice (OPP)

The OPP facilitates the experiential education and professional practice of Purdue University students within the academic environment of the institution and its global partners, participates in academic research within the field of professional practice, and assists academic units with enhanced employer engagement. Committed to fostering a holistic student experience, OPP offers formal programs in 5-Session Co-Op, 3-Session Co-Op, internship, and research formats. OPP also facilitates the Global Engineering Alliance for Research and Education (GEARE), Purdue's flagship global engineering program. Purdue designs professional practice programs to combine practical on-the-job experiences with the classroom training of a four-year college curriculum

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	3	120	360
Clerical-Workstation	3	40	120
Director	1	120	120
Grand Total	**9**	**321**	**682**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**922**

College of Engineering Honors (HONORS)

Honors at Purdue is designed to provide the highly motivated and academically successful student with a broader, more enriched educational experience during the first year of college through a variety of academic challenges. The program focuses on students looking for a higher degree of intensity in their undergraduate studies.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Advisor	3	120	360
Assistant Director	1	120	120
Associate Director	1	120	120
Director	1	120	120
Grand Total	**10**	**521**	**884**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	360	360
Grand Total	**1**	**360**	**360**
		TOTAL	**1,244**

Minority Engineering Program (MEP)

Established in 1974, MEP has been nationally recognized and replicated to advance engineering learning, discovery, and engagement in fulfillment of the Land Grant promise through outreach, recruitment, and retention of historically underrepresented students in their pursuit to become extraordinary Purdue engineers. Although our programs are open to all students without regard to race, ethnicity, or gender, underrepresented students at Purdue University include African Americans, Latino/Hispanic Americans, Native Americans, and Hawaiians/Pacific Islanders.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Assistant Director	1	120	120
Associate Director	1	120	120
Data Analyst	1	120	120
Director	1	120	120
Grad Student Admin. Workstation (Hotdesk)	2	64	128
Grand Total	**6**	**544**	**608**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Tutorial Center	1	900	900
Workroom	1	120	120
Grand Total	**3**	**1,140**	**1,140**
		TOTAL	**1,748**

Office of Future Engineers (OFE)

OFE aspires to inspire middle school through high school students to consider engineering as a career and is dedicated to recruiting a diverse and academically gifted student body to the First-Year Engineering (FYE) program at Purdue University.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Associate Director	1	120	120
Director	1	120	120
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Grad Assistant Workstation (Hotdesk)	2	64	128
Undergrad Staff Workstations (Hotdesk)	7	40	280
Grand Total	**13**	**584**	**888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**1,128**

Women in Engineering Program (WIEP)

From mentoring to career development, WIEP continually encourages current and future women engineering students by providing interesting and engaging programming relevant to their lives. WIEP programs are a place to learn, discover, and explore aspects of engineering and connect with others who are also interested. They are a place of encouragement, support, and positive perspectives for those who are interested in following their dreams by pursuing an engineering degree.

Space Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Associate Director	1	120	120
Communications	1	120	120
Director	1	120	120
Management	1	120	120
Professional Staff - Office	2	120	240
Program Manager	1	120	120
Research - Visiting Faculty	1	120	120
Grad Assistant Workstation (Hotdesk)	6	64	384
Grad Student Admin. Workstation (Hotdesk)	6	64	384
Grand Total	**21**	**1,009**	**1,769**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Student Organizations	1	120	120
Workroom	1	120	120
Grand Total	**3**	**360**	**360**
		TOTAL	**2,129**

SCENARIO A

SCENARIO A

OPTION A – DRC AND ATC

The intent of this option is to find a new, more accessible, home for the Disability Resource Center immediately, without waiting for the Potter Center to receive a renovation in entirety.

In addition to accommodating the DRC, it has been proposed during this study that the Assistive Technology Center also be relocated so that the two departments can be co-located under one roof. These two departments, especially the alternative format production unit of the DRC, share similar missions and work very closely together. Their co-location would increase efficiency and ease of use. Note that in this scenario, the accommodated testing functions of the DRC would be relocated alongside the DRC as a whole, as these functions would continue to be administered by the DRC.

The space which will be available in the near future is room 141 in the Potter Center. Currently, room 141 houses the Siegesmund Engineering Library. The library will be relocated to the Active Learning Center immediately east of Potter once construction is complete.

GOALS

The goals of Option A are to

- create a more accessible and functional Disability Resource Center
- increase efficiency and effectiveness by co-locating the DRC and the ATC

ARCHITECTURAL DESIGN

The greatest concern with Option A is the amount of space available. Room 141 is 3,279 sft, and an additional 1,104 sft is available in Room 135, for a total of 4,382 sft. The DRC currently occupies 3,692 sft in Young Hall. To house a properly sized DRC and its testing functions will require approximately 5,005 sft (3,850 net x 30% grossing factor). Including the Assistive Technology Center brings the total required area to 6,266 sft. The proposed plan does not include the ATC due to space restrictions. If the currently proposed program of spaces is required for the DRC to successfully perform its function, then it cannot be relocated into only the proposed space.

Of additional concern is the accessibility of the Potter Center. Because the DRC is primarily visited by students with accessibility needs, it is critical its home be not only accessible, but above and beyond code minimum requirements to reach “universal accessibility.” In its current condition, the Potter Center is not an accessible structure, and even if adequate area within Potter is provided, it will not be a suitable home for the DRC unless accessibility renovations occur. Restroom renovations are a priority. The restrooms in the building are not currently accessible, and wheelchair-bound students would likely need to leave the building to use the restroom. The next greatest accessibility concern is the northwest ramp, which is the closest access ramp to the DRC. This ramp would require aesthetic landscape improvements to create a more gracious entry, as its primary function is currently to serve the loading dock.

A secondary concern is the lack of daylight present in Room 141. During meetings with the DRC, the desire was expressed for the DRC to be bright, open, and airy. Six small windows currently exist in the space. Though the proposed layout aimed to provide as much light as possible to the DRC space, there are only two windows providing daylight to the area. While the space can be designed to be open and airy, it will not be adequately daylit without exterior renovations.

As currently proposed, Option A does not meet any of the stated goals.

STRUCTURAL DESIGN

No structural changes to the building are proposed in this option of the renovation.

CIVIL DESIGN

To properly serve the DRC, ADA parking spaces need to be created adjacent to the building. There is currently sufficient space immediately southeast of the building to create three parallel ADA parking spaces by slightly narrowing the sidewalk west of the Michael Golden Engineering Laboratories. The travel path from this space to the Potter Center meets slope requirements and is fully accessible. Note these parking spaces are not ideally located to serve the DRC. A wheelchair-bound student will travel 300' before entering the building and a total of 400' to access the DRC.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

The following revisions and analysis would need to occur to accommodate the proposed plans.

Mechanical Systems, HVAC

- No significant equipment changes are anticipated for this option.
- The proposed DRC project area is currently supplied by air handler ACB-1 in the basement, except for Room 141, which is served by ACP-8.
- Based on the VAV terminal unit replacement drawings, the proposed area is being delivered approximately 0.7 CFM/SF of cooling and ventilation air.
- The nature of the space indicates some additional air may have to be found to completely satisfy the loads in the project area.
- New VAV terminal units with current controllers will be required.
- An equipment changeout would require a ductwork changeout to meet energy code.
- The current cooling for existing BDF Room B045 will need to be evaluated to ensure the existing FCU will meet the new load requirements.

Mechanical Systems, Plumbing

- Restrooms are required to be remodeled.
- The current domestic water and sanitary systems should have the capacity to add the new loads without adverse effects.
- The only negative effect of remodeled restrooms is the infringement on the rooms below while new waste piping is being installed.

Fire Protection System

- The existing fire protection system mains will remain in place and heads will be relocated as required for the new floor plan layout.

Electrical Systems

- The lighting and receptacles in this area would be served by L-1-NW and L-1-NE.
- The panels serving this area may require replacement with higher-rated panels in order to serve the increased loads in the testing center. This would also require replacing the bus plugs or fused switches protecting the panels.
- Existing receptacles and lights will be removed. Associated conduit and conductors will be removed back to panel L-1-NW or L-1-NE.
- Remove any circuits for other systems that will be demolished in the renovated areas.
- Install new power receptacles as required in new DRC layout.
- Install new lighting as required in new DRC layout.
- Add circuits to serve any new fan-powered VAV terminal units.
- Run new circuits as required to support other systems.

Telecom System

- Add PICS and wireless APs as required in new DRC layout.
- New PICS to be wired back to the BDF room B045.
- All CAT 5e cabling for this renovation would need to be removed in the renovated area and

new CAT 6A cabling installed for new PICS and wireless APs.

- Install new 4” skeletal raceway or basket cable tray in renovated area as required.
- Install a new pathway between BDF B045 and the B060/B060A area in order to shorten the distance of cable pulls to the current B051-B059 areas.
- Add a new equipment rack and CAT 6A patch panels in BDF Room B045.
- New equipment rack will need to have 6" vertical wire management on the sides. They will not require horizontal cable management.
- Add a redundant 24-strand, single-mode fiber fed from a second campus telecom node in Brown Hall. Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.

Fire Alarm System

- Modify/add audio visual device locations as required for new DRC floor layout.
- Add any new initiating devices as required by new floor layout.

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Disability Resource Center (Accommodated Testing Function

Position	Number of Personnel	Area Per Person	Total Area
Testing Room, 30 Stations	1	700	700
Private Testing Rooms	10	48	480
Total	**11**	**748**	**1,180**
		TOTAL	**1,180**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Net Area Total, All Departments	4,852
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**6,308**

DISPLACED EXISTING AREA

Department	Area Per Department
Libraries	4,164
Grand Total	**4,164**

DEPARTMENT AREA AS DESIGNED

Department	Area Per Department
Disability Resource Center	4,374
General Building	342
Grand Total	**4,716**



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

SCENARIO A - FIRST FLOOR PLAN



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

SCENARIO A - FIRST FLOOR USE

OPTION B2

OPTION B2 – DRC, ATC, IDP, AND ONLINE COMPLIANCE

Realizing the failure to properly accommodate the Disability Resource Center in Scenario A, this option aims to create a new home for the DRC and related departments without the Centralized Testing Center.

The area proposed for this option is identical to the area proposed for Option B; however, only the first floor is utilized, resulting in a mostly empty basement in the Potter Center. With only minor re-work, this option could represent a first phase of option B, with the second phase consisting of the construction of the Central Testing Center. Some area constructed for the accommodated testing areas would require renovation to become the high-stakes testing center shown in Option B.

GOALS

The goals of Option B2 are to:

- create a more accessible and functional DRC
- increase efficiency and effectiveness by co-locating the DRC, ATC, IDP, and Online Compliance

ARCHITECTURAL DESIGN

The larger quantity of space available in this option allows the departments to be more effectively located. The DRC space is located in the easily accessible southwest portion of the first floor. The primary entrance off the main reception space creates the open and inviting atmosphere requested by DRC leadership. Daylight is captured to the greatest extent possible through windows to the south and east and is transmitted throughout the space through borrowed lights on the interior.

The accommodated testing function of the DRC is housed at the core of the building, across the hall to the north from the DRC itself. The separation from the larger DRC department would likely create some inefficiencies in staffing between the DRC and the DRC's accommodated testing function. However, this location at the core of the building, away from windows, is ideal for a testing center and would provide a suitable home for the high-stakes testing area if Option B2 were to be used as a first phase of Option B.

The ATC is located north of the DRC. It is fully integrated with the DRC's alternate format production while preserving an independent entrance. Daylight is provided to this area through windows to the west

IDP is located in the northwest portion of the first floor. Here it is permitted easy access to related departments and faculty while generally remaining out of the path of student travel as requested by IDP leadership. Light is provided to the IPD through windows along the west wall. Test-grading equipment would be located in the data processing area, acoustically separated from the remainder of the department with a high STC-rated wall.

The online compliance department is located east of the high-stakes testing area. On a whole, its location is effective because it positions the department near other departments with which it interfaces. However, it is not ideally located within the building as there are no windows serving this area. Due to footprint of the building and the space available in this option, at least one department is likely to be denied daylight.

Two conference rooms and a work café on the first floor will serve all departments. The large conference room will hold up to 100 in classroom-style seating or 50 in a hollow rectangle configuration. This space will be used most often by the College of Engineering. It is the only COE need met with this option.

A large reception/lounge space is located at the south side of the first floor. This space can flexibly serve as prefunction space for events in the large conference room and a general student study/gathering space. A reception desk is shown where a receptionist could direct students and staff to departments throughout the building. For the success of this option, this desk is optional. It will be more critical in subsequent options to allow the reception areas of multiple departments to be combined into one central location.

Though somewhat discretionary for the success of this layout, the plan includes a new main entrance to the south, abandoning the existing entrance to the southeast. The new entrance reduces travel distance for wheelchair-bound students entering the building from the south by 50 percent. Ideally, the new entrance would be executed along with the south plaza revision as shown in Option D. This combination would fully eliminate the travel distance discrepancy between wheelchair-bound and able-bodied students.

INTERIOR DESIGN

In corridors that have terrazzo, it is recommended to maintain, strip, and clean those corridors but to remove the terrazzo base. Because the existing terrazzo is difficult to match and expensive to replace, it prevents flexibility for future renovations. An accent terrazzo color can be used to separate existing and new terrazzo in corridors where they will need to be blended. All existing wallcoverings in corridors would be removed and receive epoxy paint. For high-traffic areas such as corridors, epoxy paint is long-lasting and extremely durable.

The entrance/commons will be a combination of terrazzo and carpet tile/vinyl, epoxy painted walls, and accents of wallcovering or digital wallcovering. All offices, workstations, grads, classrooms/computer labs, and conference rooms will have their finishes removed and replaced with carpet tile and wall paint. Carpet tile, rather than broadloom, is easy to replace if a tile is damaged. Administration areas will receive carpet tile, vinyl base, and wallcovering. Break rooms, lounges, and the coffee bar will have a luxury vinyl tile (LVT) that is durable and easy to maintain. LVT comes in multiple colors and patterns that can give anywhere from a warm to industrial appearance.

Labs and research center testing areas will keep their existing sealed concrete floors and exposed ceilings; walls will be repainted with epoxy paint. Restrooms will be renovated to meet ADA requirements, keeping existing floors and wall tile as much as possible and, where necessary, replacing floors and wall tile to match existing ones. Where applicable, new solid surface countertops with integral sinks will be installed. If in good shape, the 2x2 or 2x4 ceiling tile will remain; all other ceiling tile will be removed and replaced with new 2x2 tile.

STRUCTURAL DESIGN

Removing the steel-framed mezzanine is optional. However, if it is the University's intent to use the full basement area in the future, the mezzanine should be removed. Removing the mezzanine later would require demolition of a significant portion of the new first floor construction proposed in this option. To infill the mezzanine, conventional steel framing, including beams and columns, will be added to allow for slab-on-metal-deck construction at the first floor level. The addition of this steel framing will require modifications to the existing foundation system to adequately support the weight of the new system and occupancy loads of the building. The framing system must stand alone and cannot bear on the existing structure because it is not adequately sized to carry the additional load.

CIVIL DESIGN

To properly serve the DRC, ADA parking spaces need to be created adjacent to the building. There is currently sufficient space immediately southeast of the building to create three parallel ADA parking spaces by slightly narrowing the sidewalk west of the Michael Golden Engineering Laboratories. The travel path from these spaces to Potter meets ADA slope requirements and is fully accessible. In this option, the location of these spaces is much more desirable than in Option A due to the DRC shifting to the southwest portion of the building.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

The following revisions and analysis would need to occur to accommodate the proposed plans.

Mechanical Systems, HVAC

- The proposed DRC area is served by ACB-1. On average, ACB-1 is designed to deliver 1.1 CFM/SF in the proposed project area.
- ACB-1 also serves all of the rooms on the east side of the east north/south corridor.
- A renovation of this magnitude will likely require a State Plan Review, which requires the system to meet the energy code. The current system does not meet the energy code.
- The air from ACP-8 that currently serves a portion of the proposed DRC area can be redirected to the second and third floors where it is likely needed.

Mechanical Systems, Plumbing

- The scope of work for this option shows remodeled restrooms.
- The current domestic water and sanitary systems should have the capacity to add the new loads without adverse effects.
- The only negative effect of remodeled restrooms is the infringement on the rooms below while new waste piping is being installed.

Fire Protection System

- The existing fire protection system mains will remain in place and heads will be relocated as required.

Electrical Systems

- The lighting and receptacles in this area would be served by L-1-NW, L-1-NE, L-1-SW, and L-1-SE.
- Due to the age of the panel boards and number of circuits that will be demolished and re-used, the panel boards should be replaced.
- Some of the panels may require replacement with higher rated panels in order to serve the increased loads in the testing center. This would also require replacing the bus plugs or fused switches protecting the panels.
- Existing receptacles and lights will be removed. Associated conduit and conductors will be removed back to source panel.
- Remove any circuits for other systems that will be demolished in the renovated areas.
- Install new power receptacles as required in new DRC layout.
- Install new lighting as required in new DRC layout.
- Run new circuit as required to serve replaced air handler ACB-1.
- Run new circuits as required to support other systems.

Telecom System

- Add PICS as required in new DRC layout.
- Remove all CAT 5e cabling from the renovated area for the DRC.
- Install CAT 6A cabling to all new PICS and wireless APs.
- Add a new equipment rack and CAT 6A patch panels in BDF room B045. New rack will need to have 6” vertical wire management on the sides. They will not require horizontal cable management.
- Install new 4” skeletal raceway and/or basket cable tray in renovated areas as required.
- Install a new pathway between BDF B045 and the B060/B060A area in order to shorten the distance of cable pulls to the current B051-B059 areas.
- Add a redundant 24 strand single mode fiber fed from a second campus telecom node in Brown Hall. Purdue ITAP would install cable into the building but the cost would be absorbed by this project.

Fire Alarm System

- Modify/add audio visual device locations as required for new layout.
- Add any new initiating devices as required by new floor layout.

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Instructional Data Processing

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

Disability Resource Center (Accommodated Testing Function

Position	Number of Personnel	Area Per Person	Total Area
Testing Room, 30 Stations	1	700	700
Private Testing Rooms	10	48	480
Total	**11**	**748**	**1,180**
		TOTAL	**1,180**

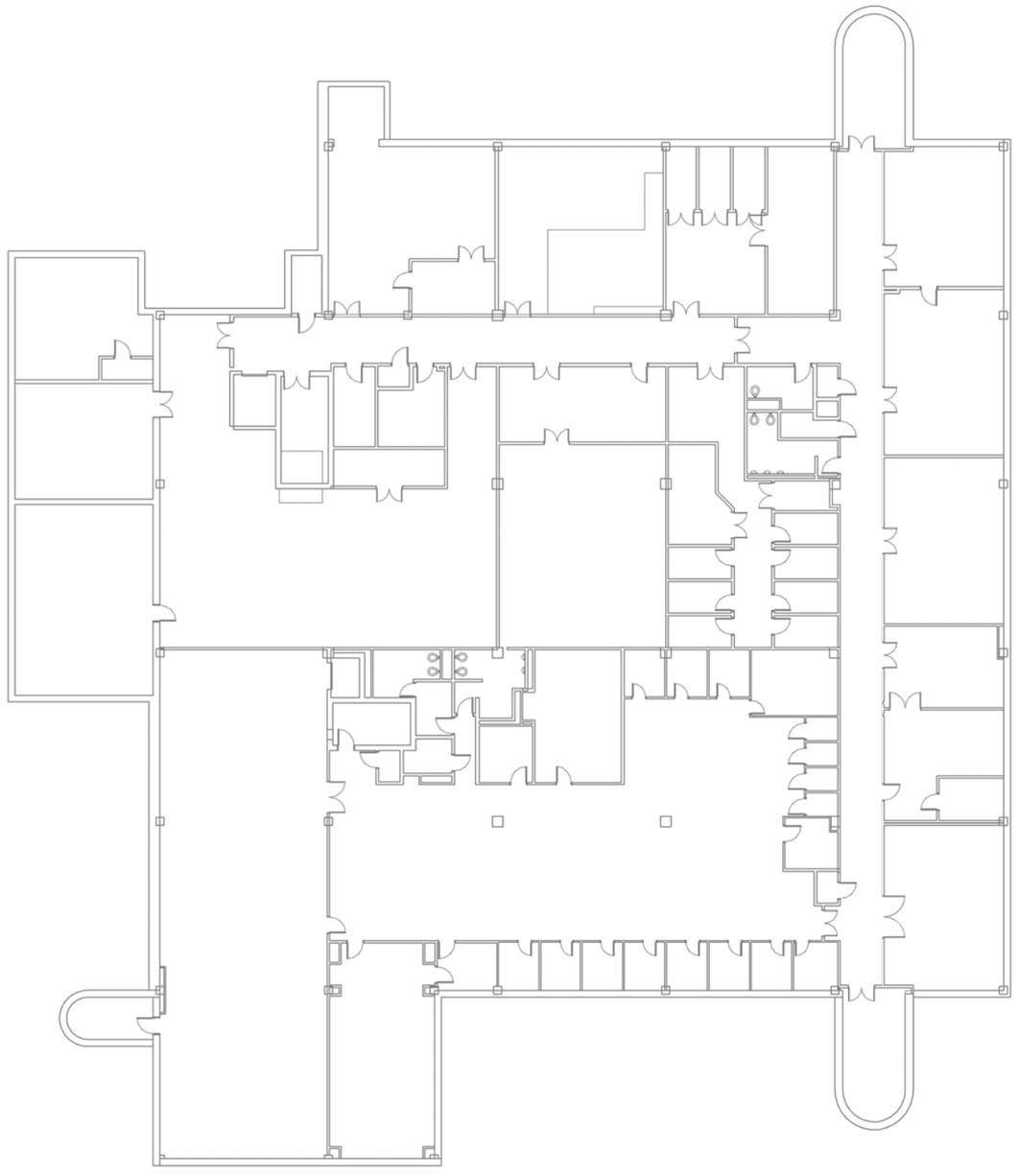
Net Area Total, All Departments	5,836
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**7,587**

DISPLACED EXISTING AREA

Department	Area Per Department
Libraries	4,164
General Building	403
Libraries	10,484
Grand Total	**15,051**

DEPARTMENT AREA AS DESIGNED

Department	Area Per Department
Disability Resource Center	6,424
Assisted Technology Center	856
Instructiona Data Processing	1,365
Online Compliance	908
General Building	5,763
Grand Total	**15,316**



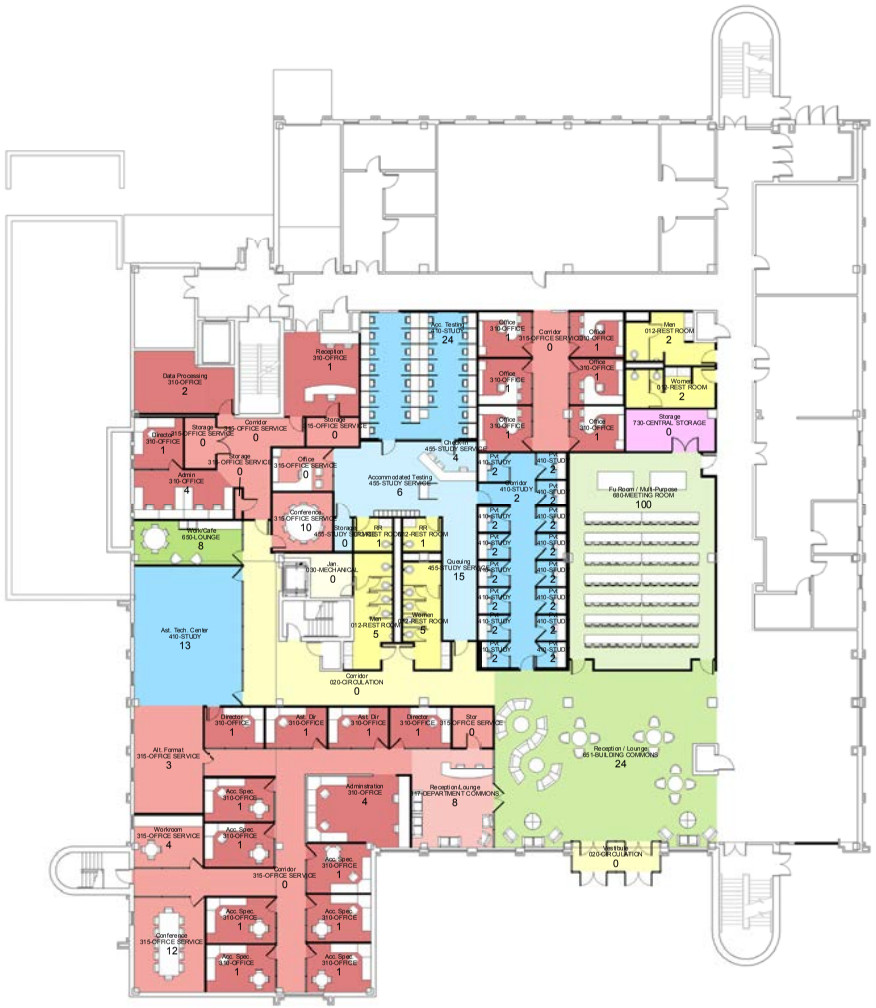
- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B2 - LOWER LEVEL



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B2 - FIRST FLOOR


Reception
310-OFFICE
1
Data Processing
310-OFFICE
2
Director
310-OFFICE
1
Storage
315-OFFICE SERVICE
0
Corridor
315-OFFICE SERVICE
0
Storage
315-OFFICE SERVICE
0
Storage
315-OFFICE SERVICE
0
Admin
310-OFFICE
4
Office
315-OFFICE SERVICE
0
Conference
315-OFFICE SERVICE
10
Acc. Testing
455-STUDY SERVICE
4
Accommodated Testing
455-STUDY SERVICE
6
Quieting
455-STUDY SERVICE
15
Writing Cafe
650-LOUNGE
8
Art. Tech. Center
410-STUDY
13
Men
012-REST ROOM
5
Women
012-REST ROOM
5
RR
012-REST ROOM
1
RR
012-REST ROOM
1
Corridor
020-CIRCULATION
0
Director
310-OFFICE
1
Asst. Dir
310-OFFICE
1
Asst. Dir
310-OFFICE
1
Director
310-OFFICE
1
Stor
315-OFFICE SERVICE
0
Art. Format
315-OFFICE SERVICE
3
Acc. Spec.
310-OFFICE
1
Workroom
315-OFFICE SERVICE
4
Acc. Spec.
310-OFFICE
1
Acc. Spec.
310-OFFICE
1
Corridor
315-OFFICE SERVICE
0
Acc. Spec.
310-OFFICE
1
Acc. Spec.
310-OFFICE
1
Conference
315-OFFICE SERVICE
12
Acc. Spec.
310-OFFICE
1
Acc. Spec.
310-OFFICE
1
Administration
310-OFFICE
4
Reception Lounge
410-DEPARTMENT COMMONS
8
Reception / Lounge
651-BUILDING COMMONS
24
Vestibule
020-CIRCULATION
0
Office
310-OFFICE
1
Corridor
315-OFFICE SERVICE
0
Office
310-OFFICE
1
Men
012-REST ROOM
2
Office
310-OFFICE
1
Office
310-OFFICE
1
Women
012-REST ROOM
2
Office
310-OFFICE
1
Office
310-OFFICE
1
Storage
730-CENTRAL STORAGE
0
Chase
455-STUDY SERVICE
4
Corridor
410-STUDY
2
Fu Room / Multi-Purpose
680-MEETING ROOM
100
410-STUDY
2
410-STUDY
2
410-STUDY
2

OPTION B

OPTION B – DRC, ATC, IDP, ONLINE COMPLIANCE, AND CTC

This option builds on the goals stated in Option A, and further proposes to create a new campus resource in the Centralized Testing Center. The space proposed for this option will become fully available in 2017, when the library spaces move into the Wilmeth Active Learning Center currently under construction.

In addition to accommodating the DRC, it has been proposed during this study that the Assistive Technology Center also be relocated so that the two departments can be co-located under one roof. These two departments, especially the alternative format production unit of the DRC, share similar missions and work very closely together. Their co-location would increase efficiency and ease of use. Note that in this scenario, the accommodated testing functions of the DRC would be relocated alongside the DRC as a whole, as these functions would continue to be administered by the DRC.

The space which will be available in the near future is Room 141 in the Potter Center. Currently, Room 141 houses the Siegesmund Engineering Library. The library will be relocated to the Active Learning Center immediately east of Potter once construction is complete.

GOALS

The goals of Option B are to:

- create a more accessible and functional Disability Resource Center
- increase efficiency and effectiveness by co-locating the DRC, ATC, IDP, and Online Compliance
- increase efficiency, cost control, academic integrity, and ease of use by creating a centralized location housing all campus testing needs, and that combines all the campus testing functions in one location
- create a new campus-wide resource that is appropriately sized and staffed to deliver better testing conditions at Purdue with increased academic integrity
- reduce the demand for academic and non-academic space during periods of high testing volume

ARCHITECTURAL DESIGN

The larger quantity of space available in this option allows the departments to be more effectively located. The DRC space is located in the easily accessible southwest portion of the first floor. The primary entrance off the main reception space creates the open and inviting atmosphere requested by DRC leadership. Daylight is captured to the greatest extent possible through windows to the south and east, and is transmitted throughout the space through borrowed lights on the interior. The southwest stair, which connects only the first floor and lower level, becomes a back entry for the DRC into the CTC. This connection allows DRC staff immediate access to the CTC private testing rooms, which primarily serve students who are receiving accommodated testing through the DRC. Though the CTC would administer the testing of students in this area, DRC staff could be immediately available via this stairway. DRC expansion could be accommodated by repurposing the workroom or half of the conference room located inside the DRC area.

The Assistive Technology Center is located north of the DRC. It is fully integrated with the DRC's alternate format production while preserving an independent entrance. Daylight is provided to this area through windows to the west.

Instructional Data Processing is located in the northwest portion of the first floor. Here it is permitted easy access to related departments and faculty, while generally remaining out of the path of student

travel (as requested by IDP leadership). Light is provided to the IDP through the windows along the west wall. Test grading equipment would be located in the data processing area, acoustically separated from the remainder of the department with a high STC-rated wall.

The online compliance department is located east of the high-stakes testing area. On a whole, its location is effective because it positions the department near other departments with which it interfaces. However, it is not ideally located within the building as there are no windows serving this area. Due to footprint of the building and the space available in this option, at least one department is likely to be denied daylight.

The High-Stakes Testing component of the Central Testing Center is located near the center of the first floor. The lack of adjacency with the General Testing Center is the major adjacency fault of this option. However, given the current program and space available, both departments cannot fit in the lower level. In itself, the high-stakes area seems to function well. Students enter through the door to the south and queue in a wide hallway leading to the test center administrator. Test center administration workstations and secure storage also open off this hallway. Thirty lockers, soft seating, and two unisex restrooms are located in the break area under supervision of the test proctors. A private testing room and the large testing room also open off this area.

Two conference rooms and a work café on the first floor will serve all departments. The large conference room will hold up to 100 in classroom-style seating or 50 in a hollow rectangle configuration. This space will be used most often by the College of Engineering. It is the only COE need met with this option.

A large reception/lounge space is located at the south side of the first floor. This space can flexibly serve as pre-function space for events in the large conference room, general student study/gathering space, and as a waiting area for students waiting for friends to complete testing in either testing center. A reception desk is shown where a receptionist could direct students and staff to departments throughout the building. For the success of this option, this desk is optional. It will be more critical in subsequent options allowing the reception areas of multiple departments to be combined into one central location.

Though somewhat discretionary for the success of this layout, the plan includes a new main entrance to the south, abandoning the existing entrance to the southeast. The new entrance reduces travel distance for wheelchair-bound students entering the building from the south by 50 percent. Ideally, the new entrance would be executed along with the south plaza revision as shown in Option D. This combination would fully eliminate the travel distance discrepancy between wheelchair-bound and able-bodied students.

The full space available in the lower level contains the General Testing Center. Due to its lack of windows and separation from major traffic patterns, the lower level is well suited for this function. The entrance to the center is via the 2-story elevator or stair just west of the restrooms. All other doors into the center are secured, exit-only doors. Students enter immediately into the locker area that could house up to 279 lockers. After storing their items, students would then wait in line, check in at the desk, and be directed toward their seat in either the general or private testing areas. Currently shown are 205 testing seats and 20 private testing rooms.

The small elevator serving the testing center is not currently ADA compliant due to its size and door opening location. There are several possible options.

- Do nothing, with the awareness that the elevator is not ADA compliant. Wheelchair-bound students would need to access the testing center through the northwest freight elevator and be permitted entrance to the center via an exit door.
- Create a custom cab to fit within the existing hoistway and apply for a variance with the state to permit a 51x46" cab size.
- Install an ADA-compliant vertical wheelchair lift in the existing shaft and apply for a variance to allow for a travel distance greater than 14'.
- Remove the east wall of the elevator shaft and construct a new shaft wall to accommodate a compliant elevator. Note that the shaft wall is cast-in-place, load-bearing concrete; therefore, demolition and construction would be complex and expensive.

Note that this issue is alleviated in Options C and D because a new elevator is proposed and the main entrance to the testing center is relocated to the east.

INTERIOR DESIGN

In corridors that have terrazzo, it is recommended to maintain, strip, and clean those corridors, but to remove the terrazzo base. Because the existing terrazzo is difficult to match and expensive to replace, it prevents flexibility for future renovations. An accent terrazzo color can be used to separate existing and new terrazzo in corridors where they will need to be blended. All existing wallcoverings in corridors would be removed and receive epoxy paint. For high-traffic areas such as corridors, epoxy paint is long lasting and extremely durable.

The entrance/commons will be a combination of terrazzo and carpet tile/vinyl, epoxy painted walls, and accents of wallcovering or digital wallcovering. All offices, workstations, grads, classrooms/computer labs, and conference rooms will have their finishes removed and replaced with carpet tile and wall paint. Carpet tile, rather than broadloom, is easy to replace if a tile is damaged. Administration areas will receive carpet tile, vinyl base, and wallcovering. Break rooms, lounges, and the coffee bar will have a luxury vinyl tile (LVT) that is durable and easy to maintain. LVT comes in multiple colors and patterns that can give anything from a warm to industrial appearance.

Labs and research center testing areas will keep their existing sealed concrete floors and exposed ceilings; walls will be repainted with epoxy paint. Restrooms will be renovated to meet ADA requirements, keeping existing floors and wall tile as much as possible and, where necessary, replacing floors and wall tile to match existing ones. Where applicable, new solid-surface countertops with integral sinks will be installed. In labs, wood casework can be relocated and refinished with new epoxy countertops. Plastic laminate casework should be replaced, as it does not withstand relocation with new plastic laminate countertops. If in good shape, the 2x2 or 2x4 ceiling tile will remain; all other ceiling tile will be removed and replaced with new 2x2 tile.

STRUCTURAL DESIGN

In order for the plans to work as shown, the steel-framed mezzanine must be removed. This will allow the area in the basement level to be utilized to accommodate the testing center. To infill the mezzanine, conventional steel framing (including beams and columns) will be added to allow for slab-on-metal-deck construction at the first-floor level. The addition of this steel framing will require modifications to the existing foundation system to adequately support the weight of the new system and occupancy loads of the building. The framing system must stand alone and cannot bear on the existing structure because it is not adequately sized to carry the additional load.

CIVIL DESIGN

To properly serve the DRC, ADA parking spaces need to be created adjacent to the building. There is currently sufficient space immediately southeast of the building to create three parallel ADA parking spaces by slightly narrowing the sidewalk west of the Michael Golden Engineering Laboratories. The travel path from this space to the Potter Center meets slope requirements and is fully accessible. The location of these spaces is much more desirable than in Option A due to the DRC shifting to the southwest portion of the building.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

The following revisions and analysis would need to occur to accommodate the proposed plans.

Mechanical Systems, HVAC

- The proposed DRC and CTC areas in the basement are served by ACB-1. On average, ACB-1 is designed to deliver 1.1 CFM/SF in the proposed project area.
- ACB-1 also serves all of the rooms on the east side of the east north-south corridor.
- The proposed DRC and CTC areas on the first floor are served by ACB-1, except as noted above in the comments on the DRC-only project.
- Due to the prosed testing center in the basement, the supply air and outside air requirements of an air handling serving this area will be elevated and highly diversified. The existing air han-

dler and ductwork system is marginal at best and should be replaced.

- In addition, a renovation of this magnitude will probably require a State Plan Review which then requires the system to meet the energy code. The current system does not meet the energy code.
- If the basement rooms on the east side of the corridor can be taken out of service for an extended period of time, a new ACB-1 sized to condition the entire DRC and CTC areas can be installed.
- The air from ACP-8 that currently serves the proposed DRC area can be redirected to the second and third floors where it is likely needed.
- Evaluate current cooling for existing BDF Room B045 and make sure the existing FCU will meet the new load requirements.

Mechanical Systems, Plumbing

- The scope of work for this option shows remodeled restrooms.
- The current domestic water and sanitary systems should have the capacity to add the new loads without adverse effects.
- The only negative effect of remodeled restrooms is the infringement on the rooms below while new waste piping is being installed.

Fire Protection System

- The existing fire protection system mains will remain in place and heads will be relocated as required for the new floor plan layout.

Electrical Systems

- The lighting and receptacles in this area would be served by L-1-NW, L-1-NE, L-1-SW, and L-1-SE.
- Due to the age of the panel boards and the number of circuits that will be demolished and reused, the panel boards should be replaced.
- Existing receptacles and lights will be removed. Associated conduit and conductors will be removed back to source panel.
- Remove any circuits for other systems that will be demolished in the renovated areas.
- Install new power receptacles as required in new DRC and CTC layout.
- Install new lighting as required in new DRC and CTC layout.
- Run new circuit as required to serve replaced air handler ACB-1.
- Run new circuits as required to support other systems.

Telecom System Changes

- Add PICS as required in new DRC and CTC layout.
- Remove all CAT 5e cabling from the renovated area for the DRC and CTC.
- Install CAT 6A cabling to all new PICS and wireless APs.
- Add a new equipment rack and CAT 6A patch panels in BDF room B045.
  - New equipment rack will need to have 6” vertical wire management on the sides. They will not require horizontal cable management.
- Install new 4” skeletal raceway and/or basket cable tray in renovated areas as required.
- Install a new pathway between BDF B045 and the B060/B060A area in order to shorten the distance of cable pulls to the current B051-B059 areas.
- Add a redundant 24 strand single mode fiber fed from a 2nd campus telecom node in Brown Hall. Purdue ITaP would install cable into the building but the cost would be absorbed by this project.

Fire Alarm System

- Modify/add audio visual device locations as required for new DRC floor layout.
- Add any new initiating devices as required by new floor layout.

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Instructional Data Processing

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Central Testing Center

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared Office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

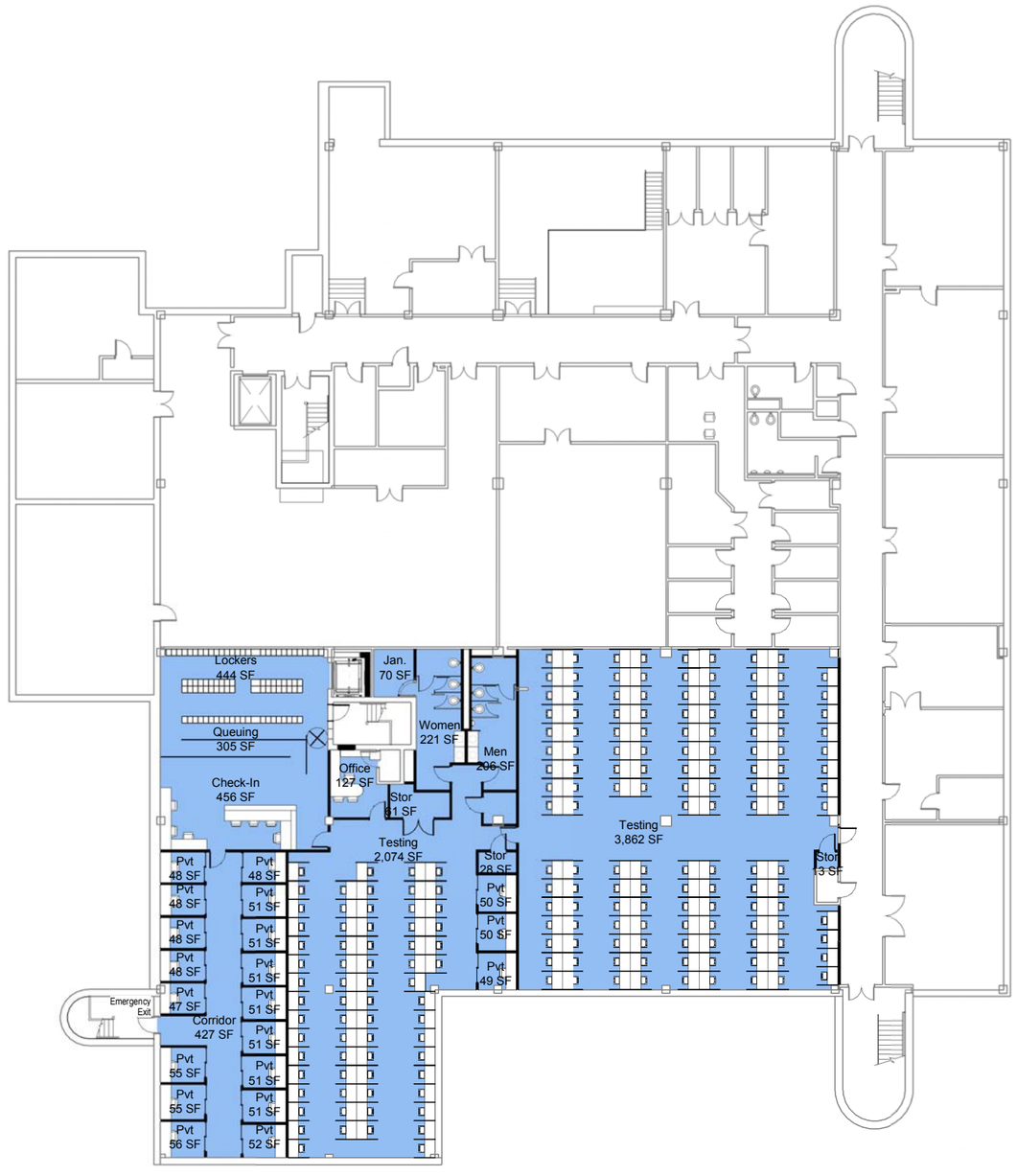
Net Area Total, All Departments	15,733
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**20,453**

DISPLACED EXISTING AREA

Department	Area Per Department
Libraries	4,164
General Building	775
It Cust Rltn	202
Libraries	19,201
Grand Total	**24,342**

DEPARTMENT AREA AS DESIGNED

Department	Area Per Department
Assistive Technology Center	865
Central Testing Center	11,871
Circulation	946
Community Resource	3,786
Disability Resource Center	3,927
General Building	275
Instructional Data Processing	1,444
Online Compliance	974
Restroom	761
Grand Total	**24,849**



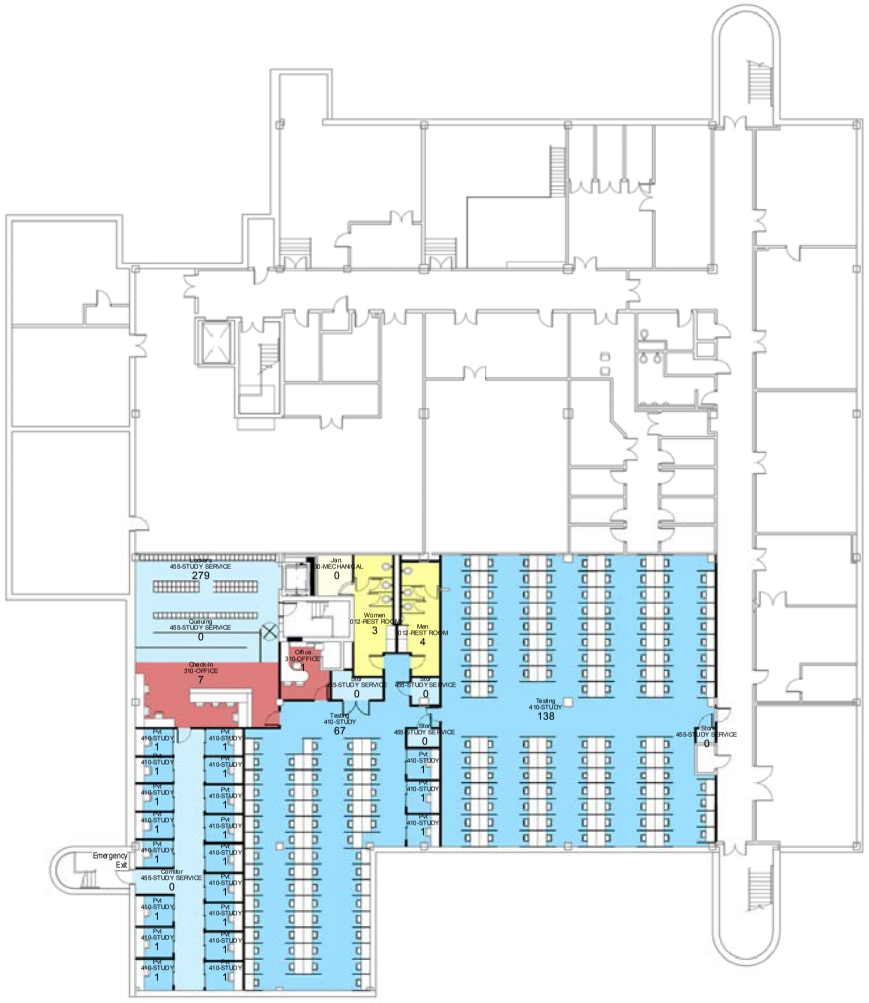
- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B - LOWER LEVEL PLAN



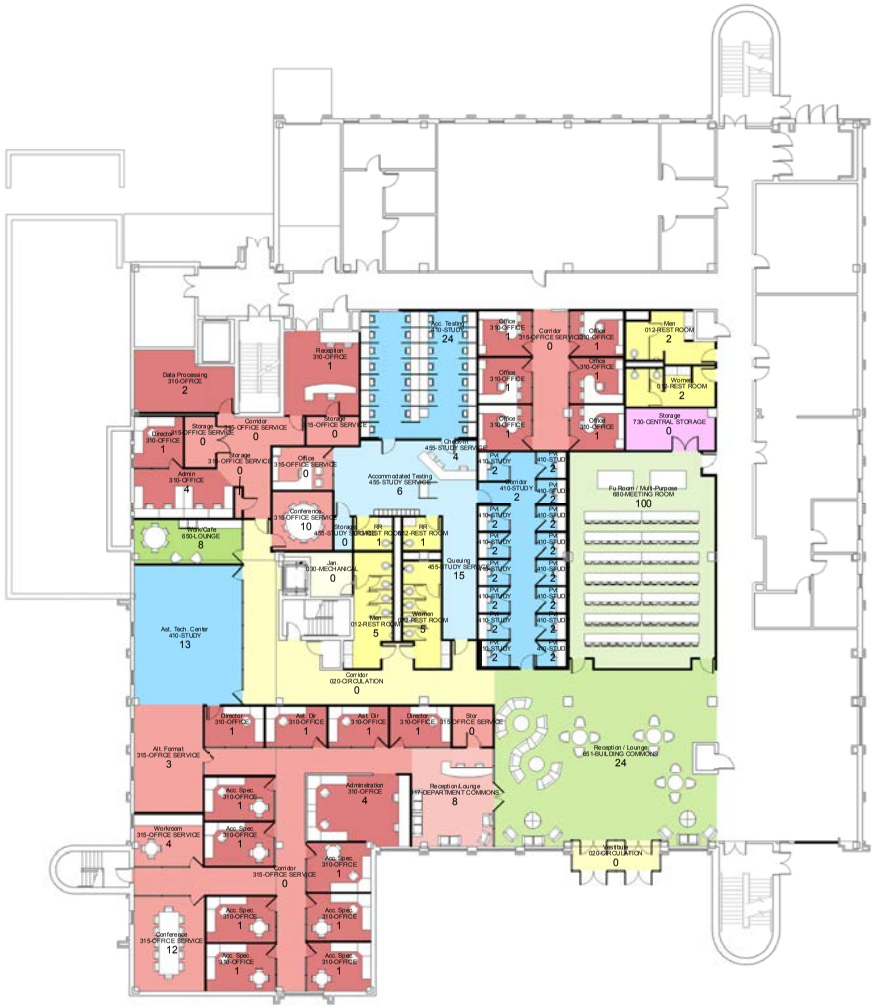
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- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
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- ENGINEERING HONORS
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- GENERAL BUILDING - PLUMBING
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- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION B - FIRST FLOOR PLAN



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION B - LOWER LEVEL USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION B2 - FIRST FLOOR USE

OPTION C

OPTION C

OPTION C – DRC, ATC, CTC, AND COE PROGRAMS

This option builds on the included departments and goals stated in Option B, while beginning to realize the goals of the College of Engineering's overall master plan for future growth.

The COE projects a 44-percent increase in staff associated with their programs, and the Potter Center will be the home for those departments. The intent of this renovation is to create a center devoted to serving associated recruitment and retention programs of First-Year Engineering. The remainder of the building will provide office and instructional space for the Division of Environmental and Ecological Engineering (EEE). However, this option proposes to stop short of a full renovation and preserve costly research lab space.

The intention of this option is to realize cost savings by not displacing costly research labs with less expensive office spaces. The assumption that the existing lab spaces are prime assets should be tempered though, as all research labs in the building have been graded between 3 and 4+ on the 1–5 scale, with 5 being labs in unacceptable condition (see the Lab Assessment in the appendix of this study for further details). There are also some safety concerns regarding the current recirculation of lab air throughout the building.

The retention of the lab spaces results in floor plans not ideal to the departments. If the intention of the University is to renovate these lab spaces in the near future, this should be conducted along with the renovation of the remainder of the building. However, if the intention is to move these lab spaces elsewhere (as proposed in the COE master plan), the resulting building would be best served by another major renovation following the decommissioning of the labs to better accommodate the COE spaces. Alternatively, the entire renovation could be held until the labs are deemed wholly unacceptable, at which time the entire building could receive a renovation.

GOALS

The goals of Option C are to:

- create a more accessible and functional Disability Resource Center
- increase efficiency and effectiveness by co-locating the DRC and the ATC
- increase efficiency, cost control, academic integrity, and ease of use by creating a centralized location housing all campus testing needs, and that combines all the campus testing functions in one location
- create a new campus-wide resource that is appropriately sized and staffed to deliver better testing conditions at Purdue with increased academic integrity
- reduce the demand for academic and non-academic space during periods of high testing volume
- begin to realize the College of Engineering's master plan by creating a center devoted to serving recruitment and retention programs associated with first-year engineering students.
- to the extent possible, maintain existing research lab spaces

ARCHITECTURAL DESIGN

The general layout of this options builds off the philosophy laid out in Option B with regard to the location of the DRC, IDP, ATC, and CTC. However, it differs greatly from Option B because it utilizes all the area in the building except research labs and research lab support spaces. In some ways this permits improvements to the plan, but in other ways it forces undesirable design solutions. This option also includes all of the desired COE departments with the exception of First-Year Engineering and Engineering Education.

All the central testing center functions are contained in one area on the lower level. This will permit a greater level of efficiency by allowing the two testing centers to easily share staff when necessary. It also allows for better wayfinding for students and staff. The general testing center operates similarly to the layout in Option B; however, the main entrance is to the southeast. This permits the small library stair to serve as a back entrance for the office functions of the testing center, allowing staff to enter and exit their offices without interrupting testing. This plan option has 192 general testing seats, 20 private testing rooms, and 265 lockers. High-stakes testing is located immediately to the north of general testing. The layout here is less effective than the layout in Option B. Less queueing space and fewer storage areas may be problematic. The high-stakes testing area contains 25 testing seats and one private testing room. No research lab spaces on the lower level are displaced.

The first floor contains IDP, DRC, and ATC in generally the same areas as Option B, except conference rooms have been moved out of individual departments and into shared community areas. A large conference room opens off the DRC area, which can be assumed to generally be for DRC and ATC student use; however, it could be used as a building-wide resource on certain occasions.

All of the College of Engineering's Program departments share the first floor. This allows these functions the greatest ease of access to students. The Office of Future Engineers are purposefully located as close to the main entrance and incoming students as possible. Other departments are spread throughout the floor. Wherever possible, open office areas are pushed against the exterior wall to permit the greatest amount of daylight into the building, and closed offices are lined along the perimeter in an effort to receive borrowed light from exterior windows. However, because these departments have a relatively small percentage of open office spaces, this strategy is only partially effective. Due to space constraints, a large amount of offices will not receive natural light on this floor.

Due to the large number of departments located on this floor, there is a deficit in shared resource spaces. Those spaces have been pushed to the second floor. Currently, only three conference rooms serve the entire floor.

The second floor primarily houses research labs and lab support spaces and, in an effort to preserve as many of those spaces as possible, has received very little new programmed space. The few programmed spaces it does contain are the shared resources for the first and third floors. While staff and students on each floor only need to travel up or down one floor to reach a conference room, focus room, or work café, it is not as desirable an arrangement as having those resources available on every floor. The business office is the only new department on the second floor. There is also 1,662 sft of unused open office space on the second floor. This space would be ideally suited for open office area. Alternatively, one program department from the first floor might be shifted to the second. This would allow more office space access to natural light, and create space for more shared resources on the first floor.

The third floor of Option C houses all of the Division of Environmental and Ecological Engineering. In order to fit the entire department on one floor, 4,000 sft of research lab space has been displaced. In an effort to displace as little research lab space as possible, no shared support functions have been included on this floor. All of EEE will be required to travel down to the second floor to reach a conference room, focus room, or work café.

Because the options are designed to progress from least to most expensive, no new vertical circulation routes are proposed for Option C. However, the existing freight elevator at the northwest corner of the building is proposed to be replaced with a more functional passenger elevator, sized to carry freight when necessary to support the research labs. This is not ideal for permitting access to the testing center. Wheelchair-bound students will need to travel to the northwest corner of the building then back down to the southeast corner of the building to access the testing center. The existing library elevator could be abandoned or retrofitted to allow for ADA access. However, if retrofitted, this elevator would open in to the testing center office space, not the entrance to the testing center. Students would then need to travel through the testing center to the front desk and back again to their testing seat. Arguably, this is no more convenient than traveling to the existing elevator.

INTERIOR DESIGN

In corridors that have terrazzo, it is recommended to maintain, strip, and clean those corridors, but to remove the terrazzo base. Because the existing terrazzo is difficult to match and expensive to replace, it prevents flexibility for future renovations. An accent terrazzo color can be used to separate existing and new terrazzo in corridors where they will need to be blended. All existing wallcoverings in corridors would be removed and receive epoxy paint. For high-traffic areas such as corridors, epoxy paint is long lasting and extremely durable.

The entrance/commons will be a combination of terrazzo and carpet tile/vinyl, epoxy painted walls, and accents of wallcovering or digital wallcovering. All offices, workstations, grads, classrooms/computer labs, and conference rooms will have their finishes removed and replaced with carpet tile and wall paint. Carpet tile, rather than broadloom, is easy to replace if a tile is damaged. Administration areas will receive carpet tile, vinyl base, and wallcovering. Break rooms, lounges, and the coffee bar will have a luxury vinyl tile (LVT) that is durable and easy to maintain. LVT comes in multiple colors and patterns that can give anywhere from a warm to industrial appearance.

Labs and research center testing areas will keep their existing sealed concrete floors and exposed ceilings; walls will be repainted with epoxy paint. Restrooms will be renovated to meet ADA requirements, keeping existing floors and wall tile as much as possible and, where necessary, replacing floors and wall tile to match existing ones. Where applicable, new solid surface countertops with integral sinks will be installed. In labs, wood casework can be relocated and refinished with new epoxy countertops. Plastic laminate casework should be replaced, as it does not withstand relocation with new plastic laminate countertops. If in good shape, the 2x2 or 2x4 ceiling tile will remain; all other ceiling tile will be removed and replaced with new 2x2 tile.

Stairwells and other existing walls with exposed aggregate finishes will receive a hard plaster coating. To negate any unevenness in the surface, the plaster will receive a slight texture and paint. This will create a long-lasting durable surface that can easily be applied over the existing wall finish.

STRUCTURAL DESIGN

In addition to infilling the mezzanine as described in Option B, certain areas of concrete floor will need to be infilled to create floors level with the adjoining hallways. Room B028 is currently depressed 4" and should be leveled with a new concrete slab. Room B018 is depressed 8" and should be leveled by infilling with 4" of rigid insulation and under a 4" concrete slab to create a level floor condition. Rooms B038 and B036 are depressed 3' and should be leveled in a similar fashion by playing 2'-8" of rigid insulation and then a 4" concrete slab to create a level floor condition. Second-floor Room 268 is depressed by 2' and should be leveled by placing 1'-8" of rigid insulation and then a 4" concrete slab to create a level floor condition.

In cases where 4" of concrete is added to an elevated structure condition, the structure should be analyzed for the additional surcharge load created by the insulation and concrete. The structure in this areas may need to be reinforced to support the additional load.

CIVIL DESIGN

No additional civil work is required outside of the new ADA parking spaces proposed for option A and B.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

Research Lab Assessment (see Appendix for Individual Lab Assessment)

- In the 2013 Lab Condition Assessment for Potter, all of the records available show the majority of the labs are classified as “3. 21-30 yrs, acceptable, some unmet needs, showing wear.” or “4. 31-39 yrs, inadequate condition hinders mission, outdated, marginal, and detrimental to recruitment.”
- The existing laboratories should not be viewed as long term spaces not requiring renovation.
- The basement laboratories are all served by small, standalone air handlers. Renovation of the remainder of the space in the basement should not impact the operation of these laboratories.

- There are no laboratories on the first floor.
- Except for the Laboratory 238 Complex, which is served by a standalone air handler, the laboratories on this floor are served by ACP-8, which also serves significant non-laboratory spaces on the first and second floor. It has been reported, but not verified, that the laboratories are not 100% exhausted, which means laboratory air is being recirculated to non-laboratory spaces. When the building was built, this was acceptable, but is generally not acceptable now, depending on the purpose and use of the laboratory.
- The entire third floor is served by ACP-7 to include laboratories. The same condition of laboratory air being recirculated to non-laboratory spaces holds true here as on the second floor.
- Repurposing the building, except for the laboratories, not only justifies new mechanical systems, but requires ones that meet the energy code.
- New mechanical systems will necessarily also have to serve the existing laboratories and it is recommended the existing laboratories be converted to 100% exhaust. This imposes a higher first cost on the project since the air handler will have to be equipped to handle large volumes of outside air that will not be required if and when the laboratories are moved.

Mechanical Systems, HVAC

- The new building function has the characteristics of not only a high thermal diversity but high ventilation requirements diversity.
- In areas such as the testing center and the computer labs, chilled beams for cooling and demand-controlled ventilation can reduce overall air handler size.
- In classroom and office areas where thermal and ventilation loads are less diverse, standard VAV reheat systems may be more appropriate.
- Preferably, the building would have one air handler to serve one floor to allow flexibility in future building modifications. However, working within an existing building and trying to utilize only existing assigned mechanical spaces may make this concept a challenge.
- While no new mechanical room space is anticipated, it is anticipated additional chase space will be required to run larger, more efficient ductwork.

Mechanical Systems, Plumbing

- The scope of work for this option shows remodeled restrooms.
- The current domestic water and sanitary systems should have the capacity to add the new loads without adverse effects.
- The only negative effect of remodeled restrooms is the infringement on the rooms below while new waste piping is being installed.

Fire Protection System

- The existing fire protection system mains will remain in place and heads will be relocated as required for the new floor plan layout.

Electrical Systems

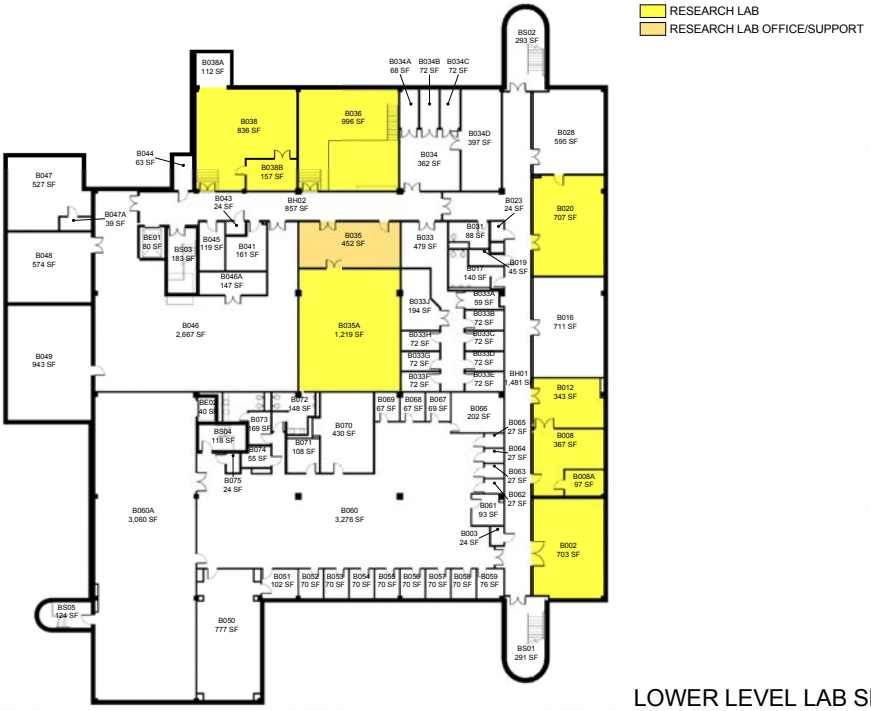
- For the DRC and CTC, the lighting and receptacles in this area would be served by L-1-NW, L-1-NE, L-1-SW, and L-1-SE.
- Due to the age of the panel boards and the number of circuits that will be demolished and reused, the panel boards should be replaced.
- For the intermediate level renovation for COE, the panels in locations to be determined should be replaced. Any circuits to remain intact from the existing panel shall be served by the new panel.
- Existing receptacles and lights will be removed. Associated conduit and conductors will be removed back to source panel.
- Remove any circuits for other systems that will be demolished in the renovated areas.
- Install new power receptacles as required in new DRC, CTC, and COE layouts.
- Install new lighting as required in new DRC, CTC, and COE layouts.
- Run new circuits as required to support other systems.

Telecom System changes

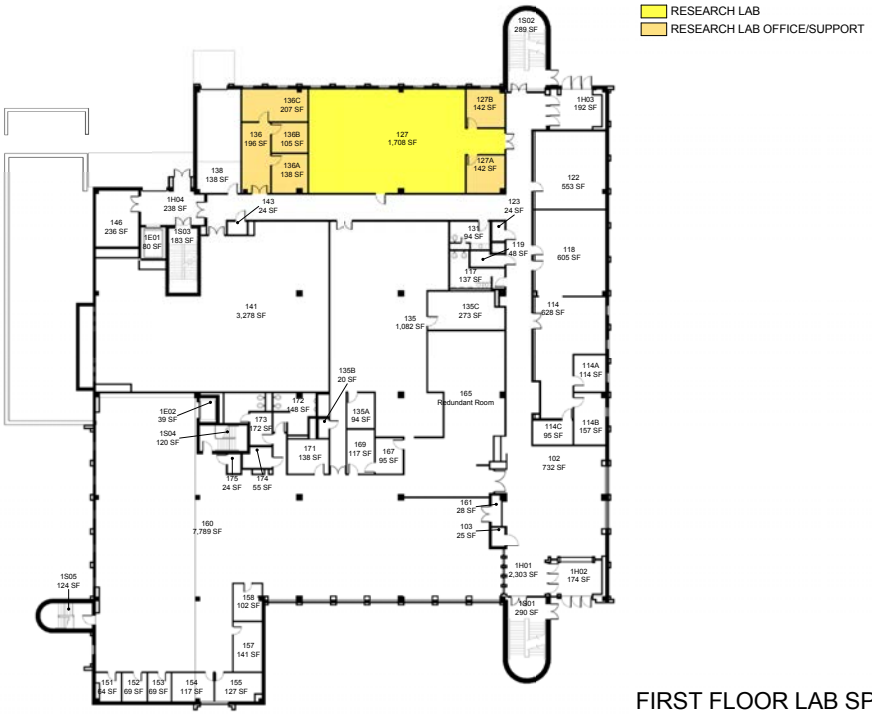
- Add PICS as required in new DRC, CTC, and Intermediate level renovation for COE layouts.
- Remove all CAT 5e cabling from the renovated area for the DRC, CTC, and renovated COE areas.
- Install CAT 6A cabling to all new PICS and wireless APs.
- Add a new equipment rack and CAT 6A patch panels in BDF room B045.
- New equipment rack will need to have 6” vertical wire management on the sides. They will not require horizontal cable management.
- Install new 4” skeletal raceway and/or basket cable tray in renovated areas as required.
- Install a new pathway between BDF B045 and the B060/B060A area in order to shorten the distance of cable pulls to the current B051-B059 areas.
- Add a redundant 24-strand, single-mode fiber fed from a second campus telecom node in Brown Hall. Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.

Fire Alarm System

- Modify/add audio visual device locations as required for new DRC, CTC, and renovated COE floor layouts.
- Add any new initiating devices as required by new floor layout.



LOWER LEVEL LAB SPACES



FIRST FLOOR LAB SPACES



SECOND FLOOR LAB SPACES



THIRD FLOOR LAB SPACES

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Potter Center Business Unit

Position	Number of Personnel	Area Per Person	Total Area
Business Manager	1	120	120
Business Staff Workstation	6	41	246
Grand Total	**7**	**161**	**366**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	80	80
Grand Total	**1**	**80**	**80**
		TOTAL	**446**

Central Testing Center

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared Office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

Community Resources

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	16	120	1,920
Reception	1	250	250
Conference Room - 12 People	6	300	1,800
Conference Room - 20 People	2	470	940
Conference Room - 6 People	8	200	1,600
Work Café	4	500	2,000
Conference Room - 100 People	1	1,000	1,000
Grand Total	**38**	**2,840**	**9,510**
		TOTAL	**9,510**

Instructional Data Processing

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

Office of Professional Practice

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	3	120	360
Clerical-Workstation	3	40	120
Director	1	120	120
Grand Total	**9**	**321**	**682**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**922**

Office of Future Engineers (OFE)

Position	Number of Personnel	Area Per Person	Total Area
Associate Director	1	120	120
Director	1	120	120
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Grad Assistant Workstation (Hotdesk)	2	64	128
Undergrad Staff Workstations (Hotdesk)	7	40	280
Grand Total	**13**	**584**	**888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area

Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**1,128**

Women In Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Associate Director	1	120	120
Communications	1	120	120
Director	1	120	120
Management	1	120	120
Professional Staff - Office	2	120	240
Program Manager	1	120	120
Research - Visiting Faculty	1	120	120
Grad Assistant Workstation (Hotdesk)	6	64	384
Grad Student Admin. Workstation (Hotdesk)	6	64	384
Grand Total	**21**	**1,009**	**1,769**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Student Organizations	1	120	120
Workroom	1	120	120
Grand Total	**3**	**360**	**360**
		TOTAL	**2,129**

College of Engineering Honors

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Advisor	3	120	360
Assistant Director	1	120	120
Associate Director	1	120	120
Director	1	120	120
Grand Total	**10**	**521**	**884**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	360	360
Grand Total	**1**	**360**	**360**
		TOTAL	**1,244**

Global Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Assistant Director	1	120	120
Associate Director	2	120	240
Director	1	120	120
Grad-Thesis Desks (160 Students)	1	26	26
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Program Manager	1	120	120
Undergrad Assistant	2	40	80
Grad Student Admin. Workstation (Hotdesk)	3	64	192
Grand Total	**14**	**891**	**1,179**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Storage	1	120	120
Student Ambassador Workroom	1	360	360
Workroom	1	120	120
Grand Total	**4**	**720**	**720**
		TOTAL	**1,899**

Minority Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Assistant Director	1	120	120
Associate Director	1	120	120
Data Analyst	1	120	120
Director	1	120	120
Grad Student Admin. Workstation (Hotdesk)	2	64	128
Grand Total	**6**	**544**	**608**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Tutorial Center	1	900	900
Workroom	1	120	120
Grand Total	**3**	**1,140**	**1,140**
		TOTAL	**1,748**

Division of Environmental and Ecological Engineering (EEE)

Position	Number of Personnel	Area Per Person	Total Area
Academic Dean / Department Head	1	120	120
Admin/Operations Staff, Workstation	2	41	82
Clerical-Workstation	1	40	40
Grad-Thesis Desks (64 Students)	45	26	1170
Hoteling Offices	4	120	480
Management	2	120	240
Professional Staff - Office	3	120	360
Research - Visiting Faculty	4	120	480
Tenured/TT Faculty	10	120	1200
Visiting Scholars	4	64	256
Grand Total	**76**	**891**	**4,428**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Open Lab - Project Engineering	2	300	600
Reception	1	180	180
Student Organizations	1	120	120
Teaching Lab - Wet	1	1200	1,200
Teaching Lab - Wet/Dry	1	1200	1,200
Teaching Support - 25%	1	300	300
Mediascape Collaboration Space	1	150	150
Grand Total	**9**	**3570**	**3,870**
		TOTAL	**8,298**

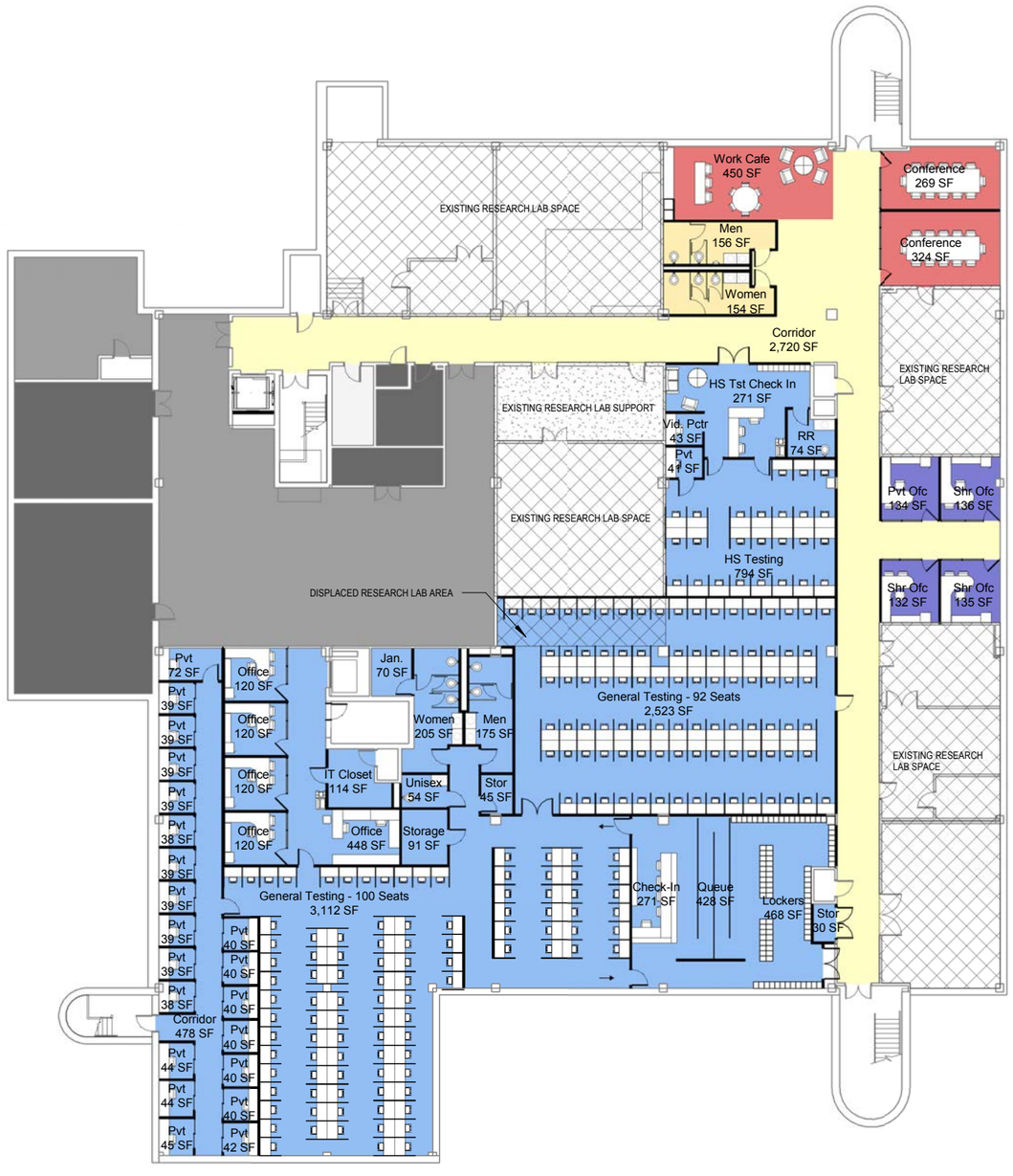
Net Area Total, All Departments	43,057
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**55,974**

DISPLACED EXISTING AREA

Department	Area Per Department
College of Engineering	2,149
Corporate and Foundation Relations	700
Electrical and Computer Engineering	4,713
Engineering	2,417
Engineering Computer Network	711
Engineering Research Center	229
Environmental and Ecological Engineering	3,387
General Building	1,327
IEEE	479
Industrial Engineering	914
IT Cust Rltn	202
Laboratory of Renewable Resources Engineering	1,027
Libraries	23,365
Nuclear Engineering	3,889
Operations and Maintenance	161
Grand Total	**50,277**

DEPARTMENT AREA AS DESIGNED

Department	Area Per Department
Assistive Technology Center	808
Business Office	631
Central Testing Center	11,192
Circulation	9,278
Community Resource	7,515
Disability Resource Center	3,761
Engineering Honors	875
Environmental & Ecological Engineering	9,143
General Building	199
Global Engineering Program	1,200
Instructional Data Processing	1,141
Minority Engineering Program	1,582
Office Of Future Engineers	712
Online Compliance	537
Open / Flex	1,662
Professional Practice	790
Restroom	709
Women In Engineering	1,699
Grand Total	**53,434**



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION C - LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION C - FIRST FLOOR PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION C - SECOND FLOOR PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION C - THIRD FLOOR PLAN



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION C - LOWER LEVEL USE



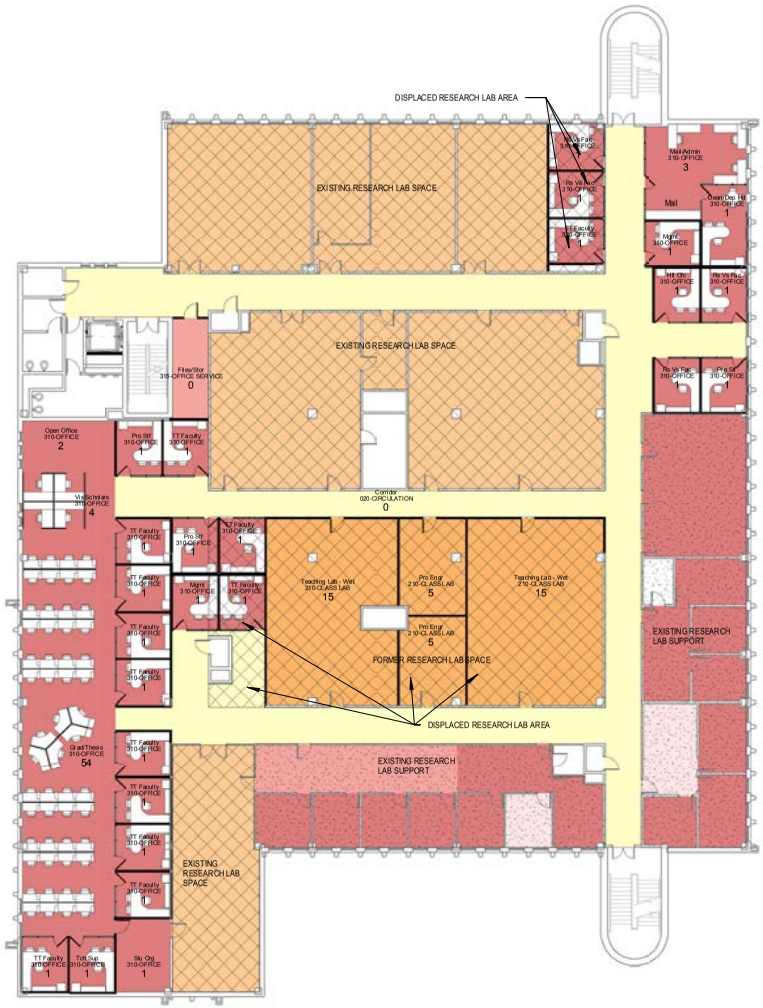
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- 012-REST ROOM
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- 030-MECHANICAL
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- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION C - FIRST FLOOR USE



010-BUILDING SERVICE
012-REST ROOM
020-CIRCULATION
030-MECHANICAL
210-CLASS LAB
250-NONCLASS LAB
255-NONCLASS LAB SERVICE
310-OFFICE
315-OFFICE SERVICE
317-DEPARTMENT COMMONS
350-CONFERENCE
410-STUDY
455-STUDY SERVICE
650-LOUNGE
651-BUILDING COMMONS
680-MEETING ROOM
730-CENTRAL STORAGE

OPTION C - SECOND FLOOR USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION C - THIRD FLOOR - USE

OPTION D

OPTION D – DRC, ATC, CTC, COLLEGE OF ENGINEERING PROGRAMS, AND ENGINEERING IN EDUCATION

This option builds on the goals stated in Option C while fully realizing the College of Engineering's plans for the building, which include locating Engineering Education and First-Year Engineering spaces in Potter.

Here, the building will receive a complete renovation, including exterior renovations aimed at updating the aesthetics and functionality of Potter, permitting more daylight into and better views from the building. Landscape renovations will fully integrate the building into the overall campus by affecting the Centennial Mall project to fully recognize and react to the Potter building, with the goal of creating a welcoming and inviting building and surrounding plaza spaces.

GOALS

The goals of Option D are to:

- create a more accessible and functional Disability Resource Center
- increase efficiency and effectiveness by co-locating the DRC and the ATC
- increase efficiency, cost control, academic integrity, and ease of use by creating a centralized location housing all campus testing needs and combining all campus testing functions in one location
- create a new campus wide resource that is appropriately sized and staffed to deliver better testing conditions at Purdue with increased academic integrity
- reduce the demand for academic and non-academic space during periods of high testing volume
- realize the College of Engineering's master plan for the Potter Center by creating a center devoted to serving First-Year Engineering students and associated recruitment and retention programs
- locate ENE alongside FYE and COE Programs to realize the natural synergies between these departments and buildings
- update the interior and exterior of the Potter Center and fully integrate the building into the surrounding campus

ARCHITECTURAL DESIGN

Option D represents the ideal scenario for all departments with a stake in this study. Because the building has been assessed as a blank slate, each department can be ideally located where it will operate at maximum efficiency and realize synergies with its surrounding departments. Additionally, the skin and areas surrounding the building itself are proposed to be renovated, creating an updated and more inviting image for the Potter Center. Because many of the departments involved in this study focus on the recruitment and retention of COE students, the aesthetic upgrades may prove to be beneficial to the mission of these departments, in addition to benefiting the occupants of Potter.

As in Option C, the lower level houses the Central Testing Center. The layout of the General Testing Center is nearly identical to Option C. However, the High-Stakes Testing Center has been moved to the east side of the floor. This location allows the testing center more space to effectively accommodate periods of high volume. It is also easily accessible via the building's main stairway or primary elevator. The CTC's secure storage needs are also better met in this option. Two large classrooms, the i2i Classroom and i2i Design/Project Studio, are located on the lower level to allow the max-

imum amount of regularly occupied office spaces access to natural light. The Online Compliance Department, Instructional Support, and shared conference and breakrooms round out the floor. Both of these departments have been placed here by request; however, consideration should be given to relocating these offices to the open space on the third floor since space is now available.

The departments housed on the first floor are the same as those in Option C; however, the removal of the lab space to the north allows the entirety of the core of the building to be dedicated to support spaces, including a 100-person conference room and a large work/café. All open open offices spaces are located along the perimeter, and closed offices are daylit through glass storefront along the narrow face of each office wall.

The second floor houses all of Engineering Education and First-Year Engineering. The consolidation of this department to one floor will greatly contribute to enhanced security, allowing the entire floor to be locked off after hours. The i2i/Honors Teaching Lab occupies the core of the floor plate, allowing regularly occupied offices areas to be located nearer to the perimeter. The lab will receive borrowed light through the south wall across the grad/thesis desk area. Shared resource spaces are generally located in areas without access to natural light.

The third floor houses all of the Division of Environmental and Ecological Engineering. As this Division seeks to become a School, it is critical it has its own space and identity. This floor has a separate reception area, immediately adjacent to the elevator. Large teaching labs and project engineering spaces occupy the core. The north end of the floor houses the business office. Approximately 4,100 sft of open office space and 10 closed offices are available on this floor for flexible future use. Shifting the offices from the lower level to this floor may be a good use of these spaces.

In addition to installing a new elevator in the existing freight elevator shaft, a new elevator is proposed to be installed in the southeast portion of the building. This location allows the new elevator to serve all floors with easy access from the main entrance. The small library elevator is shown as remaining; however, the building may be better served by reclaiming that space for other uses and removing the elevator.

In addition to interior renovations, this option further proposes to reskin the building's south façade. Here, the existing precast concrete panels are removed and replaced with a glass curtainwall system. Reskinning this face will allow for the maximum penetration of daylight; however, the light must be modulated to prevent overheating the interior space. A simple system of solar shades (designed to effectively reduce solar heat gain during the warm months) will provide shading as well as an updated, high-tech aesthetic for this face of the building. Thought not studied here, a similar treatment could be applied elsewhere on the façade at other areas of precast panels.

INTERIOR DESIGN

Corridors will have new terrazzo, with vinyl base flexibility for future renovation. The existing terrazzo is both difficult to match and expensive to replace. In the long term, new terrazzo is extremely cost-effective and durable.

The entrance/commons will be a combination of terrazzo and carpet tile/vinyl, epoxy-painted walls, and accents of wallcovering or digital wallcovering. Administration areas will receive carpet tile, vinyl base, and wallcovering. All offices, workstations, grads, classrooms/computer labs, and conference rooms will have carpet tile and wall paint with accent colors. Carpet tile, rather than broadloom, is easy to replace if a tile is damaged. Break rooms/lounges/coffee bar/commons will have luxury vinyl tile (LVT), which is durable and easy to maintain. LVT comes in multiple colors and patterns that can give anything from a warm to industrial appearance. Teaching labs and workrooms will have sealed concrete floors, epoxy painted walls, and exposed ceilings.

Stairwells and other existing walls with exposed aggregate finishes will receive a hard plaster coating. To negate any unevenness in the surface, the plaster will receive a slight texture and paint. This will create a long-lasting durable surface that can easily be applied over the existing wall finish.

STRUCTURAL DESIGN

As a result of the reimaging of the building, an elevator will be added within the existing floor plan. At the location of the elevator, an opening will be required at each of the floor framing levels and at the roof framing level. The floor framing may require reinforcing around the opening to support the design floor/roof load. At the basement level, an opening will be required in the slab-on-grade to accommodate an elevator pit. The elevator pit construction will consist of reinforced cast-in-place concrete walls supported on a reinforced concrete mat slab that makes up the bottom of the elevator pit. The elevator shaft itself can either be framed with light-gauge metal studs and gypsum board sheathing or framed with full-height reinforced concrete masonry units. One advantage to framing with reinforced concrete masonry units would be the possibility of using the elevator shaft walls to assist in supporting the floor and roof systems that had holes cut in them to accommodate the elevator construction.

CIVIL DESIGN

Option D proposes to renovate all of the plaza space south of the Potter Center to serve two goals. First, it will provide an easy, directly accessible route to Potter Center's main entry. More than any other building on campus, Potter Center will be used by wheelchair-bound and disabled students. Creating an entry that allows these students to walk directly toward the door with their peers is a critical aspect of the success of the DRC within Potter. Additionally, renovation of the existing plaza can create more welcoming, inviting, and useable space outside Potter. In its current configuration, the building itself is obscured by very large trees and the patio area is difficult to access. The proposed configuration of Centennial Mall only addresses the Wilmeth Active Learning Center and makes no gestures of inclusion toward Potter. The revised design pulls from the distinctive geometry of Potter's curved staircase, radiating out through the plaza and creating a welcoming avenue toward the main entry. A terraced outdoor classroom creates a semi-private outdoor seating area, an interesting option for outdoor teaching, and serves as a subtle reminder that this is a place of learning.

MECHANICAL, ELECTRICAL AND PLUMBING DESIGN

Mechanical Systems, HVAC

- Repurposing the building not only justifies new mechanical systems but requires systems that meet the energy code.
- The new building function has the characteristics of high thermal and high ventilation requirements diversity.
- In areas such as the testing center and computer labs, chilled beams for cooling and demand-controlled ventilation can reduce overall air handler size.
- In classroom and office areas where thermal and ventilation loads are less diverse, standard VAV reheat systems may be more appropriate.
- Preferably, the building would have one air handler to serve one floor to allow flexibility in future building modifications. However, working within an existing building and trying to use only existing assigned mechanical spaces may make this concept a challenge.
- While no new mechanical room space is anticipated, it is anticipated additional chase space will be required to run larger, more efficient ductwork.

Mechanical Systems, Plumbing

- The existing utilities to the building should be adequate for the proposed building usage, but all systems need to be evaluated for capacity once the usage is better defined.
- The age of the domestic water heater is not known. If it is found to be the original water heater or if it has significant aging, it can be replaced with a more efficient steam-to-water heater that sub cools the steam condensate.
- Additions/modifications to the existing plumbing system will be made as required to accommodate floor plan changes.

Fire Protection Systems

- It is assumed the existing incoming service is adequate and no changes are required.
- It is anticipated the existing fire mains can remain in place and new laterals run as required. If a main is found to be in conflict with a requirement of the new floor plan, the main will be rerouted or offset for the new work.

Electrical Systems

- Remove all receptacles and lights in renovated area. Associated conduit and conductors will be removed back to source.
- Remove all conduit and conductors serving HVAC, plumbing, and other equipment to be demolished.
- Maintain four bus ducts that run vertically in the northwest, northeast, southwest, and southeast corners of the building.
- Replace existing fused-style panel boards.
- Some of the panels may require replacement with higher-rated panels to serve the increased loads in the testing center.
- Replace 208/120V fused-switch distribution panels with 400A breaker-style panel boards.
- Install new lighting throughout the building.
- Install new power receptacles throughout the building.
- Provide circuits to new HVAC and plumbing equipment as required.
- MCC-G-1 and MCC-PH may be reused depending on the final determined needs of the building systems.

Telecom System changes

- Remove all existing CAT 5e cabling from entire building, coordinating timing so that critical equipment does not go down.
- Add PICS as required for reimaging of entire building
- Install CAT 6A cabling throughout building to all PICS and wireless access points.
- Install new CAT6A patch panels in existing BDF and IDF rooms. Equipment racks will need to have 6" vertical wire management on the sides. They will not require horizontal cable management.
- Establish new minimum 8x8' IDF rooms on first, second, and third floors, ideally stacked.
- Install new telecom grounding system throughout building.
- Install new 4" skeletal raceway and basket cable tray in renovated areas as required.
- Demo all existing ECN cabling out of building and back into utility tunnel.
- Install voice over IP (VOIP) telephone system.
- Add a redundant 24-strand single-mode fiber fed from a second campus telecom node in Brown Hall. Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.
- Demo the existing 250-pair OSP copper cable and replace with a 25-pair into the building. Purdue ITaP would install cable into the building, but the cost would be absorbed by this project.

Fire Alarm System

- Redesign of complete system, but main fire alarm panel could be retained.

SPACE PROGRAM

Disability Resource Center

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	2	120	240
Director	1	120	120
Undergrad Assistant	2	40	80
Accommodation Specialist	7	160	1,120
Total	**14**	**481**	**1,642**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Alternative Format Production	1	350	350
Files/Storage	1	60	60
Reception	1	300	300
Work Café	1	200	200
Workroom	1	150	150
Total	**5**	**1060**	**1,060**
		TOTAL	**2,702**

Assistive Technology Center

Position	Number of Personnel	Area Per Person	Total Area
Director	1	120	120
Total	**1**	**120**	**120**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Assistive Technology Center	1	850	850
Total	**1**	**850**	**850**
		TOTAL	**970**

Potter Center Business Unit

Position	Number of Personnel	Area Per Person	Total Area
Business Manager	1	120	120
Business Staff Workstation	6	41	246
Grand Total	**7**	**161**	**366**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	80	80
Grand Total	**1**	**80**	**80**
		TOTAL	**446**

Central Testing Center

Position	Number of Personnel	Area Per Person	Total Area
HS Testing - Office	1	120	120
HS Testing - Video Proctor	1	90	90
HS Testing - Work Stations	4	90	360
Online Compliance - Private Office	1	120	120
Online Compliance - Shared office	3	120	360
Testing - Administration Counter	1	90	90
Testing - Director	1	120	120
Testing - Office	2	120	240
Testing - Video Proctor	1	90	90
Grand Total	**15**	**960**	**1,590**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
General Testing Stations	22.5	230	5,175
High Security Testing Stations	26	25	650
Lockers, 250	300	1	300
Lockers, 35	30	1	30
Private Testing Room	48	21	1,008
Queueing for 60	540	1	540
Secure Break Area	150	1	150
Secure Restrooms	64	6	384
Secure Storage	200	1	200
Storage	50	1	50
Testing Check-in	400	2	800
Testing Concierge	200	1	200
Grand Total	**2,030.5**	**291**	**9,487**
		TOTAL	**11,077**

Community Resources

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	16	120	1,920
Reception	1	250	250
Conference Room - 12 People	6	300	1,800
Conference Room - 20 People	2	470	940
Conference Room - 6 People	8	200	1,600
Work Café	4	500	2,000
Conference Room - 100 People	1	1,000	1,000
Grand Total	**38**	**2,840**	**9,510**
		TOTAL	**9,510**

Instructional Data Processing

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Director	1	120	120
Check-In/Reception	1	100	100
Grand Total	**6**	**261**	**384**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Data Processing Equipment	1	200	200
Files/Storage	1	100	100
Reception	1	300	300
Grand Total	**3**	**600**	**600**
		TOTAL	**984**

First-Year Engineering (FYE)

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Advisor	10	120	1,200
Instructional Support Coordinators	3	120	360
Instructional Support Manager	1	120	120
Instructional Support Specialist	1	120	120
Assistant Head	1	120	120
Student Advisor Workstations (Shared)	4	40	160
Total	**22**	**681**	**2,162**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	200	200
Teaching Lab Support	1	200	200
i2i Classroom	1	2,800	2,800
i2i Honors Teaching Lab ENGR 141/142	1	3,000	3,000
i2i Design / Project Studio	1	3,000	3,000
Total	**5**	**9,200**	**9,200**
		TOTAL	**11,362**

Office of Future Engineers

Position	Number of Personnel	Area Per Person	Total Area
Associate Director	1	120	120
Director	1	120	120
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Grad Assistant Workstation (Hotdesk)	2	64	128
Undergrad Staff Workstations (Hotdesk)	7	40	280
Grand Total	**13**	**584**	**888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**1,128**

Office of Professional Practice

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	2	41	82
Assistant Director	3	120	360
Clerical-Workstation	3	40	120
Director	1	120	120
Grand Total	**9**	**321**	**682**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Workroom	1	120	120
Grand Total	**2**	**240**	**240**
		TOTAL	**922**

College of Engineering Education (ENE)

Position	Number of Personnel	Area Per Person	Total Area
Adjuncts/Joint Appointments	2	120	240
Communications Director	1	120	120
Director of Project Assessment	1	120	120
Director, Development	1	120	120
Emeritus	2	120	240
Grad-Thesis Desks (160 Students)	112	26	2,912
Instructor/Lecturer	2	120	240
Managing Director	1	120	120
Post Doctoral Assistants - Workstation	9	40	360
Teaching Assistant	20	40	800
Tenured/TT Faculty	28	120	3,360
Visiting Scholars	4	64	256
Grand Total	**183**	**1,130**	**8,888**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Focus/TA Consulting/Help/Advising	14	100	1,400
Student Organizations	1	120	120
Student Societies	3	120	360
MS Distance Learning Classroom (Opt.)	1	1,440	1,440
Workroom	1	120	120
Grand Total	**20**	**1,900**	**3,440**
		TOTAL	**12,328**

Women In Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Associate Director	1	120	120
Communications	1	120	120
Director	1	120	120
Management	1	120	120
Professional Staff - Office	2	120	240
Program Manager	1	120	120
Research - Visiting Faculty	1	120	120
Grad Assistant Workstation (Hotdesk)	6	64	384
Grad Student Admin. Workstation (Hotdesk)	6	64	384
Grand Total	**21**	**1,009**	**1,769**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Student Organizations	1	120	120
Workroom	1	120	120
Grand Total	**3**	**360**	**360**
		TOTAL	**2,129**

College of Engineering Honors

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	4	41	164
Advisor	3	120	360
Assistant Director	1	120	120
Associate Director	1	120	120
Director	1	120	120
Grand Total	**10**	**521**	**884**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	360	360
Grand Total	**1**	**360**	**360**
		TOTAL	**1,244**

Global Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Admin/Operations Staff, Workstation	1	41	41
Assistant Director	1	120	120
Associate Director	2	120	240
Director	1	120	120
Grad-Thesis Desks (160 Students)	1	26	26
Management - Office Manager	1	120	120
Management - Staff	1	120	120
Program Manager	1	120	120
Undergrad Assistant	2	40	80
Grad Student Admin. Workstation (Hotdesk)	3	64	192
Grand Total	**14**	**891**	**1,179**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Storage	1	120	120
Student Ambassador Workroom	1	360	360
Workroom	1	120	120
Grand Total	**4**	**720**	**720**
		TOTAL	**1,899**

Minority Engineering Program

Position	Number of Personnel	Area Per Person	Total Area
Assistant Director	1	120	120
Associate Director	1	120	120
Data Analyst	1	120	120
Director	1	120	120
Grad Student Admin. Workstation (Hotdesk)	2	64	128
Grand Total	**6**	**544**	**608**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Tutorial Center	1	900	900
Workroom	1	120	120
Grand Total	**3**	**1,140**	**1,140**
		TOTAL	**1,748**

Division of Environmental and Ecological Engineering (EEE)

Position	Number of Personnel	Area Per Person	Total Area
Academic Dean / Department Head	1	120	120
Admin/Operations Staff, Workstation	2	41	82
Clerical-Workstation	1	40	40
Grad-Thesis Desks (64 Students)	45	26	1170
Hoteling Offices	4	120	480
Management	2	120	240
Professional Staff - Office	3	120	360
Research - Visiting Faculty	4	120	480
Tenured/TT Faculty	10	120	1200
Visiting Scholars	4	64	256
Grand Total	**76**	**891**	**4,428**

Support and Collaboration	Number of Spaces	Area Per Space	Total Area
Files/Storage	1	120	120
Open Lab - Project Engineering	2	300	600
Reception	1	180	180
Student Organizations	1	120	120
Teaching Lab - Wet	1	1200	1,200
Teaching Lab - Wet/Dry	1	1200	1,200
Teaching Support - 25%	1	300	300
Mediascape Collaboration Space	1	150	150
Grand Total	**9**	**3570**	**3,870**
		TOTAL	**8,298**

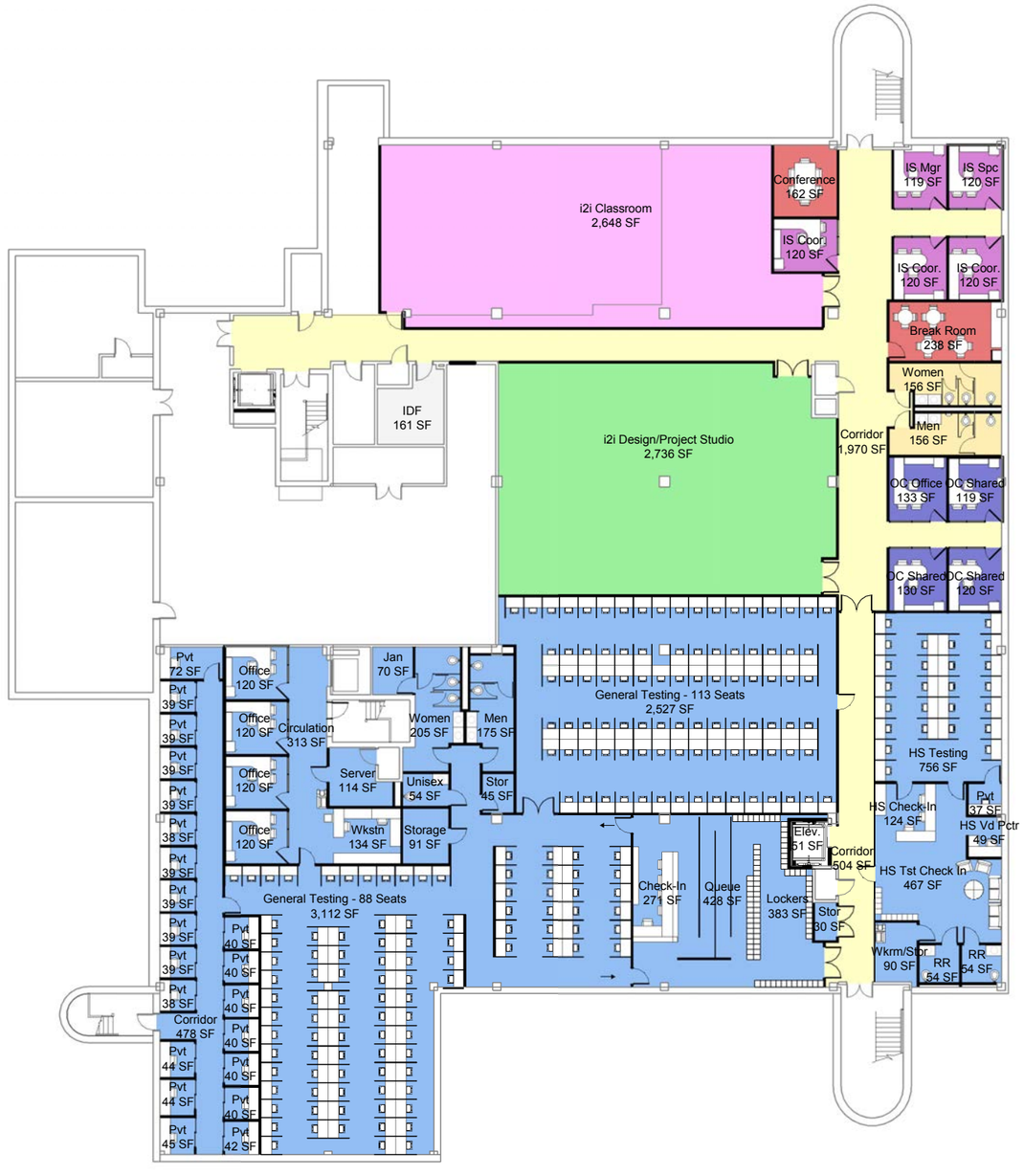
Net Area Total, All Departments	66,747
Grossing Factor	30%
NET AREA, ALL DEPARTMENTS	**86,771**

DISPLACED EXISTING AREA

Department	Area Per Department
COE Performance Space	541
College of Engineering	2,149
Corporate and Foundation Relations	700
Electrical and Computer Engineering	5,295
Engineering	2,863
Engineering Computer Network	711
Engineering Research Center	7,909
Environmental and Ecological Engineering	3,865
General Building	2,132
IEEE	479
Industrial Engineering	3,795
IT Cust Rltn	202
Laboratory of Renewable Resources Engineering	10,265
Libraries	23,365
Mechanical Engineering	703
Nuclear Engineering	9,817
Operations and Maintenance	2,943
Grand Total	**93,866**

DEPARTMENT AREA AS DESIGNED

Department	Area Per Department
Assistive Technology Center	808
Business Office	630
Central Testing Center	11,417
Circulation	17,625
Community Resource	12,197
Disability Resource Center	3,761
Engineering Education	14,038
Engineering Honors	7,581
Environmental & Ecological Engineering	8,615
First-Year Engineering	2,093
General Building	72
General Building - Telecommunications	112
Global Engineering Program	1,288
Instructional Data Processing	1,108
Minority Engineering Program	1,807
Office of Future Engineers	656
Online Compliance	502
Open / Flex	6,441
Professional Practice	897
Restroom	1,464
Women In Engineering	1,818
Grand Total	**94,930**



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION D - LOWER LEVEL PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION D - FIRST FLOOR PLAN



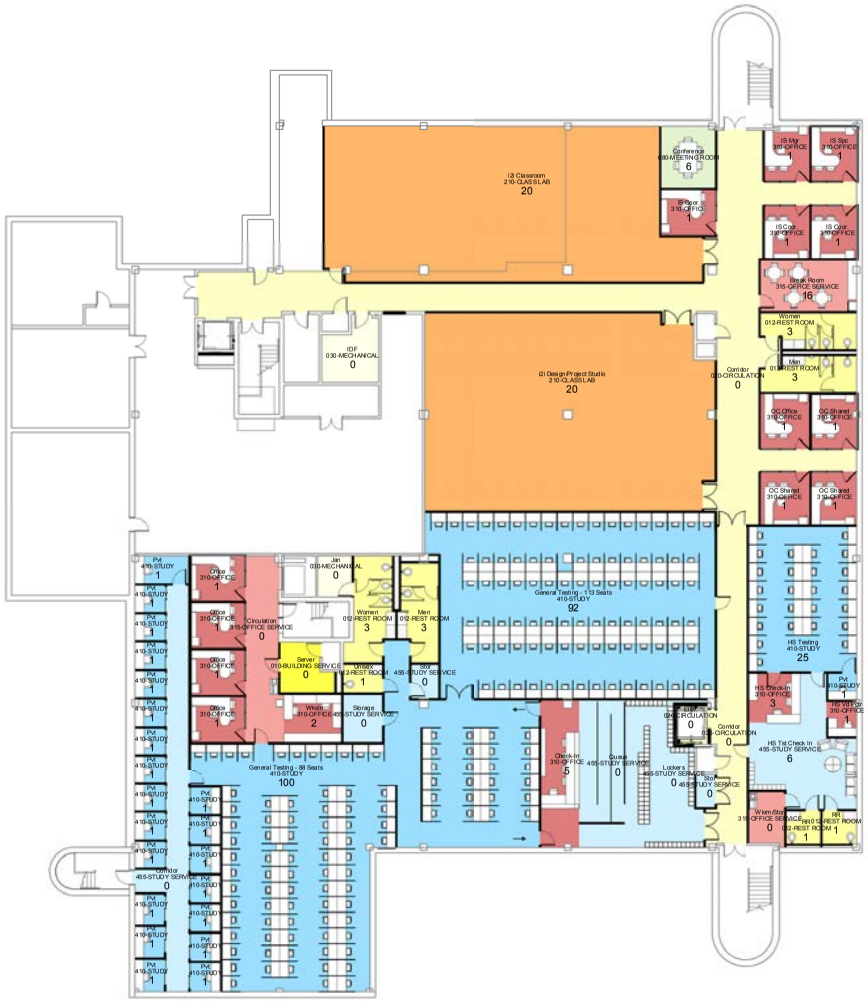
- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION D - SECOND FLOOR PLAN



- ASSISTIVE TECHNOLOGY CENTER
- BUSINESS OFFICE
- CENTRAL TESTING CENTER
- CIRCULATION
- COLLEGE OF ENGINEERING
- COMMUNITY RESOURCE
- DISABILITY RESOURCE CENTER
- ENGINEERING EDUCATION
- ENGINEERING HONORS
- ENVIRONMENTAL & ECOLOGICAL ENGINEERING
- FIRST YEAR ENGINEERING
- GENERAL BUILDING
- GENERAL BUILDING - ELECTRICAL
- GENERAL BUILDING - MECHANICAL
- GENERAL BUILDING - PLUMBING
- GENERAL BUILDING - TELECOMMUNICATIONS
- GLOBAL ENGINEERING PROGRAM
- INSTRUCTIONAL DATA PROCESSING
- MECHANICAL ENGINEERING
- MINORITY ENGINEERING PROGRAM
- OFFICE OF FUTURE ENGINEERS
- ONLINE COMPLIANCE
- OPEN / FLEX
- OPERATIONS & MAINTENANCE
- PROFESSIONAL PRACTICE
- RESTROOM
- WOMEN IN ENGINEERING

OPTION D - THIRD FLOOR PLAN



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION D - LOWER LEVEL USE



- 010-BUILDING SERVICE
- 012-REST ROOM
- 020-CIRCULATION
- 030-MECHANICAL
- 210-CLASS LAB
- 250-NONCLASS LAB
- 255-NONCLASS LAB SERVICE
- 310-OFFICE
- 315-OFFICE SERVICE
- 317-DEPARTMENT COMMONS
- 350-CONFERENCE
- 410-STUDY
- 455-STUDY SERVICE
- 650-LOUNGE
- 651-BUILDING COMMONS
- 680-MEETING ROOM
- 730-CENTRAL STORAGE

OPTION D - FIRST FLOOR USE

Conceptual Aerial of South Facade and Plaza - Option D



Conceptual South Facade and Plaza - Option D



Conceptual Plaza Redesign - Option D



APPENDIX

APPENDIX A

LAB ANALYSIS

Lab Information	
Room Number: 205,209,257.263,265,277	ASF:
Date: 04/09/2013	Photo Numbers:
Department Priority:	
Description:	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top		4	
Flooring		4	4+ in 209
Walls			
Ceilings			
Lighting		4	
Fume Hood	#	4+	
Emergency E/S	#		
Sinks	#	5	
Ductwork			
Air Handler			
Utility Fixture			
Power/Data			
Overall		4+	

Other Comments:

Open up space, enlarge rooms

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.



Room 205



Room 205



Room 209 – water leak



Room 209



Room 209



Room 209



Room 209



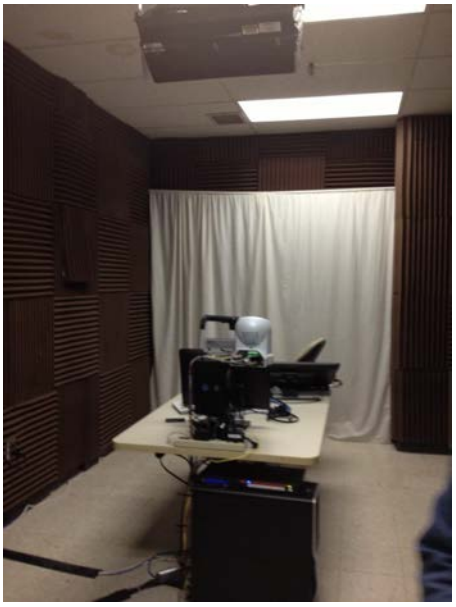
Room 257



Room 257A



Room 263



Room 265



Room 277

Lab Information	
Room Number: 231,233,235,237,252,254,256	ASF: 3134
Date: 04/09/2013	Photo Numbers:
Department Priority:	
Description:	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top		4	Counter tops: 4+ casework: 3
Flooring		4+	
Walls		3	
Ceilings		-	
Lighting		4	
Fume Hood	#1	5	
Emergency E/S	#1		
Sinks	#1	4	
Ductwork			
Air Handler			
Utility Fixture			
Power/Data			
Overall		4	

Other Comments:

Open up labs, from 11 to 2-3 spaces

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.



Room 231



Room 231



Room 235



Room 235

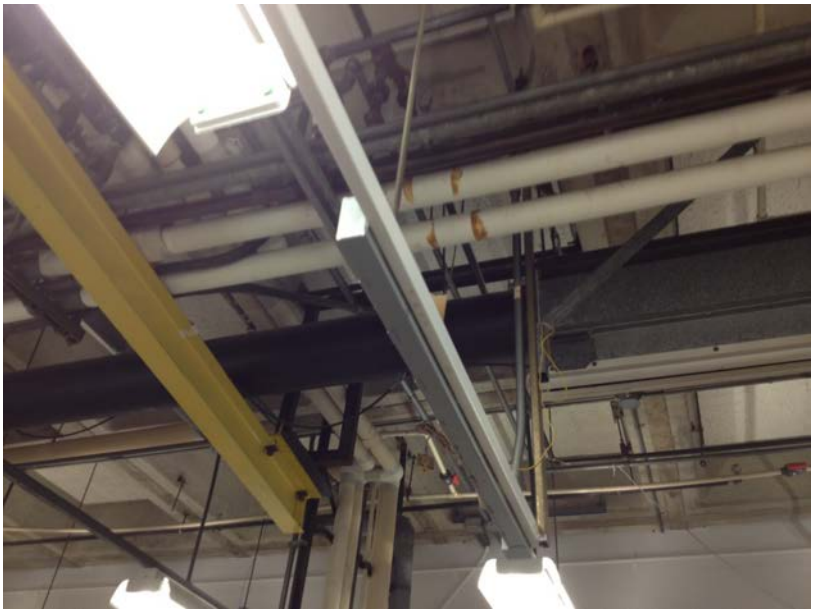


Room 252


Room 252



Room 254



Room 254



Room 256



Room 256



Room 256



Room 256



Room 256

Lab Information	
Room Number: 238	ASF: 634
Date: 04/09/2013	Photo Numbers:
Department Priority:	
Description: Human subj. research lab – engg. education	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top		3	
Flooring		3	
Walls		3	
Ceilings		4+	Needs ceiling
Lighting		4+	Needs appropriate lighting, remove surgical lighting
Fume Hood	#	-	
Emergency E/S	#	-	
Sinks	#1	3	
Ductwork			
Air Handler			
Utility Fixture			
Power/Data			
Overall			

Other Comments:

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.





Needs ceiling



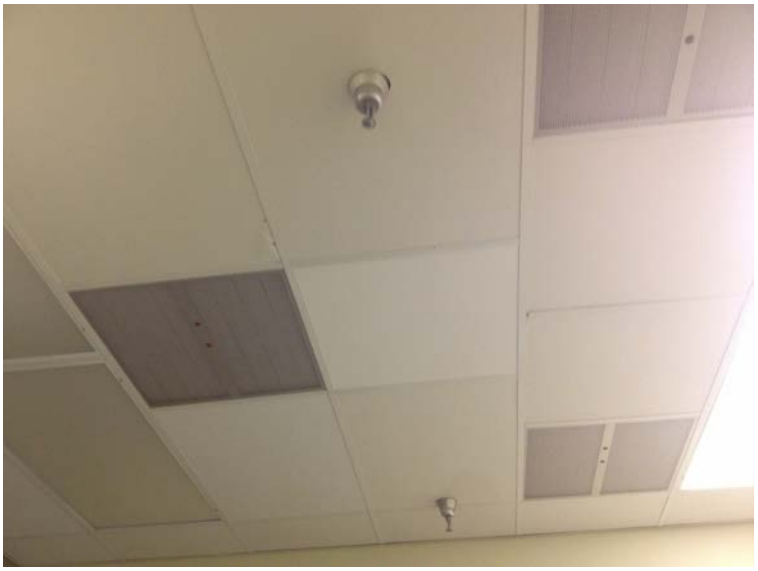




Sink



238B



238B

Lab Information	
Room Number: 337, 337A	ASF: 1418
Date: 04/09/2013	Photo Numbers:
Department Priority: less than 5 years, med-high	
Description:	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top		4+	
Flooring		4	
Walls			
Ceilings		-	
Lighting		4	
Fume Hood	# 1	4+	Not working
Emergency E/S	# 1		
Sinks	# 1	4	
Ductwork			
Air Handler			
Utility Fixture			Gas(remove)
Power/Data			No data connection
Overall			

Other Comments:

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.







Fume hood







337A



337A

Lab Information	
Room Number: B028	ASF: 589
Date: 04/09/2013	Photo Numbers:
Department Priority: less than 5 years	
Description: computational research lab	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top		3	
Flooring		3	
Walls		3	
Ceilings			
Lighting		3	
Fume Hood	#	-	
Emergency E/S	#	-	
Sinks	#	-	
Ductwork			
Air Handler			
Utility Fixture			
Power/Data		3	
Overall		3	

Other Comments:

Remove liebert CRAC
Remove raised floor

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.



Raised floor





Lab Information	
Room Number: B033	ASF: 1240
Date: 04/09/2013	Photo Numbers:
Department Priority: 5-10 years	
Description: future lab	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top			
Flooring			
Walls			
Ceilings			
Lighting			
Fume Hood	#		
Emergency E/S	#	4	
Sinks	#		
Ductwork			
Air Handler			
Utility Fixture			
Power/Data			
Overall		3	

Other Comments:

No animals in POTR now
MJIS uses cage wash daily

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.





Cage wash



Emergency eye wash/ shower



Lab Information	
Room Number: B034 suite	ASF: 952
Date: 04/09/2013	Photo Numbers:
Department Priority: high	
Description: bio-animal space, RPS in progress	

Evaluation/Scope			
		level*	Comments
Casework/Counter Top			
Flooring		4	Remove protrusions
Walls			Remove kennel walls, convert to lab
Ceilings			
Lighting		4	
Fume Hood	# 0		
Emergency E/S	# 0		
Sinks	# 1	4	
Ductwork			
Air Handler			
Utility Fixture			
Power/Data			
Overall		3	

Other Comments:

Future lab

***1**: New-10yrs, state of art, major asset
2: 11-20 yrs, adequate, asset to recruitment
3: 21-30 yrs, acceptable, some unmet needs, showing wear
4: 31-39 yrs, inadequate, condition hinders mission, out dated, marginal, and detrimental to recruitment
5: >40 yrs, inadequate, major detriment to recruitment, neglected cond.







APPENDIX B

MEETING MINUTES

AMERICAN
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MEETING MINUTES

Location: FREH 2-2 Conference
Date: 1-08-16
Project Name: Purdue Potter Renovation Study
Project No.: 2015.02663 PID#15413
Owner: Purdue University – Aaron Hunley
Attendees: See attached sign in sheet
Minutes By: Scott Marchisin
Minutes By: Brandon Farley

The following notes reflect our understanding of the discussions and decisions made at this meeting.
If you have any questions, additions, or comments, please contact the issuer of these minutes.

ITEMS DISCUSSED:
This meeting was the initial project kick-off, with the purpose of gathering information and more thoroughly defining the scope of the project. The meeting started with introductions and participants discussing their connection to the project.

1. Disability Resource Center (DRC) – Dan Carpenter spoke on behalf.
    a. The president's office believes this is a high priority and needs a short term solution.
        i. Dan representing the DRC would prefer to complete this area alonside the rest of the project and "do it right" rather than rush forward with a short term solution.
    b. Students of today do not see help centers in a negative light as prior generations. Students are willing to use the resources if they are provided. There has been a steady increase in the use of these centers.
    c. Currently two testing units (Yong 8th floor, Schleman Hall)
        i. First space – run by two people currently
            1. Used for those taking midterms and final exams
            2. Small testing facility
            3. Small number of private rooms (6 individual rooms)
            4. Has larger small testing room (space for 25 to 30 carrels)
            5. Has cameras at each testing seat
            6. Lockers
        ii. 2nd test center
            1. Small staff for 3 people
            2. Set of lockers – used for check-in
            3. Administrates national exams
            4. Seats about 20 to 30 students

7TER

5. Has cameras at each testing seat
6. Has video display
7. Computers here used for on-line testing
8. During finals week they need to use or borrow spaces (In Hicks they use spaces with 50 people in a room

d. Success is defined as:
   i. Ground floor access. Currently 1 of 2 schools in big ten that does not have the DRC space located on the first floor.
      1. No elevators – issues with breaking down, as well as negative perception
   ii. Welcoming
e. 1,100 students are served by the organization
f. Need 8-9 closed offices for private interactions
g. Production facility is needed which produces alternative format text etc.
h. Space for Growth – this unit will get larger over time
i. Overall – need larger testing site all in one place
j. The testing spaces
   i. Semi-Private testing room,
      1. Limited visual and acoustic distractions
      2. Used by students who need additional time to take tests
      3. Should hold 25-30 students
   ii. Private Testing Rooms
      1. Distraction Free environment – visual and acoustic distractions should be reduced to the greatest extent possible
      2. Need 6 individual rooms
      3. Would need a small window into the space for monitoring
k. Disable parking for students adjacent to the building. A couple of spots.

2. Centralized Testing Center (CTC) – Frank Dooley, Dan Carpenter, Jon Harbor)
   a. Foreign language placement testing could use same space
   b. Testing for 50 to 150 people
   c. Digital test center needs 25 computers for online testing (Current Stewart Centers)
      i. Used for online classes – some professors want the students to come in for the tests
      ii. Teaching online course – such as Chinese 101 online – demand is going up
   d. Currently a lot of students on campus who are in STEM. Students take online courses to manage time and space.
   e. How to give the test for online testing?
      i. A lot more demand than 25 seats.
      ii. Enrollment of classes in the 1000's.
   f. Need to confirm who the test takers are. Proctoring is the key.
   g. Overall a few private offices, with a large testing room.
      i. Large room is to be flexible.
   h. The new space will need a large number of cameras to tape testing, as well as monitor testing in progress
      i. Prometric is a client
   i. White noise will likely be required
   j. Needs a secure internal restroom
   k. May need card swipe access to computers
   l. Testing entry procedure
      i. Sign rules agreement
      ii. Form of ID
   m. Open lab space during non-high testing times could use space as large lecture or other uses.

7TER

i. Office of future engineers is looking for space during the middle of the semester and not just a lecture space

n. Might need both full and half sized lockers

o. Room should hold 175 to 200 students

i. (Based on graph provided by Kim from space management based on Evening Exams enrollment in Fall 2014.

p. Deborah Howard –(Director, Facilities Resources and Planning at Penn State) provided a floor plan of their Pollock Testing Center.

i. Total Capacity – 161

ii. Setup with security zone similar to airport

iii. The Purdue staff have not evaluated the space in person

3. Instructional Data Processing (IDP)
   a. The space should be evaluated to see if this function can also move to the building
   b. This is the location for grading "scantron" type testing forms
   c. 2 people full time currently work there
   d. Need 1 additional person and possibly up to 3 more, for a total of 5 employees
   e. Currently the space is equal to 3 times a 19x26 space.
   f. Has specific humidity requirements
   g. Works regularly with the DRC and Digital education
   h. This would make the Potter testing center a "one stop shop" for all testing services.

4. Re-imaging of Potter (CoE)
   a. Would like the space to be similar to the recent Grissom renovation
   b. Look at relocating entry from the corner to a more prominent central location with a connection to the student success corridor
      i. Located off of future green plaza
      ii. Renovation should create a student centric building
         1. Office of professional development
         2. Student connections with programs (ex.Women in Engineering)
   c. Building believed to be about 120,000 sq. ft.
   d. Relocate elevator – access is poor
   e. Need library to move out to start work. They move out during the summer of 2017.

ACTION ITEMS:

1. Aaron to get application for digital files to American Structurepoint (Completed 1/8/16)
   a. American Structurepoint to complete and return (Completed 1/11/16)
   b. Purdue to upload pdf and cad files they have regarding Potter. (Completed 1/11/16)
2. American Structurepoint and Purdue to discuss scope of study desired
   a. Schedule
   b. Deliverables
      i. Written summary
      ii. Floor plans
         1. Discuss extent of reimaging
      iii. Elevation –of new entry
      iv. Opinion of Probable Cost
      v. Time Frame of study
3. Discuss Proposal
4. Need information from Slam Study for organization of Potter

ITER

NEXT MEETING:
TBD

cc: Attendees

Very truly yours,
American Structurepoint, Inc.

Scott A. Marchisin, AIA, CSI, LEED-AP, BD+C
Account Manager / Project Manager

SAM:bf

ITER

AMERICAN STRUCTUREPOINT INC.

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Meeting Minutes CTC Study

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 01/29/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Jason Baumer	Purdue
Madeleine Henry	Purdue

ITEMS

1. Introductions
2. Current space is windowless, distraction free
3. Future demand is unknown with the testing no longer giving credits after this semester, could see a drop in numbers.
    a. Want people to take placement tests right before they take the course not semesters in advance.
4. 5 testing sessions that take place throughout the year.
    a. Summer (4 weeks of 10am to 4pm)
    b. Beginning fall semester (first two nights of semester)
    c. Beginning spring semester (First two nights of semester)
    d. Late fall semester (November)
    e. Late spring semester (April)
5. Currently just using an ITap lab, no dedicated space.
    a. If they had dedicated space, more open testing.
    b. Currently 4000 students through the course of an academic year.
6. Current about 36 stations in the computer lab.
7. There is a push to move towards online testing.
8. Currently run by Jason and 3 to 5 TA's.
    a. It is possible that TA's could run testing center.
9. Biggest demand for the space is summer.
    a. Have 150 to 300 students a day.
    b. Placement exams from 10 to 4 during the summer.
    c. Current setup in summer has que line in hallway outside of room. Students are brought in and screened. (takes about 7 to 10 minutes per student) This is very important to the testing.
    d. Test can take from 15 minutes to 1 hr.
    e. Student receive immediate results of pass or fail.
    f. Once complete students raise their hand when done to discuss their score at the station and escorted out.
    g. Extra noise from other students hanging around waiting fro rest of group to finish testing.
10. Ideal if they could more to another place to debrief test scores.
11. Interaction with students needs to be maintained
12. Currently no lockers, so students put belongs under desk.
13. Currently no carrels for testing, feel they are needed.

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.

14. They test around 4000 students during the year.
15. Enter ID to take test.
16. Some equipment carry over to lab, for headphones.
17. For summer placement time is not controlled, students just show up.
18. Understand surround spaces it effects.
19. During the year they test 90 to 100 students.
20. Need a space in the summer for those who are part of the orientation group but are not taking the test. You may have 2/3's of the group testing and the rest sitting around.
21. Currently they do not have accommodations for disabilities other than 2 ADA stations.
22. Currently a server is used, but not aware of where it is.
23. Problems they face is that they need to screen all students to make sure they are both who they say they are and that the language they are testing for is not their primary language.
24. They use a form 231 to show that they have been screened. These forms need to be scanned, currently just hangs onto the archive. Ideal if scanned at screening.
25. They currently assign a student to a station and exam, but there is no registration for users, just show up.
26. Need TA's who speak different languages for sign-in review.
27. Dedicated testing space would allow them to setup more testing possible throughout the year.
28. Since it is not a dedicated testing area, it takes approximately an hour and a half to setup the space for testing purposes.
29. They feeling having the testing in another building could be an issue, you would be outside the environment of the students in the language and culture, and missing out on advertisements etc.
30. They need a dedicated storage space Head phones.
31. If one testing center they will need some home office over there to keep stuff since it would be off site from the language building where the staff/TA running the test would reside.
32. Jason's office is currently in close proximity to the testing lab

Respectfully,
Scott Marchisin

cc: Those present

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.

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Meeting Minutes Data Processing

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 01/29/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
David Nelson	Purdue
Laura Miller	Purdue

ITEMS

1. Introductions
2. Five activities run from this space:
    a. Course evaluations
    b. Proctoring
    c. Scantron
    d. Specialized grade book work for special departments (math and bio)
    e. Use scantron for specialized forms (for research)
3. They handle the bubble scantran
    a. 400,000 to 500,000 scantran are used a semester through their area.
    b. 75,000 to 85,000 per finals week
    c. They have a high capacity scanner. (their old machines sits unused as a backup)
    d. The machine is loud (requires ear protection for the users)
    e. They have a wall installation scheduled to dilute the noise
    f. The machine is very climate sensitive. If it is too humid the sheets swell and machine does not run properly.
    g. The space needs to be climate controlled and dust free. (Currently bad humidity control. There is a work order for them to replace the current HVAC system for the space)
4. They handle walk in business.
    a. The scantron machine needs to be secure.
    b. Currently there is keypads to all doors for security.
    c. Counter service for grading area. Needs to be secure but quick access to the machine.
    d. All scantron are recorded electronically.
    e. They have to keep all scantrons for a month after the semester.
    f. Current space is big enough, actually larger than needs.
    g. Secure office to maintain 4 staff with workstation. Open office collaborative space
    h. One secure office with meeting space.
    i. Current space has small break room.
    j. Storage supplies
    k. Needs to have proper database hookup – semester evaluations they provide 650 distinct surveys each semester.
    l. Conference rooms space for 6 or 7. Training for 10 or less. A space of 12x21 would work.
5. TA proctoring is a service that they have entering and exiting the space.

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.

6. Location can be in the same building but not in busy student area.
7. Based on a note after the meeting, they have never had access to natural light and would like natural light.

Respectfully,
Scott Marchisin

cc: Those present

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www.structurepoint.com

Meeting Minutes Digital Test Center

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 01/29/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Kevin Flanagan	Purdue
Maricel Lawarence	Purdue
Michael Eddy	Purdue

ITEMS

1. Introductions
2. Biggest issue is the noise from everywhere else. It is even louder than the white noise they have in the space.
3. Kevin runs the day to day activates.
4. Walk-ins discouraged
5. The really big issue is the half glass wall is not low enough to allow full vision into the space.
6. Cameras are in the room, old cameras with limited range.
7. This space does have student's schedule to use it.
8. Final exams that are done here are make-up exams, especially for online courses.
9. The common tests are a default of an hour and a half in length.
10. Need to make sure when other tests are provided here that they have been instructed on what is allowed.
    a. Vendor tests have different requirements, involving seating, glass, camera's etc.
11. 40% of the people who come are taking paper tests, remainder is online.
12. They verify the student by their id and have a sign in/sign out log.
13. The handle restroom breaks by telling them if timed test, the timer does not stop. They make sure they do not go to the locker.
14. When it is an online exam they use a lock down browser. Only if instructor tells them not to lock down the computer will they leave it fully accessible.
15. Student's rarely use restroom during testing.
16. They keep video from the testing users for 30 days. ITEP keeps the access to video.
17. There are currently 25 stations. Never really full. Only on 80% full on Saturday. Average is 5 or 6 at a time.
18. 1 student desk is wheel chair accessible.
    a. It is mostly used for tall students, example basketball player.
19. They can handle disability students depending on the need.
20. Test parameters Pearson, Casa, did not meet requirements.
21. Offer 4 types of tests, Purdue students are not charged. Pearson exams are charged by Pearson. Casa exams are charged. They collect cash elsewhere in building and provide receipt. Also can collect via credit card.
22. Scheduling is done in outlook.
23. They had only 4 students the day we visited with as the semester goes on maybe 10 to15 a day.

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.

24. They currently have the software on only one company from a vendor. If multiple vendors, than multiple computers needed.
25. They avoid scheduling make-up exams.
26. Medium size locker is used most often, almost never use larger locker. Student holds possession of the key. Student sign out and give key back.
27. Students receive a pass or fail on Pearson and Casa test.

Respectfully,
Scott Marchisin

cc: Those present

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.

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Meeting Minutes DRC Testing Unit (revised)

Project Name: Potter Center Renovation study
Project Number: 2015.02663.0001
Meeting Date: 01/29/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Dan Carpenter	Purdue
Mary Ferkis	Purdue
Caren Hanley	Purdue

ITEMS

1. Introductions
2. The DRC has 1200 registered students, about 1/3 of those students use the testing services.
3. They currently have 22 seats for testing in the bullpen. It is divided into cubicles. They have 2 private rooms. The most students they can handle currently is 24 unless they borrow other space.
    a. All exams are scheduled. The scheduling system is online home built system. Students usually have to put a request in 3 days in advance.
    b. In the course of one week they could have 90 students use the services.
    c. They are currently 5 days a week, with two days during the week being from 8 to 8:30pm. They do not work weekends, holidays or breaks.
    d. During finals week they proctor the tests in four different locations and gave over 650 exams. All these exams are paper and pencil.
    e. 2500 exams are proctored for a semester.
    f. All testing is done in support of academic term.
    g. Some students need private rooms for testing, they cannot have any distractions. They prefer these rooms to be video recorded.
    h. During finals testing 10 to 12 private rooms are needed for testing.
    i. If they give 90 exams in one day they are unable to process for the next day.
4. Process for a student to come take a test at the DRC once the test is scheduled:
    a. Check in with ID
    b. Student told what items they are allowed to have with them for the test.
    c. Belongings put in a locker
    d. Assigned a seat number
    e. Go into testing center and take exam
5. Reader scribes are hired students.
    a. Almost all testing for the DRC is done as paper tests. There are a couple of laptops being used recently.
6. The Braille emboss equipment is very noise.
7. Spaces needed:
    a. Room to que students.
    b. Room to prepare for tests
    c. Room to take a break.

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.

   d. Internal rest facilities
   e. Private test rooms
   f. 30 to 35 student test rooms. (Currently 22)
   g. Need an isolation room for students who have meltdown.
8. DRC offices etc.
   a. 10 total offices
      i. 6 accommodation specialists' offices with desk and small group meeting for four. (staff, student, and parents)
      ii. Director office
      iii. 2 assistant directors
      iv. 1 academic service provider coordinator office
      v. All offices to have panic button
   b. Reception for 2 staff.
      i. Area secure but separate from offices
      ii. Copier in a separate work room behind the reception desk.
      iii. There is concern about confidential documents.
      iv. Currently their storage consists of 4 five drawer filing cabinets.
      v. Reception to have a panic button.
      vi. Area needs to be seen as welcoming. Current space is tight.
      vii. Need ability to have discreet conversations.
      viii. Should be accommodating to all with ADA at reception
9. Need to have universal accessibility beyond code.
10. Sound is a big issue.
11. Alt format production – has 2 staff and 3 student workers. Need to be cooled. Work area currently too tight. Students come and go from format area.
12. Math would like DRC to handle all the accommodated testing, but DRC doesn't have the capacity.
13. The need for a conference room large enough to hold all staff. It should be set up with ability for computer access/projection for webinars and such.
14. Break room for staff with kitchen area.
15. New space: Room for a commons area for students (like Veteran's Success Center)

Respectfully,
Scott Marchisin

cc: Those present, design team

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Meeting Minutes Purdue Testing Center

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 01/29/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Dan Carpenter	Purdue
Cindy Fields	Purdue

ITEMS

1. Introductions
2. They started mapping a centralized testing center in the early 2013. There has been a lot of changes since. Cindy was part of the 2013 discussion.
3. Making sure that we have adequate lockers and queing space is required.
4. Prometric and Pearson testing requires a separate space during breaks.
5. Pearson is a testing Vendor.
6. Pearson has different requirements for space, processing, security, camera angles.
7. Distance learning took different tests because of there not being enough room.
8. They provide an engineering test that takes 6 hours. The break area would need to serve snacks. They receive a 25 minute break.
9. Pearson if you have 15 or lease people you have to have 2 proctors. If over you require 3.
10. For prometric or Pearson they are only allowed to use their own cameras at each station. May require multiple camera's at each station due to this.
11. Pearson you are required to have a picture, a palm vein scan, and wanding.
12. Need location for pictures to be taken. If you are going to allow 40 or more to test you would likely need two photo stations.
13. The engineering exam can run the full 15 seats and could use more and would likely fill them.
14. CASA tests are year round.
15. MCAT selected dates.
16. LSAT paper pencil test.
17. Microsoft is year round.
18. PTE exam has speaking and hearing exam.
19. Highest day is 28 to 32 students. Typical start times are 8:30 or 12:30.
20. Pearson's test are 9:30 to 4:30 M, W, TH 12:30 to 8:30pm, and Friday afternoon.
21. Outside of the facility advance credit Chem – paper pencil.
22. Use Hampton hall for the paper pencil test of 40 to 80 students.
    a. It needs to be classroom style, no tiers.
23. Prefer no carrels for test, need sight line to the person, with all students facing one way.
24. Pearson has specific test center requirements of Carrol there and okay to face each other.
25. The tests here are prepaid.

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26. Worst case scenario if 15 students show up, line up in hallway and unload all items. Reading materials to read before testing. Get in line for admittance. Give signature, palm vein, photo, check if need to use restroom, lead into room.
    a. To pass through security it is lucky if done in 5 minutes.
    b. If restroom is currently needed proctor has to walk to restroom with student. When coming back they have to perform a self-pat down and wand.
    c. Students told to report 30 minutes prior to exams.
    d. For the Pearson English test all students have to take at same time.
27. All users schedule the test.
28. They have their own server for Pearson and it has to be lock in closet. It currently in ITEP with other servers.
29. Check person has personal computer, 3 monitors, 2 DVR's and two towers, scanner, camera for pictures, signature pad, thumb print, plus swipe card.
30. Supposed to be able to accommodate scooter, wheel chair, scribe, or reader in private rooms.
31. Secure room storage required for national storage. No ceiling tile or windows.
    a. Contains LSATS and paper pencil national exams and secure files and maybe equipment
    b. Should be in close proximity, but does not need to be next door.
32. GMAT has need for multiple bathrooms with limited break. They are only aloud 8 minute break so need for multiple restroom fixtures for quick use.

Respectfully,
Scott Marchisin

cc: Those present

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Meeting Minutes CoE

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 02/05/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Robert Frosch	Purdue
Donna Ahlen	Purdue
Brad Bowen	Purdue

ITEMS

1. Introductions
2. Priorities of this study
    a. Support resource programs
    b. New welcoming front door that engages centennial mall area
    c. Way to update building – natural light, more aesthetically appealing
    d. Efficiency of space through renovation
    e. Synergy and consolidation of resources.
3. Accuracy of SLAM study – still correct.
4. ENE might have to be housed in Wang instead of Potter.
5. EEE should go in Potter.
    a. With the wet/dry teaching labs which the space
    b. Will need business office – for all engineering business office support
        i. Space with Manager
        ii. Several support staff (4 to 6) staff in open cubicles
        iii. Not expecting business office to handle money.
6. Want student centered activities all to be kept together.
7. Resource organizations:
    a. OPP – Office of Professional Practice.
    b. ENE – Engineering Education
    c. EEE – Environmental Ecological Engineers
    d. GEP – Global Engineering Program.
    e. OPP – Office of Professional Practice (Coop program)
    f. FYE – First Year Engineering (same as Freshman engineering)
    g. OFE – Office of future Engineers (Recruiting arm)
    h. WIEP – Women in Engineering Program
    i. HON – Honors in Engineering
    j. MEP – Minorities engineering program
    k. ENE Adv – Engineering education Advising (managing over 2000 kids)[most stay in building]
8. Potter is seen as a good spot team for these organizations. Wants to build and emphasize the relationships between organizations.
9. The enrollment growth continues to increase yearly for the organizations.

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10. Honors College is mostly engineering from campus but a program within Honors for campus. Schrieve holds honors classroom space and could not house in lab.
11. Needs a reception space for all the groups located on the first floor that could then send individuals to their correct location in the building.
12. Having the resource organizations in Potter is priority one.
13. Office spaces in the building are mainly to be setup as open with focus booth spaces for privacy. Similar to Grissom.
14. Look at shared research.
15. Need to have a named conference room.
    a. Name for conference room is:"K.S. Fu".
    b. It should be large with seats 50 people. (max people 80)
    c. To have video conferencing
16. Commons to be named:
    a. Malott Commons
    b. Considered a laptop space
    c. Little tables for formal teaming
17. ENE will just have lab and offices, no research here.
18. Offices should belayed out at 120 sq ft for advisor with sliding door.
19. Grissom for grads had either a 6x6 or 6x8 space.
20. Staff is to have 8x8 space
21. Hot desking for students. Have 2 lockers for every mobile seat. Ration 1.7 for people to desk.
22. There are resident stations are held by one person.
23. Third floor was meant for EEE offices in the SLAM study. Around the perimeter would be open space similar to Grissom on the second and third floor.
24. Everything currently in Potter will be displaced.
25. All research is being displaced. Labs could be setup for teaching wet labs.
    a. Currently Potter has at least 24 hoods.
    b. Labs have reached their useful life currently in Potter. It has great electrical power hoods need updated ductwork. The exhaust ductwork may have been eaten through.
    c. It is their belief that it is cheaper to put new lab building up vs renovate. They could more economically modify Potter into an academic building.
26. Study will capture the places displaced from Potter.
27. Reimaging
    a. Could see entry at two different levels with grade changes.
    b. Open atrium area connecting the two floors would allow organizations on second floor to still have connection to first floor.
    c. Open feeling
    d. Natural light
    e. Welcoming
    f. Act as a front door to engineering from Centennial mall & Potter Quad
    g. Multiple doors
    h. Can take cures from Active Learning.
28. Partial reimaging
    a. Not preferred
    b. It would only be surgical approach – pick and choose.
    c. Entry way is important
    d. Basement and first floor should be full gut
    e. If you don't get the entire building it will bail, will not get the right environment. They will not have focus booths etc.
    f. Elevator should be more pedestrian friendly closer to entry.
    g. Existing restroom seem inadequate and fixture count is too low, no equality between men and wome

29. Privacy room also used as lactation room.
30. Unisex restroom (one for building not per floor, also it should have a changing table has become campus standard)
31. Brad wants to keep lab space as lab space where possible since it's infrastructure is in place.
32. In masterplan the plan was to move all the research outwards.
33. Is the age and condition of the labs affecting rectruiting? Not really.

Respectfully,
Scott Marchisin

cc: Those present

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Meeting Minutes Pre 50% meeting

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 02/05/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Robert Frosch	Purdue CoE
Frank Dooley	Purdue Provost Office
Maricel Lawarence	Purdue Digital Education
Phil Knobloch	Purdue AV
Ryron Reed	Purdue ITAP
Brad Bowen	Purdue
Dan Carpenter	Purdue Student Success/DRC
Cindy Fields	Purdue Test Center
Kim Rechkemmer	Purdue Space Management
Madeleine Henry	Purdue SLC

ITEMS

1. Brandon reviewed the bubble diagram of the testing center as to our understanding of the process.
    a. It was noted that the locker area may need to move further into the sequence do to Pearson people are allowed to have access to some belongings accessible on breaks.
    b. After breaks Pearson requires you to check back in after leaving the room.
    c. It was discussed looking at a second set of lockers near the Pearson testing area.
    d. It was noted that the private testing rooms could be used for de-briefing rooms.
    e. Pearson needs its own break room.
2. Pearson testing requires people entering testing needing to be led to their seat by a proctor. If the student needs to get up or finish the exam they need to raise their hand and wait for a proctor to lead them out.
    a. Students if they go to lockers need to be supervised so that they are not accessing items not allowed.
3. Pearson tests take place at designated time. No one else can test in that room while the Pearson tests are going on.
4. Pearson is 15 students at a time currently. If they have 15 people testing they need to proctors. If they have 16 or more they need 3 proctors.
5. Testing demand information was provided to team from 3 years ago.
6. For standard testing a room for 200 seats from 8 to 5, five days a week they could keep the space full.
    a. This would relieve an academic classroom being over 200 seats.
7. Pearson tests during finals currently so this could be an issue of scheduling in the future.
8. DRC wells during finals.
9. It was determined that we design for weekly needs and we would not be designing.

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10. We need to think about the faculty.
    a. Need a concierge setup for professors to come and drop off their test information.
    b. Needs to be separate from the student area because the professors do not want to wait in same lines, something quick and simple. One stop shop.
11. Need area for walk in students for career access which could tie in with concierge.
12. Concierge should have computer to allow teachers to sign in for tests.
13. Digital assistance technology has two people from the DRC along with other departments that could possible all be combined. That would be for 6 people plus equipment.
14. To make the space flexible need computer carrels that allow the computer to recess and secure away.
    a. Look into see if there is a system that the walls of carrels could recess or remove.
        i. This would allow the computer area to become a meeting place once a week for 100 kids and parents for meeting and discussion.
        ii. Look at Steelcase to see if they have anything like what is described.
15. ENE needs to at least have advising in Potter.
16. ENE facility offices to stay in Wang. There are also more in There Armstrong building.

Respectfully,
Scott Marchisin

cc: Those present

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Meeting Minutes Testing

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 02/16/2016 (Conference Call)

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Frank Dooley	Purdue Provost Office
John Harbor	Purdue
Dan Carpenter	Purdue Student Success/DRC
Kim Rechkemmer	Purdue Space Management

ITEMS

1. Frank's high level view is one stop testing center instead of multiple around campus.
    a. This will help faculty understand which facility to use.
    b. Creates a one stop shop for both students and professors.
    c. One place of service
        i. For professors it is currently hard to determine which testing center they should be sending students too.
    d. Should allow for the opportunity to provide better service with all the similar services in one location.
    e. Create a great opportunity for more support
    f. Allow testing to concentrate on academic integrity.
2. This is seen as a test center for standard academic classes too.
    a. Need a space of 150 to 200.
    b. They had talked about test center being able to handle 1500 students
    c. The 200 student mark would allow them to address some of the largest courses on campus.
    d. Could be in separate rooms next to each other and IT link them together.
3. Data processing moving to be part of Potter would continue the one stop shopping with their involvement in testing.
    a. If the scan tron machine does not move over, at least need a space that acts as satellite location to collect tests etc. and then deliver them to the main location a couple times a day.
    b. Allow to drop off testing at front door
    c. There is an understanding of the limited space and the whole facility may not move over but needs some type of representation.
4. The combination of all the centers would allow them to have one manager
    a. Be accessible
    b. Shared proctoring
    c. In the heart of campus

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   d. Near the active learning center
5. Testing on computers is the future
   a. Currently some professors are having to proctoring with computers being automated, with the computer's camera monitoring them at their remote locations.
      i. Some online students have to pay $15 for the online proctoring service while if on campus and the testing center was open it could be free.
   b. There has been a movement to online testing, DRC actually has had to get a few laptops for testing this year.
6. The 200 seats is equal to a standard class size of rooms on campus and would help alleviate the academic demand on space.
7. First spaces that go during scheduling are the larger size rooms.
8. Math has started to try to do testing online.
9. The 100 plus student academic spaces currently on campus are not setup for testing, they are a lecture environment.
   a. A test environment has more privacy, reduces ability of cheating
   b. Need dividers on the desks
10. Someone should ask Penn State if they had to do it over again would they have 161 students in one huge room.
    a. Talk to Deb Howard.
    b. It took them a few years to get to use the facility at capacity.
11. If the space was sized for 100 it would cover at least half of the requests on campus for testing space.

Respectfully,
Scott Marchisin

cc: Those present

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Meeting Minutes 50% meeting

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 02/05/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Maricel Lawarence	Purdue Digital Education
Phil Knobloch	Purdue AV
Ryron Reed	Purdue ITAP
Brad Bowen	Purdue
Dan Carpenter	Purdue Student Success/DRC
Cindy Fields	Purdue Test Center
Kim Rechkemmer	Purdue Space Management
Deb Johnson	Purdue
Randall Ward	Purdue DRC
Donna Ahlen	Purdue

ITEMS

1. Goals of the meeting
    a. Confirm the spaces of the program
    b. Confirm CTC testing area size and spaces
2. Option A
    a. Presented where DRC would go if it was to go in early
        i. Using loading dock ramp as main entry
        ii. Program space vs. available space, more program space than space available.
    b. Noted more negatives than positives with the early scenario.
    c. Everyone was told that the President had relaxed the idea of them moving in early, so based on that and the negatives involved option A can be dropped from any further study. We will keep it in the final study but will not that it does not accomplish the goals and for what reasons.
    d. Should concentrate on Option B and C.
3. DRC to provide trend data concerning growth for Purdue and nationally to include in the study.
    a. We need to maximize the use today, but prepare for future growth.
4. Next Thursday the testing units are going to get together to define the testing area and how many staff it will require to run.
5. It was discussed where proctors need to be in the testing center.
    a. It was noted that in the special testing area that this was defined by each type of test and number of students.
    b. In regular test center it would be what was felt to be adequate to cover the room.
6. All campus testing to occur through the central testing center is the goal.
7. During the meeting the mezzanine and associated conditions were discussed.

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   a. If the mezzanine is removed it affects the basement, mezzanine, and first floor. The system in place supports the first floor above it. When removed, we would have to look at putting in a new independent structural system with columns to support any new floor going back in.
   b. If the mezzanine left in, than there would be closely spaced columns in the basement and mezzanine level. The mezzanine level would not be useful since it has less than 7'-6" clear head room and it would be a change of use.
   c. Purdue would like to understand the cost comparison and impact of leaving it in, vs taking it out.
8. Testing center will need a lot of acoustical treatment.
9. Concern was expressed over 200 people with computer load in the space. The team will address.
10. The assistive technology center – we know nothing about it. Dan was supposed to get us connected with this group (Dean and Dave).
   a. They have a lot of computers, a braille printer, they do not do any testing. They provide the format. The spaces is used to (Alternative format) actually create instructional material.
   b. One private office should be included in the alternative format testing space.
11. Outside of the testing center there will be the need for a compliance office needed in close relationship to the DRC.
   a. Where is this group now? They do not exist yet.
   b. There is one person starting March 14th.
12. Reviewed testing diagrams again.
   a. Overall testing diagram okay
   b. Specialty testing
      i. Adjust check-in desk to L-shape configuration.
      ii. Locate break area, lockers all visible from desk.
      iii. Bathrooms should be located right by entry to testing rooms since if someone needs to use they will have to check out and re-check in again.
      iv. The area by the desk will also be used to stand and take photos for the tests that need it.
      v. The secure zone is the center of everything.
13. In the concierge station for the testing center it was talked about this would be a good place for career testing stations.
   a. Currently have 2 stations would like 4 to 6.
14. Option C was reviewed.
   a. The reception area they want could be combined and as simple as a desk.
15. For the Potter building we need to look at an option D which is look at Lab space vs. current program.
   a. Try to keep the lab space since so expensive.
16. We need someone to review the square footages in the Slam study for the programs to validate since the information is a couple years old that it is still good.

Respectfully,
Scott Marchisin

cc: Those present

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Meeting Minutes 75% meeting

Project Name: Potter Center Renovation Study
Project Number: 2015.02663.0001
Meeting Date: 03/21/2016

ATTENDEE:	ORGANIZATION:
Scott Marchisin	American Structurepoint
Brandon Farley	American Structurepoint
Aaron Hunley	Purdue
Maricel Lawarence	Purdue Digital Education
Phil Knobloch	Purdue AV
Byron Reed	Purdue ITAP
Dan Carpenter	Purdue Student Success/DRC
Kim Rechkemmer	Purdue Space Management
Randall Ward	Purdue DRC
Donna Ahlen	Purdue
Robert Frosch	Purdue CoE
Michael Gulich	Purdue Campus Master Plan

ITEMS

a. A question raised by the design team was does option "B" act as a phase of C or D or is it independent?
    i. After discussion it was preferred as a phase but with the understanding that once the testing center is in the building it would be unlikely to have construction around them since it would disrupt the quiet testing environment.
b. Robert noted that he looked at the renovation to Potter being a complete floor renovation vs just portion of the floors. Example would be "whole first floor".
c. Describe Options B, C, D
    i. Option B – is test center and DRC.
    ii. Option C – is now reusing labs and renovate everything else.
    iii. Option D – is renovate everything more in line with Slam study.
d. It was asked if we are including a new elevator for the whole building as part of option B? Design team will look into this.
e. Option "B" how do we get ADA access to the lower level. Brandon is going to verify the size of the current Library elevator to see if the cab meets ADA requirements.
    i. Purdue expressed some concern over the fact that some of their students use oversize chairs etc so the minimum ADA requirements may not be enough.
f. In option "B" – adding the compliance office as part as the testing center is ideal. The compliance office needs to be setup in a suite instead of separate rooms along a corridor. Allow for connection between offices.
g. Remember to add a large classroom for CoE for large meeting space.

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h. Then year DRC expansion needs to be accounted for by adding another office or two. This space would be used for accommodation specialist.
i. Need to create a schematic floor plan for option "B" for final study, showing walls and doors.
j. B035 in existing building is Lab office support.
k. Purdue expressed a concern of power in the potter building, based on previous work done in the space.
    i. Is it capable of powering all those computers (around 200)?
l. Include proctor stations in computer room layouts
m. When we look at new elevations Robert see's entry as glass area of Library.
    i. Brandon at this meeting presented it at the southwest corner and explained why.
    ii. We will explore the south recessed face of the library for the entry point.
n. Brandon has requested a grading plan from Aaron of the area outside of Potter.
o. Robert see's 2 story space as a way to make entry seem larger and create more spaces connected visually to the entry allowing some space on the second floor to act as an extension of first story. (Per program there may be about 2,000 of unassigned space)
p. Robert would like to see total reskinning of the building.
q. We were directed to continue designing around occupancy figures in the SLAM master plan. Even though they are a couple years old the figures should be close enough. The only unit that potentially would have had some change is EEE since it is continuing the migration to being a school.
r. ENE still could keep faculty offices in Wang if needed. Counselors would need to be in Potter though.
    i. Potter is seen to have the teaching labs for ENE and undergraduate advisors.
s. Robert re-emphasized the SLAM open office spaces shown around the perimeter is what he wants to maintain.
t. Option C was discussed and it is understood that it is not desired by CoE but is a request by Purdue, so it will continue to be part of the study.
u. For the CoE the reception space is a nebulous situation, it could be as much as a ceiling defining the different reception spaces, they are not seen as separate rooms but as an open office environment.
v. The upper floors in Potter may be setup as security zones similar to Wang where after the hours of 8 to 5 you need a card to access. Limit accessibility for the open office spaces. Cannot have open office totally accessible and unsecure.
w. May need to have receptionist station per floor to assist with security.
x. Based on the program, we should be able to get all groups except EEE and ENE on a floor together, they are different animals based on their needs.
y. Counselors are separate from Labs.
z. Conference space between groups can be shared but EEE would want more branding with its movement towards a school.
aa. Space should have a Grissom/Wang feel.
bb. Provide a separate cost to reskin the building. Potter has the same exterior as Psych.

Respectfully,
Scott Marchisin

cc: Those present

American Structurepoint will proceed in reliance on this report. Any discrepancies should be brought to our attention in writing within seven (7) days.



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